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Macroeconomic Conditioning of Loss Given Default under IFRS9: Evidence from the Italian Banking System

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Introduction

Credit risk is one of the main sources of risk for financial institutions, as it arises from the possibility that a counterparty does not honor its contractual obligations. Since the global financial crisis of 2007–2008, this area has become increasingly important in the evolution of the banking regulatory framework, which revealed significant limitations in the measurement models and accounting practices adopted until that moment. In this context, international regulators introduced new rules and methodologies to make financial institutions more resilient and to detect signs of credit deterioration earlier.

From a prudential perspective, the Basel Committee on Banking Supervision (BCBS) responded with the Basel III framework, focused on capital: banks were required to hold more and better-quality capital, so as to better absorb unexpected shocks (Signorini (2021)).

A parallel weakness, however, concerned accounting practices. Existing models relied mainly on historical information and losses already incurred, and they therefore struggled to capture forward-looking information and expected future macroeconomic developments (AIFIRM (2016)). To address this gap, the International Accounting Standards Board (IASB) issued the International Financial Reporting Standard IFRS 9 – *Financial Instruments*, replacing the previous IAS 39 standard.

The latter, based on an *incurred loss* approach, allowed losses to be recognised only when objective impairment events occurred, resulting in excessive delays in the recognition of credit deterioration. By contrast, IFRS 9 requires financial institutions to estimate expected losses over the entire lifetime of financial instruments, incorporating forward-looking macroeconomic scenarios into the entire risk assessment process (AIFIRM (2025)). Consequently, the transition from the *incurred loss* model to the *expected credit loss* framework has led to a profound revision of credit risk measurement models, strengthening the integration between accounting processes, risk management functions, and planning and control activities. It has also made it necessary to develop tools capable of consistently linking macroeconomic variables to credit risk dynamics (Canton et al. (2021)). Within this context, satellite models have emerged as key instruments, specifically designed to translate macroeconomic scenarios, often provided by supervisory authorities or internally developed by banks, into forward-looking estimates of the underlying credit risk parameters.

In light of these considerations, this thesis aims to analyse the role of satellite models within the IFRS 9 framework. The underlying idea of the research is inspired in particular by the study of

Crook & Bellotti (2012), which analyses a real dataset relating to retail credit card exposures in the United Kingdom, and highlights how the inclusion of macroeconomic variables, together with exposure-specific characteristics, can improve the explanatory and predictive performance of Loss Given Default models. While their analysis combines this macroeconomic component with micro-level information on individual borrowers and exposures, the present thesis focuses specifically to the macroeconomic dimension of LGD, relying exclusively on aggregated, publicly available data.

Furthermore, the methodological approach adopted has been influenced by the experience gained during my internship at PwC Italy as a *Risk Capital and Reporting Intern*. The purpose is therefore to adapt the main industry practices adopted by financial institutions for the construction of satellite models to a different application context.

Building on this narrower, macro-only perspective, the thesis sets out to answer three closely related research questions. *Which macroeconomic factors drive the dynamics of LGD for Corporate, SME, Retail, and Financial Institutions exposures? Does each of these segments exhibit its own distinct sensitivity to the economic cycle? Are macroeconomic variables alone, without any borrower-specific information, sufficient to explain a meaningful share of LGD dynamics?*

Particular attention is devoted to the key methodological choices underlying the model construction, including the selection of explanatory variables, the transformations applied to the data, and the definition of forward-looking scenarios used for LGD projections consistently with the forward-looking requirements introduced by IFRS 9. The objective is to reconstruct, step by step, the various stages of the modelling process, using SAS software, which is widely employed in banking industry practice for these quantitative analyses. As a result, the final model specification is not defined *a priori* but emerges from a progressive process of selection, validation, and comparison among alternative solutions, reflecting the fact that building an IFRS 9 satellite model is a complex process, shaped by trade-offs between explanatory power, statistical robustness, and economic interpretability.

Underlying this is one of the key characteristics of the current IFRS 9 framework: the absence of a universally recognised optimal methodology. Despite this flexibility, important implementation challenges remain: the increasing volatility of the macroeconomic environment and the emergence of new sources of risk, including climate-related and geopolitical risks, require satellite models to be continuously integrated, refined, and updated.

For this reason too, the topic remains one of the most relevant and continuously evolving areas within the IFRS 9 framework (AIFIRM (2025)).

1. Literature Review

1.1 From IAS 39 to IFRS 9

Before examining the theoretical and methodological aspects related to LGD estimation and its adjustment through satellite models, it is useful to review the evolution of the accounting framework and the main parameters used in credit risk measurement.

1.1.1 From "Incurred Loss" to "Expected Credit Loss": The New Forward-Looking Approach

As previously mentioned, IAS 39 was based on an "incurred loss" model, according to which credit losses could only be recognised once objective evidence of impairment had emerged, such as a default event or a significant delay in repayment. In other words, no impairment could be recognised until a trigger event occurred, so losses were recognised later than the actual deterioration in credit risk. This delay had two distinct implications. At the system level, since impairment could not be anticipated, the mechanism tended to underestimate risk during periods of economic expansion and to concentrate loss recognition during downturns, reinforcing the procyclicality of the banking system. At the level of individual institutions, provisions, once recognised, were often too small relative to the deterioration that had already taken place before the trigger event – a pattern known as the "too little, too late" phenomenon (AIFIRM (2025)), which left institutions with inadequate provisions when facing a crisis.

This concern was not merely academic: following the global financial crisis, the G20 summit of April 2009 (G20 (2009)) explicitly called on prudential and accounting regulators to introduce measures capable of reducing the procyclicality of loss-absorbing capacity in the banking system. Responding to this mandate, the International Accounting Standards Board (IASB) issued IFRS 9 in July 2014 as a replacement for IAS 39. The new accounting standard officially came into effect on 1 January 2018.

The new approach, based on the forward-looking estimation of expected credit losses (*Expected Credit Losses*, ECL), requires financial institutions to account for expected future economic developments when assessing credit risk. Impairment allowances are no longer based only on a borrower's current financial condition; instead, they require a probabilistic assessment that considers multiple macroeconomic scenarios and their impact on credit risk parameters (AIFIRM

(2025); Canton et al. (2021)). In this context, the traditional parameters used in credit risk measurement, namely Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD), play a central role in the estimation of expected losses. The so-called satellite models derive from the need to project these parameters forward and link them with changes in macroeconomic conditions.

At the same time, as briefly discussed earlier, the *principle-based* nature of IFRS 9 gives financial institutions considerable discretion in defining estimation methodologies, establishing criteria for the identification of a Significant Increase in Credit Risk (SICR), and developing forward-looking scenarios. This flexibility reflects the fact that IFRS9 defines the objectives of the assessment process without prescribing a specific methodology. Instead, institutions are required to make use of all reasonable and supportable information available, including expectations regarding future economic conditions. The Basel Committee has also clarified that the reference to "information available without undue cost or effort" should not be interpreted restrictively and does not exempt banks from investing in adequate data, systems, and processes to support forward-looking assessments (AIFIRM (2016, 2025); Canton et al. (2021)). Over time, this approach has resulted in considerable heterogeneity in the practices adopted by European financial institutions: the EBA's 2023 monitoring report on IFRS 9 implementation across EU institutions explicitly documents how the latitude embedded in the standard has led institutions to adopt markedly different approaches in estimating ECL (AIFIRM (2025)).

This heterogeneity became even more pronounced during periods of heightened macroeconomic uncertainty. A clear example can be found during the COVID-19 pandemic, when many institutions relied not only on quantitative models but also on *management overlays* and other management adjustments aimed at incorporating risk factors that were not fully captured by statistical methodologies. This experience highlighted the need to supplement model-based evidence with an expert judgement component that is appropriately documented and subject to effective governance, a pattern also documented empirically through a dedicated survey of practitioners (AIFIRM (2025)).

Subsequently, the European energy crisis, the sharp rise in inflation and interest rates, persistent geopolitical tensions, and the growing importance of climate-related and environmental risks have further increased the complexity of the assessment process. Incorporating these factors into forward-looking estimates has become one of the main challenges in the implementation of IFRS 9 and requires models capable of capturing increasingly complex relationships between the macroeconomic environment and credit risk parameters (AIFIRM (2025)).

1.1.2 The Main Credit Risk Parameters

Central to the IFRS 9 framework is the concept of *Expected Credit Loss* (ECL), defined as the present value of the cash flows that a financial institution does not expect to collect, estimated through a probabilistic methodology that considers both the probability of default and the

magnitude of the associated loss. In practical applications, this quantity is expressed as a function of three key parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD) and the precise expression depends on the specific credit exposure under consideration, on the corresponding time horizon applied, and, more generally, on the institution's own implementation choices. In every case, to ensure that loss estimates are not based solely on historical performance but also incorporate current and anticipated information about exposures, ECL is calculated as a probability-weighted average across different macroeconomic scenarios:

$$ECL = \sum_{s=1}^{S} w_s \cdot ECL^{(s)} \tag{1.1}$$

where w_s represents the probability assigned to scenario s, and $ECL^{(s)}$ denotes the expected loss estimated under that specific scenario.

ECL values play a role in several respects. Directly, they determine the impairment allowances recognised on credit exposures in the financial statements, and are therefore central to the IFRS 9 accounting framework for credit risk. Indirectly, by affecting the level of these allowances, ECL estimation also has an impact on financial institutions' capital and prudential indicators (Canton et al. (2021); AIFIRM (2016)).

From a quantitative perspective, to provide the most realistic estimate possible of the losses that an institution expects to incur over the relevant time horizon, expected credit losses must be recognised from the initial recognition of a financial instrument, regardless of whether a default event has occurred. They are then updated throughout the life of the exposure as the counterparty's credit risk evolves.

Consistent with this approach, IFRS 9 introduces a credit risk classification mechanism known as *staging*, which determines the time horizon used to calculate expected losses:

- **Stage 1** includes exposures that have not experienced a significant deterioration in credit quality since initial recognition; in this case, expected losses are calculated over a 12-month horizon;
- **Stage 2** includes assets that already exhibit a Significant Increase in Credit Risk (SICR); for these exposures, expected losses are estimated over the instrument's remaining lifetime;
- **Stage 3** includes credit-impaired assets: exposures for which default events have already occurred, or whose expected cash flows have been adversely affected, for example by missed or delayed payments. For these exposures, expected losses continue to be estimated on a lifetime basis, though typically with a higher probability of default.

The transition between stages is driven by the assessment of a *Significant Increase in Credit Risk* (SICR), the identification of which moves an exposure from Stage 1 to Stage 2, shifting the measurement basis from a 12-month expected loss to an expected loss calculated over the entire remaining life of the instrument – the *lifetime ECL* (AIFIRM (2016)). The allocation in the

three stages, so, doesn't look at default risk in its absolute level only, but considers also changes in default risk relative to the exposure's origination date. This mechanism is supported also by qualitative indicators that can flag deterioration not yet visible in statistical risk measures – payment delays, *forbearance* measures, and early warning signals among them (AIFIRM (2016)). Precisely because this assessment combines quantitative and qualitative indicators in this way, IFRS 9 does not define a single set of operational thresholds for identifying a significant increase in credit risk. Instead, it requires institutions to draw on all reasonable and supportable information available, *without undue cost or effort*, to assess collectively which exposures should be reclassified and measured on a lifetime expected credit loss basis (International Accounting Standards Board (2014)).

The *Probability of Default* (PD) is the fundamental link between a counterparty's risk profile, the staging process, and the measurement of expected losses. It represents the probability that a counterparty will enter a state of default within a given time horizon.

The estimation of this parameter itself is one of the main differences between the IFRS 9 accounting framework and the prudential regulatory framework.

Within the latter, internal IRB models are typically calibrated with reference to *Through-the-Cycle* (TTC) probabilities of default, based on long-term average default rates (*Long-Run Average*). These are designed to remain relatively stable over time, thereby limiting excessive volatility in regulatory capital. IFRS 9 takes a different approach: since its objective is to estimate expected losses in light of current and future economic conditions, the standard requires *Point-in-Time* (PiT) probabilities of default, which promptly reflect changes in both the economic environment and the risk profile of counterparties. The PDs used for accounting purposes are, as a result, more sensitive to the economic cycle than their regulatory counterparts, a requirement made explicit by the Global Public Policy Committee (GPPC): *"A PD used for IFRS 9 should reflect management's current view of the future and should be unbiased (i.e. it should not include any conservatism or optimism)"* (Global Public Policy Committee (2016)). The probability of default used for IFRS 9 purposes should therefore represent the best possible estimate of expected risk, free of regulatory prudence, overly optimistic assumptions, conservatism margins, regulatory floors, or other prudential adjustments.

In general terms, PD can be defined as:

$$PD(t) = P(\tau \leq t) \tag{1.2}$$

where τ denotes the time of default of the counterparty and t the relevant time horizon. To incorporate the macroeconomic environment, PD can be expressed as:

$$PD_t = f(C_t, X_t) \tag{1.3}$$

where:

- C_t represents the counterparty-specific characteristics, typically available only to the financial institution holding the exposure;
- X_t represents the set of relevant forward-looking macroeconomic variables.

The distinction between counterparty-specific characteristics and macroeconomic drivers applies equally to LGD, the parameter at the centre of this thesis, and is developed in detail in the following chapters.
In banking practice, satellite models are used to estimate this relationship (Equation 1.3) empirically, and the correct identification of the relevant macroeconomic drivers X_t is central to ensuring that PD estimates genuinely reflect the point-in-time logic required by IFRS 9, rather than merely replicating a through-the-cycle average under a different name.

Loss Given Default

As discussed in the previous section, IFRS 9's forward-looking requirement extends beyond the probability of default of the counterparties: financial institutions must also estimate the losses they may incur once a default has occurred, in light of future economic conditions.
To this end, similarly to what happens for PD, financial institutions use satellite models to capture the relationship between macroeconomic variables and LGD and to incorporate the expected effects of economic cycle developments into recovery rate estimates.
As with PD, a key distinction exists between the LGD used for prudential regulatory purposes and the LGD required under IFRS 9. The LGD used within the IRB framework for capital adequacy purposes is calibrated on a through-the-cycle basis, incorporates a prudential *downturn* component reflecting loss severity under stressed economic conditions, and is subject to regulatory floors. The LGD required under IFRS 9, by contrast, must be point-in-time and forward-looking, must exclude the downturn adjustment, and remains unconstrained by regulatory floors (AIFIRM (2016, 2025)). This has a direct implication for the way LGD is estimated for accounting purposes: rather than reflecting a single stressed scenario applied uniformly over time, IFRS 9 LGD must be conditioned on current and future macroeconomic conditions, and updated as these conditions evolve. Since a single point forecast cannot adequately capture the range of plausible future outcomes, this typically requires LGD to be estimated under multiple macroeconomic scenarios rather than a single one.
The accounting standard does not prescribe specific realization probabilities to be attributed to each scenario, leaving institutions free to define these on the basis of their own portfolios.
In this regard, Canton et al. (2021) show that the methodology most commonly adopted by institutions involves using three macroeconomic scenarios (baseline, adverse and positive) and, for the banks analysed, the following occurrence probabilities: 58% for the baseline scenario and 21% for each of the adverse and positive scenarios. Consistently with this evidence and with the probability-weighted logic underlying IFRS 9, a synthetic, forward-looking measure of LGD can therefore be constructed, for each future period t, as a probability-weighted average of the LGD

values projected under the different macroeconomic scenarios for that period:

$$LGD_t^{Weighted} = \sum_{s=1}^{S} w_s \cdot LGD_t^{(s)}, \tag{1.4}$$

where w_s denotes the probability assigned to scenario s, held constant across the projection horizon, and $LGD_t^{(s)}$ the LGD projected under that scenario at time t, according to the general probability-weighted formulation of ECL already introduced in Equation 1.1.
LGD modelling, however, presents additional challenges compared with PD. While PD reflects only the likelihood that a counterparty will default, Loss Given Default incorporates the complexity of the credit recovery process and is therefore influenced by a variety of factors, including:

- the type of counterparty and exposure (e.g. Corporate, SME, Retail, or Financial Institutions);
- borrower-specific characteristics, such as income, employment status, and credit history;
- the existence, quality, and seniority of collateral, together with the loan-to-value ratio and the size of the exposure;
- recovery procedures and recovery times;
- the efficiency of the legal and judicial system in the relevant jurisdiction;
- prevailing macroeconomic conditions.

For this reason, LGD modelling is more complex and requires sufficiently long historical datasets to observe the entire recovery process.
Before examining the methodologies used to estimate Loss Given Default (LGD), it is worth noting that, as a measure of the proportion of an exposure a financial institution expects to lose if a counterparty defaults, it can be defined as:

$$LGD = \frac{EAD - Recoveries}{EAD} \tag{1.5}$$

where:

- EAD represents the exposure outstanding at the time of default;
- $Recoveries$ denotes the present value of the cash flows recovered after default.

Defining the *Recovery Rate* (RR) as the proportion of the exposure that is recovered, similarly, gives:

$$RR = \frac{Recoveries}{EAD} \tag{1.6}$$

and therefore:

$$LGD = 1 - RR \tag{1.7}$$

This relationship highlights the complementary nature of LGD and the Recovery Rate: as expected recoveries increase, the expected loss associated with the exposure decreases. This distinction is introduced here because much of the empirical literature reviewed in this thesis, including Crook & Bellotti (2012), models the Recovery Rate directly rather than LGD, so that the relationship in Equation (1.7) will be used repeatedly when comparing results across studies.

Recovery Times

Among the main determinants of Loss Given Default listed above, recovery times are certainly the most complex, and it is important to be sure to estimate them in the correct way. For example, obtaining a recovery after several years is different from obtaining an immediate one, for the same amount recovered. In the first case, a lower economic value must be recorded, due to the discounting of cash flows and the costs incurred throughout the recovery process.
The European Banking Authority defines *Time to Recovery* as the period between the initiation of formal recovery proceedings and the creditor's final recovery of the non-performing exposure (European Banking Authority (2025)). Together with the recovery rate and legal costs, this indicator is one of the main measures used to assess the efficiency of insolvency procedures across European Union countries. The EBA's analysis shows that longer recovery procedures are generally associated with lower recovery rates: longer periods lead to a greater decline in collateral value, higher administrative and legal costs, and increased uncertainty around the final recovery outcome, and therefore tend to push LGD upward (European Banking Authority (2025)).
The EBA also identifies several institutional factors that shorten recovery times and improve recovery outcomes: the possibility of enforcing collateral without court intervention (*out-of-court enforcement*, meaning creditors can seize and sell collateral directly), the absence of long payment suspensions (*moratoria*, i.e. periods during which debtors are allowed to delay repayments), greater creditor participation in insolvency proceedings through creditor committees, and the availability of preventive restructuring tools for SMEs, such as *pre-pack* regimes, which allow a restructuring plan to be agreed before formal insolvency proceedings begin (European Banking Authority (2025)). Empirical evidence therefore suggests that LGD depends not only on macroeconomic conditions and the characteristics of credit exposures, but also on the efficiency of the legal system and of recovery procedures themselves.

1.1.3 Satellite Models

The considerations presented in this section are primarily based on AIFIRM Position Paper No. 46 (AIFIRM (2025)). As discussed above, satellite models are the main tool financial institutions use to incorporate the *forward-looking* component required by IFRS 9 into credit risk parameters. In the case of LGD, modelling is particularly challenging because the actual loss associated with a defaulted exposure is only observed at the end of the recovery process, which may take

several years. Unlike default rates, observed LGD cannot therefore be directly linked to the macroeconomic conditions prevailing at the time of default. For this reason, some methodologies use observed recovery rates as the target variable and subsequently derive a forward-looking measure of LGD from them, exploiting the complementary relationship between the two quantities already introduced in Equation 1.7.

Developing satellite models also requires selecting macroeconomic variables consistent with both the portfolio under analysis (e.g., SME-Corporate, Retail, or Mortgage portfolios) and the relevant geographical area (Italy, Europe, or global markets), since exposures that appear similar may generate very different recovery rates due to differences in collateral, collection strategies, or recovery timelines.

The choice of explanatory variables should therefore not rest solely on statistical considerations, but should also be supported by a plausible economic relationship with the risk parameter being modelled. To preserve interpretability and limit overfitting, practitioners generally favour models with a limited number of explanatory variables. The time horizon used for estimation matters as well: longer historical samples make it possible to observe different phases of the economic cycle and tend to improve model robustness, but may be less representative of the current composition of the portfolio. Model development therefore requires a balance between historical depth, data quality, and portfolio representativeness.

Data quality is another critical aspect. Relying on consistent, comprehensive, and representative datasets should be a fundamental prerequisite for obtaining robust and reliable estimates of credit risk parameters. In recent years, financial institutions have increasingly relied on internal data rather than industry-wide databases. This approach certainly allows for a more accurate capture of the vulnerability and sensitivity of their own specific credit portfolios. However, it also comes with its own challenges: the availability of sufficiently long historical time series may be limited, and changes in prudential regulation and regulatory definitions, such as the definition of default, may create discontinuities in historical data that need to be properly addressed during model development and validation.

Even when extensive and consistent datasets are available, nonetheless, models are estimated on historical observations and must then be applied to future economic scenarios that may differ significantly from those observed in the estimation sample. This issue is particularly relevant for LGD: as noted in Section 1.1.2, recovery rates are strongly influenced by the macroeconomic conditions prevailing during the recovery process, and during economic downturns, declines in collateral values and the deterioration of borrowers' financial conditions tend to reduce recovery rates and increase observed LGD. A model calibrated exclusively on periods of economic growth may therefore fail to adequately capture LGD behaviour under stressed conditions – a risk that reflects, at the level of individual model calibration, the broader procyclicality concerns that originally motivated the shift away from the incurred loss framework (Section 1.1; G20 (2009)).

The need to properly capture these dynamics has encouraged a wide range of modelling approaches, spanning from traditional econometric methods for time series analysis to more advanced machine

learning techniques. These methods can be used either directly to develop satellite models or to estimate the underlying risk parameters – PD, LGD, and EAD – when these are derived from an institution's internal models. Linear and non-linear regressions, together with ARIMA, VAR, and ECM models, remain among the most widely used approaches, valued for their interpretability and well-established acceptance by supervisory authorities. Machine learning techniques such as logistic regression, random forests, neural networks, and support vector machines have become increasingly popular in more recent years, especially where relationships are complex and non-linear and data volumes are large. Their lower interpretability and the greater challenges they pose for validation and governance, however, mean that econometric models remain the predominant approach within the IFRS 9 framework.

Whichever methodology is adopted, however, its effectiveness ultimately depends on how well it accounts for the heterogeneity that characterises different portfolios. The increasing volatility of the economic environment and the emergence of new risk factors – energy shocks, inflationary pressures, climate-related risks, geopolitical tensions – have further encouraged the development of increasingly granular models, differentiated by customer segment, economic sector, and geographical area. This segmentation is particularly relevant for Corporate portfolios, where sensitivity to macroeconomic factors can vary significantly across sectors: the COVID-19 pandemic, for instance, affected the manufacturing sector very differently from the pharmaceutical industry, and the general rise in prices and interest rates has had a more pronounced impact on firms in sectors characterised by high fixed costs and elevated indebtedness. The need for greater segmentation has also been driven by recent ECB stress testing exercises (European Central Bank (2022), European Central Bank (2023)), which have required an increasingly granular breakdown of results.

A further important aspect in defining the *forward-looking* component concerns the choice of macroeconomic scenarios used in the estimates. IFRS 9 does not prescribe a specific number of scenarios, leaving institutions free to define both the number of scenarios and the probabilities attributed to each, as already discussed in relation to LGD in Section 1.1.2. In practice, institutions typically use between three and five scenarios – a baseline, one or more adverse scenarios, and a favourable scenario – each assigned a specific probability of occurrence. In some cases, more advanced simulation-based approaches, such as Monte Carlo simulations, provide a more detailed representation of macroeconomic uncertainty.

IFRS 9 also requires institutions to explain the assumptions underlying this component adequately: the scenarios considered, their associated probabilities, and the sensitivity of expected losses to the main macroeconomic assumptions are useful to understand the degree of uncertainty embedded in the estimates and to assess their robustness.

The role of satellite models, however, is not confined to the estimation of ECL under IFRS 9. The same models used to project risk parameters under different macroeconomic scenarios are also widely used for other purposes within a financial institution's risk management and planning activities. For example, they are commonly used in stress testing exercises, both

internal (ICAAP and Recovery Plans) and regulatory (such as the EBA Stress Test), to estimate the evolution of risk parameters under adverse macroeconomic scenarios. These models also support strategic planning and budgeting by enabling forecasts of metrics such as default rates, impairment allowances, and Risk-Weighted Assets (RWAs). More broadly, satellite models act as the link between the macroeconomic environment and risk measurement, contributing to the estimation of economic capital, climate and environmental risk materiality assessments, and, in some cases, the pricing of credit products and the evaluation of securitisation transactions.

1.1.4 Determinants of LGD: A Review of the Empirical Evidence

As discussed in the previous section, the effectiveness of a satellite model depends on its ability to correctly identify and capture the relationships between macroeconomic factors and the risk parameter under analysis. In the case of LGD, this first requires understanding the main determinants of loss given default and the extent to which they are affected by economic conditions. The literature generally distinguishes between microeconomic and macroeconomic determinants of LGD.

Microeconomic determinants include all exposure-specific and borrower-specific characteristics and they directly affect both the likelihood and the amount of recovery following a default event. Clear examples include the presence and quality of collateral, the loan-to-value ratio, credit seniority, exposure size, loan type, and the borrower's socio-demographic characteristics.

Grippa et al. (2005) develop empirical LGD models using a sample of more than 20,000 Italian exposures belonging to both households and small firms. Their results highlight significant differences in recovery rates and recovery durations across geographical areas and recovery channels. The multivariate analysis also reveals a negative relationship between the presence of real or personal guarantees and LGD, whereas loan amount is found to be positively associated with loss given default. The set of variables considered, however, is much more limited than the information typically available to a financial institution providing credit cards or other forms of retail lending, and the study does not address the forecasting of LGD.

Querci (2005) similarly analyse LGD using data from an Italian bank and highlight the importance of regional differences in explaining recovery rates. Like Grippa et al. (2005), however, their analysis remains confined to microeconomic and geographical determinants, without incorporating dynamic macroeconomic variables or explicitly considering the effects of the economic cycle.

On the other hand, for what concerns the macroeconomic dimension, numerous studies have shown that a significant share of the variability in recovery rates can be explained by aggregate economic conditions particularly GDP growth, unemployment rates, interest rates, wage growth, inflation, and developments in the real estate market, with these variables affecting mainly borrowers' repayment capacity and the value of collateral. An important contribution in this area is provided by Dermine & Neto de Carvalho (2006), who develop an LGD model for loans granted to Portuguese small and medium-sized enterprises using survival analysis (*mortality*

analysis) techniques and including GDP growth among the explanatory variables. Although the authors do not find a statistically significant relationship between economic growth and LGD, they attribute this result to the limited size of the sample analysed (1995–2005) and to the absence of a particularly severe recession during the period under consideration.

Evidence supporting the existence of a systematic component in recovery risk emerges more clearly in the study of Figlewski et al. (2007). Analysing several macroeconomic variables – including the unemployment rate, inflation, GDP growth, and the industrial production index – in relation to default and rating transitions among rated US corporate issuers, they find that many of these variables are statistically significant drivers of credit risk.

Similar findings emerge in the literature on Probability of Default. Using data from Spanish residential mortgage portfolios, for example, Saurina & Trucharte (2007) show that GDP growth is a significant cyclical variable negatively related to default risk. Although the study focuses on PD rather than LGD, it further supports the view that macroeconomic conditions exert a systematic influence on credit risk parameters.

A further notable contribution is provided by Crook & Bellotti (2012), who analyse a database containing more than 55,000 defaulted credit card accounts in the United Kingdom, drawn from four credit card products and covering the period 1999–2005 – a study examined in detail in Chapter 3, where it is used as a direct methodological benchmark for the present thesis. Their model includes both borrower- and exposure-specific characteristics – such as housing status, income, and credit bureau score – as well as macroeconomic variables, namely UK retail bank interest rates, the unemployment level, and year-on-year earnings growth, tested at leads and lags of six months relative to the date of default. The results show that increases in interest rates and unemployment reduce recovery rates, consistent with a reduced ability of borrowers to repay their obligations, whereas earnings growth, despite carrying the expected positive sign, does not reach conventional statistical significance.

To investigate whether the impact of macroeconomic variables varies with the specific characteristics of borrowers and exposures, the authors introduce interaction terms between macroeconomic factors and individual variables. This assumption has a plausible theoretical basis: for instance, borrowers with higher outstanding balances might be more sensitive to interest rate rises than others. The empirical results, however, tell a different story: the interaction model achieves a marginally higher in-sample fit, but its out-of-sample forecasting performance is actually slightly worse than that of the simpler model without interactions.

Crook & Bellotti (2012) also highlight an important feature of the empirical distribution of LGD, one that recurs across most applications of the parameter and is directly relevant to the choice of estimation method discussed later in this thesis: a strong concentration of observations at the extreme values, such as cases of almost full recovery ($LGD = 0$) and cases of almost total loss of the exposure ($LGD = 1$), giving rise to a U-shaped bimodal pattern that makes the application of traditional linear regression models problematic.

A closely related methodological concern is raised by Qi & Zhao (2011), who compare six alternative estimation methods on a sample of 3,751 defaulted US corporate bonds and bank loans drawn from the Moody's Ultimate Recovery Database over the period 1985–2008. The authors find that the performance of transformation-based methods is highly sensitive to the treatment of LGD observations located exactly at the boundaries of the $[0, 1]$ interval, and that even under an optimally chosen specification, these methods perform no better than a simple OLS regression on the untransformed variable. This evidence reinforces the view that the extreme values of the LGD distribution, rather than its central range, represent the primary source of estimation difficulty, a point revisited when comparing the Logit-OLS and Beta-OLS specifications adopted in this thesis (Section 2.4.1 and Section 2.4.4).

Building on this line of research, Konečný et al. (2017) explicitly investigate the time dimension of the relationship between LGD and the macroeconomy. Their analysis is based on a dataset covering approximately 15% of the Czech consumer credit market over the period 2002–2012. The authors note that Crook & Bellotti (2012) remains, to their knowledge, the only prior study to explicitly consider time effects in LGD modelling, even if restricted to three fixed points in time, namely the date of default and six months before and after it. Konečný et al. (2017) extend this framework considerably, testing lags of up to eight quarters and using elastic net regularisation to select the relevant ones, a technique well suited to deal with the strong collinearity typically found among macroeconomic time series. Their main finding is simple: when the model only looks at current-quarter macroeconomic effects, several variables come out statistically significant, but some have signs that do not make economic sense. Once longer lags are allowed, however, current-quarter variables become largely irrelevant, and it is the delayed variables that turn out to drive LGD the most. This supports the use of lagged macroeconomic regressors in LGD satellite models, an approach also adopted in the present thesis (Section 2.4.1).

Overall, Crook & Bellotti (2012) provide evidence of a systematic relationship between macroeconomic conditions and Loss Given Default. This study, together with the time-dimension evidence discussed above, represents the main reference and focus of the analysis carried out in this thesis.

2. Objective

The literature reviewed in the previous chapter has shown that Loss Given Default largely depends on the macroeconomic environment. At the same time, empirical evidence suggests that the strength of this relationship may vary depending on the type of counterparty and the characteristics of the portfolio under consideration, raising the questions: *Which macroeconomic factors drive the dynamics of LGD for Corporate, SME, Retail, and Financial Institutions exposures? Does each of these segments exhibit its own distinct sensitivity to the economic cycle?*
To address this question, the present study draws inspiration from the work of Crook & Bellotti (2012), while adopting a substantially different approach, shaped mainly by the data available. Without combining borrower- or exposure-specific information, the analysis developed in this thesis relies entirely on aggregated data from European institutional sources, using macroeconomic and financial variables alone to explain LGD dynamics across portfolio segments. This is not just a practical limitation but the central question the thesis sets out to answer: *Are macroeconomic variables alone, without any borrower-specific information, sufficient to explain a meaningful share of LGD dynamics?*
Furthermore, this thesis extends the use of satellite models to forecasting purposes, in order to assess their ability to translate forward-looking economic scenarios into forward-looking LGD estimates, in line with the principles introduced by IFRS 9 and to evaluate how different future conditions may affect the evolution of loss given default across different portfolio segments.

2.1 Data Collection

The analysis is based on aggregated LGD series extracted from the European Central Bank (ECB) Data Portal, which provides risk parameters reported by Significant Institutions supervised under the Single Supervisory Mechanism (SSM) and applying the Internal Ratings-Based (IRB) approach for credit risk measurement (European Central Bank (2026a,b,d,c)). In particular, the analysis uses the *Exposure-Weighted Average LGD*, that is, the average loss given default weighted by exposure. The quarterly series included in the analysis are:

- Exposure-Weighted Average LGD – Corporates;
- Exposure-Weighted Average LGD – Small and Medium-Sized Enterprises (SME);
- Exposure-Weighted Average LGD – Retail;

- Exposure-Weighted Average LGD – Financial Institutions;

The data are available on a quarterly basis for the period from the third quarter of 2018 to the first quarter of 2026 and are aggregated at the national level (Italy) according to the residence of the obligor.
The recommendations set out in Commission Delegated Regulation (EU) 2021/930 (Commissione Europea (2021)) have been consulted for the selection of macroeconomic variables. The Regulation suggests a minimum set of economic indicators that should be generally considered across all exposure categories, including Gross Domestic Product (GDP), the unemployment rate, aggregate default rates, and aggregate credit losses; it is this common set of general macro-financial indicators, rather than the additional portfolio-specific variables discussed below, that directly informs the selection of candidate regressors adopted in this thesis.
Even though this is not the case in the present thesis, given its reliance on aggregated, portfolio-level data rather than exposure-level information, the Regulation, beyond these common indicators, also would recommend portfolio-specific variables tailored to the nature of the exposures under consideration. These include, for example, sector-specific indicators for corporate and SME exposures, real estate prices for property-secured exposures, household indebtedness for retail portfolios, and collateral- or commodity-related indicators for specialised lending, whose relevance is well documented in the literature reviewed in the previous chapter (Section 1.1.4).
More broadly, in this research, while the macroeconomic indicators considered are available over considerably longer periods, the historical availability of the LGD series published by the European Central Bank limits the empirical exercise to the time span for which both the dependent variable and the explanatory variables are jointly available, namely 2018Q3–2026Q1. Within these constraints, and given the objective of comparing satellite models across different portfolio segments, a common set of regressors has been adopted across all categories analysed. The selected variables are intended to capture both the evolution of the economic cycle and conditions in monetary, bond, and energy markets. More specifically, the analysis includes the growth rates of Italian and Euro Area GDP, the unemployment rate, the aggregate default rate, the Euribor rate, the yield on the 10-year Italian government bond (BTP), short-term government bond yields (BOT), and oil prices. Taken together, these variables provide information on real economic activity, labour market conditions, credit-cycle developments, monetary conditions, sovereign risk, and energy-market dynamics, allowing the main macroeconomic drivers of Loss Given Default to be captured while maintaining a consistent modelling framework across all portfolio segments.
The data sources used in the analysis are the following:

- Data on Italian and Euro Area GDP growth are obtained from Eurostat's *Gross Domestic Product (GDP) and main aggregates* database. The analysis uses the quarterly GDP series at market prices expressed in chain-linked volumes, seasonally and calendar adjusted,

measured as the percentage change compared with the previous quarter (*percentage change on previous period*) (Eurostat (2026)).

- Data on the unemployment rate derive from OECD (2026). The quarterly series for the harmonised unemployment rate in Italy are used, defined as the share of the labour force that is unemployed. It should be noted that the labour force includes all employees, self-employed workers, unpaid family workers, and unemployed individuals. Unemployed individuals are defined as working-age persons who do not have a job, are available to work, and have actively searched for employment during the four weeks preceding the survey.

- Oil price data refer to the *Global Price of Brent Crude*, available from the Federal Reserve Bank of St. Louis FRED database. The series is expressed in U.S. dollars per barrel and is sourced from the International Monetary Fund (IMF) (Federal Reserve Bank of St. Louis (2026)). It reports the average quarterly global price of Brent crude oil.

- Interest rate data refer to the gross yield of the benchmark 10-year Italian government bond (BTP). The data are obtained from the *Statistical Database (BDS)* of the Bank of Italy (Banca d'Italia (2026d)). The original series is available at a monthly frequency and represents the gross yield to maturity of the benchmark Italian government bond with a residual maturity of ten years. The data refer to securities traded on the electronic government bond market and, at monthly frequency, are calculated as the simple average of daily yields. To ensure consistency with the frequency of the other macroeconomic variables and the LGD series, the monthly observations have been aggregated to quarterly frequency using the arithmetic average of the values recorded within each quarter.

- The aggregate loan deterioration rate, used as a proxy for the aggregate default rate, is obtained from the *Central Credit Register* of the Bank of Italy (Banca d'Italia (2026a)). The series refers to all resident borrowers excluding monetary financial institutions. Moreover, the use of a deterioration rate based on the number of borrowers, rather than on the amount of credit utilised, provides a measure that is closer to a probability of default, while reducing the influence of a small number of large exposures that could distort the dynamics of the series if exposure amounts were used instead.

In addition, a dummy variable, referred to as *Covid Dummy*, has been included to account for the structural break associated with the COVID-19 pandemic and the extraordinary support measures implemented during the emergency period. More specifically, the variable takes a value of one for all quarters between 2020 and 2021 and zero for the remaining periods of the sample. Its inclusion is consistent with common IFRS 9 modelling practices, where dummy variables are frequently used to capture the effects of the pandemic, including moratoria, public support measures, management overlays, and structural changes in the relationship between macroeconomic variables and credit risk parameters (Giordano (2026)).

2.2 Methodology

Building on the data described in the previous section (2.1), this chapter sets out the methodology used to link the dynamics of Loss Given Default to the macroeconomic environment. Rather than being defined *a priori*, the specification presented below results from the progressive selection and validation process discussed throughout this chapter, and it is applied separately to each of the four portfolio segments in order to allow their sensitivity to macroeconomic conditions to be compared directly.
In general terms, the model can be written as:

$$Y_{j,t} = \beta_{0,j} + \sum_{i=1}^{n} \beta_{i,j} X_{i,t} + \varepsilon_{j,t} \tag{2.1}$$

where:

- $Y_{j,t}$ represents the dependent variable (appropriately transformed observed LGD) for segment j at time t;
- j identifies the counterparty category under consideration (Corporate, SME, Retail, or Financial Institutions);
- $X_{i,t}$ represents the i-th macroeconomic variable observed at time t, with $i = 1, \ldots, n$;
- n denotes the number of explanatory variables included in the model;
- $\beta_{0,j}$ represents the intercept of the model estimated for segment j;
- $\beta_{i,j}$ measures the sensitivity of the dependent variable for segment j to the macroeconomic variable X_i;
- $\varepsilon_{j,t}$ represents the error term.

Since there is no universally recognised methodology considered optimal for LGD modelling, the analysis is based on a comparison of two alternative econometric approaches, both particularly well suited to the bounded nature of the dependent variable. The first approach applies a Fractional Logit Transformation to the observed LGD prior to estimation, using the logit function and subsequently performing linear regression on the transformed variable; consistently with the terminology adopted by Crook & Bellotti (2012), this specification is referred to as Logit-OLS. The second approach instead transforms LGD through the cumulative distribution function of a fitted Beta distribution, followed by the same linear estimation step, and is referred to as Beta-OLS. Comparing the two specifications makes it possible to assess whether the choice of transformation materially affects the estimated relationship between LGD and the macroeconomic regressors, or whether the results are instead robust to this methodological choice.
The research process consists of the following steps:

- Preliminary analysis of the statistical properties of the LGD series and macroeconomic regressors, with particular attention to their distribution, normality, serial autocorrelation, and stationarity;
- Development of an initial set of satellite models using the Fractional Logit Transformation (Logit-OLS). The most relevant macroeconomic drivers are identified through iterative regressions, selecting regressors on the basis of statistical significance, Adjusted R^2, the absence of multicollinearity (VIF), and the economic plausibility of the estimated relationships. To further assess the robustness of the selected specifications, an additional variable selection procedure based on LASSO regression is subsequently applied;
- Development of an alternative approach based on a Beta Transformation of LGD followed by linear regression estimation, with the aim of comparing its performance with the first one;
- Comparative evaluation of the two models in terms of explanatory power, forecasting accuracy, and the statistical properties of the residuals;
- Economic interpretation of the estimated relationships, comparing the empirical evidence with the theoretical expectations associated with the selected macroeconomic drivers and assessing the role of macroeconomic conditions in shaping LGD dynamics;
- Application of forward-looking macroeconomic scenarios and analysis of their expected effects on LGD across the different portfolio segments, culminating in the construction of a probability-weighted LGD measure consistent with the forward-looking principles of the IFRS 9 framework;
- Comparison of the methodology and results obtained with those of Crook & Bellotti (2012), used as a direct benchmark to situate the macroeconomic-only framework developed in this thesis within an application that also incorporates borrower- and exposure-level information.

2.3 Variable Analysis

2.3.1 Descriptive and Diagnostic Analysis of the Target Variables

The target variables considered in the satellite models are the historical *Loss Given Default* (LGD) series relating to four different categories of credit exposures reported by *Significant Institutions* under the *Internal Ratings-Based* (IRB) approach: Corporate, Small and Medium-Sized Enterprises (SME), Retail, and Financial Institutions.
Building on the 31 quarterly observations described in Section 2.1, descriptive analyses of the four time series have been conducted, together with tests for normality (Shapiro-Wilk and

Kolmogorov-Smirnov), autocorrelation (Ljung-Box), and stationarity (Augmented Dickey-Fuller, applied both to the series in levels and in first differences), in order to assess the statistical and dynamic properties of the data used in the model estimation stage and to identify any features that may be relevant for the subsequent econometric specification. Figures 2.1, 2.2, 2.3, and 2.4 show the evolution of the four LGD series considered.

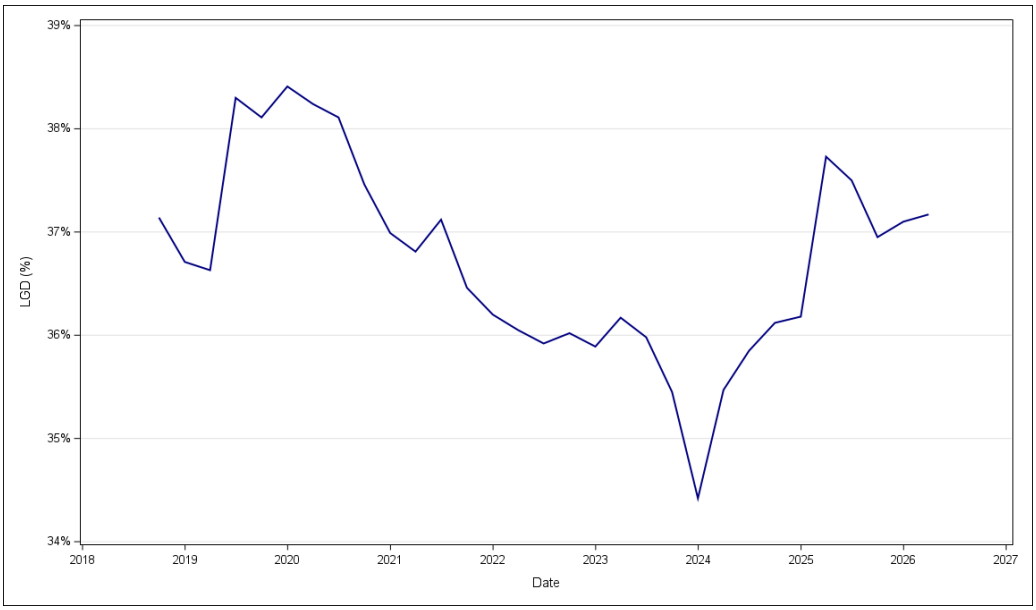


Figure 2.1: Exposure-weighted LGD (%) for Corporate exposures, 2018Q3–2026Q1

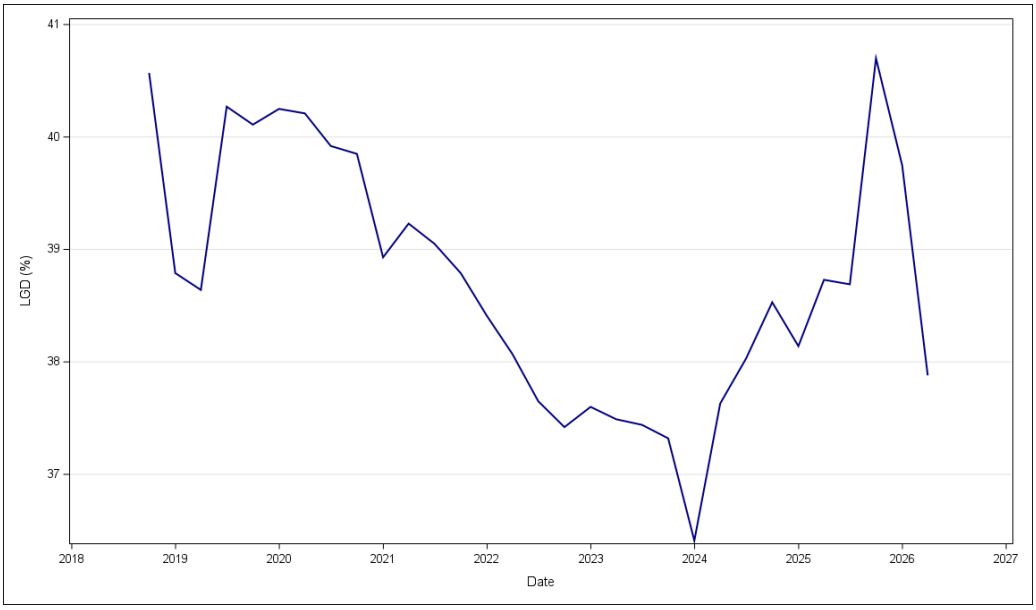


Figure 2.2: Exposure-weighted LGD (%) for SME exposures, 2018Q3–2026Q1

As shown in Figure 2.1, the LGD associated with Corporate exposures exhibits the most stable pattern among all the series considered. After remaining at levels close to 37–38% between 2018 and 2020, the series gradually declines until the beginning of 2024, when it reached its lowest

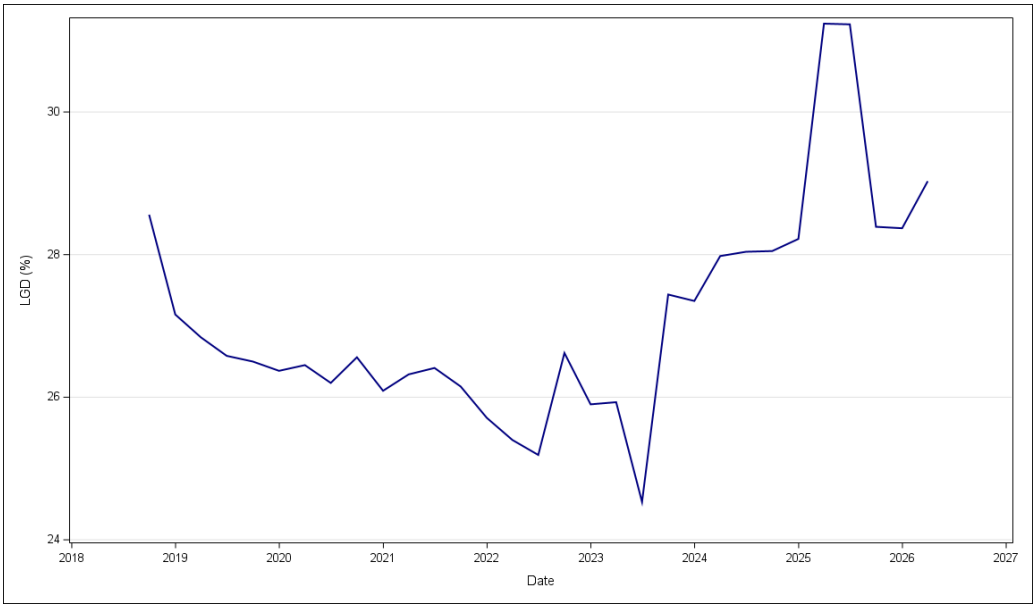


Figure 2.3: Exposure-weighted LGD (%) for Retail exposures, 2018Q3–2026Q1

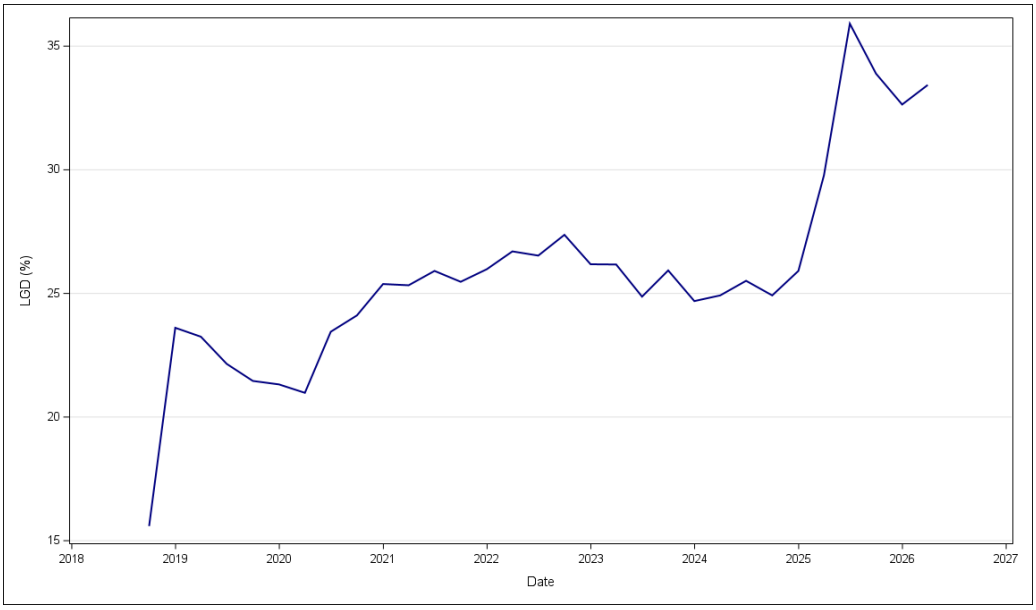


Figure 2.4: Exposure-weighted LGD (%) for Financial Institution exposures, 2018Q3–2026Q1

value over the sample period. In the most recent quarters, a moderate recovery can be observed, bringing LGD back to slightly higher levels.

A similar pattern characterises the SME series shown in Figure 2.2. The series displays a gradual decline between 2020 and 2024, falling from values above 40% to levels close to 36%. From 2024 onwards, however, the trend reverses and LGD starts to increase steadily, reaching its sample maximum during 2025. Overall, the time series appears smooth and does not exhibit any major structural breaks.

The LGD associated with Retail exposures, reported in Figure 2.3, shows a higher degree of variability. Following an initial decline, which brings the series from levels close to 28.5% to values below 26%, LGD begins to rise gradually from the second half of 2023 onwards. The increase becomes particularly pronounced during 2025, when the series reaches its highest value over the entire sample period, followed by a slight correction in subsequent quarters. This pattern suggests greater sensitivity to changing economic conditions compared with the other exposure categories.

The most pronounced dynamics are observed for Financial Institutions exposures (Figure 2.4). After an initial value of approximately 16%, the series rises rapidly during 2019 and subsequently stabilises within a range of roughly 24–27% for several years. From 2025 onwards, however, a significant shift can be observed, characterised by a sharp increase in LGD to levels exceeding 35%. This development results in substantially higher volatility than that observed in the other series and contributes to a more asymmetric distribution.

Overall, all four series display a high degree of persistence and relatively gradual movements, with no evidence of erratic short-term fluctuations. However, while the Corporate and SME LGD series appear particularly stable over time, the Retail and, above all, the Financial Institutions series show greater variability in the final part of the sample, accompanied by a marked increase in observed loss levels.

In order to further investigate the distributional characteristics of the target variables, the main descriptive statistics have been computed for each of the four LGD series analysed. The results are reported in Table 2.1 and highlight several differences across the four series considered.

Table 2.1: Descriptive statistics of the LGD series

Variable	Mean (%)	SD (%)	Min (%)	Max (%)	Skewness	Excess Kurtosis
Corporate	36,73	0,96	34,42	38,41	-0,028	-0,271
SME	38,73	1,13	36,41	40,70	0,114	-0,888
Retail	27,12	1,54	24,53	31,24	1,123	1,581
Institutions	25,79	4,07	15,59	35,91	0,515	1,596

The Corporate and SME LGD series exhibit average values of 36.73% and 38.73%, respectively, together with relatively low variability, as indicated by their small standard deviations (0.96% and 1.13%). The skewness and excess kurtosis values are low for both series, though somewhat more

pronounced for SME (0.114 and −0.888) than for Corporate (−0.028 and −0.271), suggesting distributions that remain broadly symmetric and reasonably close to normality.
By contrast, the Retail series displays greater dispersion, with a standard deviation of 1.54% and a range of 6.71%. The distribution is characterised by a marked positive skewness (1.123) and excess kurtosis above that of a normal distribution, indicating a greater concentration of observations around the mean and a more pronounced right tail.
The highest variability is observed in the Financial Institutions series, which exhibits a standard deviation of 4.07% and a range exceeding 20%. In this case, the skewness and excess kurtosis coefficients indicate moderate positive asymmetry and a leptokurtic distribution. These results reflect the presence of unusually high values recorded in the final part of the sample, which contribute to the overall dispersion of the series.
In order to examine the distributional shape of the target variables and assess whether the data are consistent with the assumption of normality, the Shapiro-Wilk and Kolmogorov-Smirnov tests have been applied. Both tests evaluate the null hypothesis that the observations originate from a normal distribution.
The results are reported in Table 2.2.

Table 2.2: Shapiro-Wilk and Kolmogorov-Smirnov normality test statistics and p-values for the four LGD series, by portfolio segment

	Shapiro-Wilk		Kolmogorov-Smirnov	
Variable	Statistic (W)	p-value	Statistic (D)	p-value
Corporate	0,9658	0,4124	0,1286	$> 0,1500$
SME	0,9577	0,2537	0,1082	$> 0,1500$
Retail	0,9080	0,0115	0,1764	0,0152
Financial Institutions	0,9053	0,0098	0,2175	$< 0,0100$

For the Corporate and SME series, the tests return p-values above the conventional 5% significance level, meaning that the null hypothesis of normality cannot be rejected. This evidence is consistent with the findings of the descriptive analysis presented earlier.
By contrast, the Retail and Financial Institutions series exhibit p-values consistently below the 5% significance level across all the tests considered. In these cases, the null hypothesis of normality is therefore rejected.
Subsequently, since the presence of serial correlation is a typical feature of economic and financial time series and represents an important aspect in the specification of satellite models, autocorrelations for the first six lags have been computed for each series. In addition, the Ljung-Box test has been performed to assess the null hypothesis of no joint autocorrelation.
The results reported in Table 2.3 indicate a high degree of persistence over time across all the series considered. In particular, the first-order autocorrelations are high for all exposure categories, with values ranging from 0.693 to 0.813. The Corporate series exhibits the highest coefficient, followed by the Retail, SME, and Financial Institutions series. Autocorrelations remain positive at higher lags, although they gradually decline as the lag increases. This pattern

Table 2.3: Autocorrelation coefficients (lags 1 to 6), Ljung-Box statistics and p-values, and Durbin-Watson statistics for the four LGD series, by portfolio segment

Variable	ACF(1)	ACF(2)	ACF(3)	ACF(4)	ACF(5)	ACF(6)	LB χ^2	p-value	DW
Corporate	0,813	0,644	0,513	0,359	0,142	0,044	52,56	$< 0,0001$	0,361
SME	0,695	0,543	0,530	0,432	0,327	0,164	49,50	$< 0,0001$	0,503
Retail	0,721	0,553	0,489	0,429	0,251	0,162	47,75	$< 0,0001$	0,479
Financial Institutions	0,693	0,539	0,387	0,206	0,098	0,062	34,22	$< 0,0001$	0,288

suggests that shocks affecting LGD tend to persist over several quarters, resulting in a highly persistent dynamic over time. The results of the Ljung-Box test confirm these findings. For all the series analysed, the null hypothesis of no autocorrelation is rejected with a high degree of statistical significance, as the p-values are consistently below the 1% significance level. In addition, the test statistics are particularly high for the Corporate, SME, and Retail series, indicating strong serial dependence. These results are further supported by the values of the Durbin-Watson statistic, which are considerably lower than the theoretical value of 2 associated with the absence of autocorrelation, providing additional evidence of positive serial correlation. This phenomenon appears to be particularly pronounced for the Financial Institution series, which records a value of 0.288.

Finally, in order to investigate the presence of potential non-stationary components in the target variables, a stationarity analysis of the LGD series has been conducted using the Augmented Dickey-Fuller (ADF) test. The test has been applied both to the series in levels and to their first differences under three different specifications: without an intercept (*Zero Mean*), with an intercept (*Single Mean*), and with both an intercept and a deterministic trend (*Trend*). The results provide insight into the nature of the stochastic processes underlying the target variables and allow the possible order of integration of the series to be identified. The results of the ADF test applied to the LGD series in levels are reported in Table 2.4.

Table 2.4: Augmented Dickey-Fuller test statistics and p-values (Zero Mean, Single Mean, and Trend specifications) for the four LGD series in levels, by portfolio segment

Variable	Zero Mean		Single Mean		Trend	
	Tau	p-value	Tau	p-value	Tau	p-value
Corporate	-0,03	0,6642	-1,67	0,4367	-1,60	0,7699
SME	-0,67	0,4177	-2,36	0,1599	-2,39	0,3778
Retail	-0,02	0,6679	-1,90	0,3267	-2,98	0,1553
Financial Institutions	1,19	0,9368	-2,13	0,2355	-3,06	0,1341

The outputs reveal a broadly homogeneous pattern across the different exposure categories considered. In particular, for all the series analysed, the ADF test applied to the variables in levels does not allow the null hypothesis of a unit root to be rejected, regardless of the specification adopted. Both under the *Single Mean* specification and under the specification including a deterministic trend (*Trend*), the p-values are consistently above the conventional 5% significance level. These findings therefore suggest that the LGD series may contain a

Table 2.5: Augmented Dickey-Fuller test statistics and p-values (Zero Mean, Single Mean, and Trend specifications) for the first differences of the four LGD series, by portfolio segment

Variable	Zero Mean		Single Mean		Trend	
	Tau	p-value	Tau	p-value	Tau	p-value
Corporate	-8,64	$< 0,0001$	-8,48	0,0002	-8,31	0,0001
SME	-7,98	$< 0,0001$	-7,83	0,0002	-7,89	0,0001
Retail	-8,64	$< 0,0001$	-8,48	0,0002	-8,31	0,0001
Financial Institutions	-7,83	$< 0,0001$	-7,63	0,0002	-7,43	0,0001

non-stationary component. This result is consistent with the evidence from the autocorrelation analysis, which highlighted a high degree of persistence over time across all target variables.
By contrast, applying the test to the first differences leads to the opposite conclusion. For all exposure categories considered, the test statistics are strongly negative and the corresponding p-values are well below the 1% significance level, both under the *Single Mean* specification and under the specification including a trend. In all cases, as shown in Table 2.5, the null hypothesis of a unit root can therefore be decisively rejected. Overall, the results suggest that the four LGD series can be treated as integrated processes of order one, $I(1)$.
This shared dynamic structure does not, however, mean that the dependent variable has to be differenced for estimation purposes: doing so would make LGD harder to interpret directly as a bounded measure of loss severity. Instead, what the evidence calls for is an econometric specification that can handle both the bounded nature of LGD, confined to the interval $[0, 1]$, and its persistent, non-stationary behaviour in levels. This is exactly the role of the Logit and Beta transformations introduced in Section 2.4.1, which map the observed LGD series onto an unbounded scale while keeping its meaning as a measure of expected loss.

2.3.2 Descriptive and Diagnostic Analysis of the Macroeconomic Variables

Following the same approach adopted for the target variables in the previous section, descriptive analyses of the seven candidate macroeconomic and financial regressors have been conducted, together with tests for normality (Shapiro-Wilk and Kolmogorov-Smirnov), autocorrelation (Ljung-Box), and stationarity (Augmented Dickey-Fuller, applied both to the series in levels and in first differences), in order to assess the statistical and dynamic properties of these variables prior to their use as explanatory drivers in the satellite models.
The time-series behaviour of the macroeconomic variables considered in the analysis is presented in Figures 2.5, 2.6, 2.7, 2.8, 2.9, 2.10 and 2.11. As a reminder, the variables are observed over the period from the third quarter of 2018 to the first quarter of 2026, consistently with the availability of the LGD data. Overall, the series exhibit heterogeneous patterns, reflecting both the impact of the macroeconomic shocks that characterised the observation period and the specific dynamics of each indicator. The main descriptive statistics are reported in Table 2.6.

The GDP growth rates for the Euro Area and Italy, shown in Figures 2.5 and 2.6, respectively, display average values close to zero but considerable variability, mainly driven by the large fluctuations observed during the COVID-19 crisis. This feature is also reflected in their wide ranges and in the particularly high kurtosis values, which indicate the presence of extreme observations.

By contrast, the Italian unemployment rate (Figure 2.7) exhibits a more stable pattern, with an average value of 8.22% and relatively limited volatility.

Similarly, the aggregate default rate (Figure 2.11) shows modest variability, consistent with the more persistent nature of the underlying phenomenon.

The oil price series (Figure 2.8) exhibits the highest volatility in absolute terms, with a range exceeding 75 U.S. dollars per barrel, reflecting the cyclical nature of the energy market and its sensitivity to economic and geopolitical shocks.

With regard to the benchmark 10-year Italian government bond yield (BTP), originally available at monthly frequency, the series have previously been converted to quarterly frequency in order to ensure consistency with the other variables included in the analysis. The aggregation has been performed by computing the arithmetic average of the monthly observations within each quarter. The resulting series is reported in Figure 2.10. It is worth noting that this variable, similarly to the three-month Euribor rate shown in Figure 2.9, displays movements that are mainly attributable to the shift in the stance of European monetary policy from 2022 onwards. Both variables experienced a rapid increase in response to the inflationary pressures observed across the Euro Area.

From a distributional perspective, most macroeconomic regressors exhibit relatively low skewness values, suggesting limited asymmetry in their distributions. Greater differences emerge with respect to kurtosis. The GDP growth rates for both the Euro Area and Italy display strongly leptokurtic distributions, while oil prices exhibit moderately positive kurtosis. By contrast, the Italian unemployment rate, the three-month Euribor rate, the 10-year BTP yield, and the aggregate default rate are characterised by distributions that are flatter than the normal distribution.

Table 2.6: Descriptive statistics of the macroeconomic and financial variables

Variable	Mean	SD	Min	Max	Skewness	Excess Kurtosis
EU GDP Growth (%)	0.34	2.88	-10.80	10.70	-0.436	12.623
IT GDP Growth (%)	0.29	3.67	-12.50	14.00	0.213	10.901
Unemployment Rate (%)	8.22	1.59	5.20	10.50	-0.220	-1.187
Oil Price (USD/barrel)	72.71	16.38	33.37	111.99	-0.151	0.782
3M Euribor (%)	1.10	1.75	-0.57	3.96	0.645	-1.530
10Y BTP Yield (%)	2.72	1.31	0.64	4.38	-0.403	-1.460
Default Rate (%)	0.31	0.06	0.24	0.42	0.523	-0.271

To formally assess the normality assumption, both the Shapiro-Wilk and the Kolmogorov-Smirnov tests have been applied. The results are reported in Table 2.7 and show a substantial degree of consistency between the two approaches. In particular, the GDP growth rates of both the

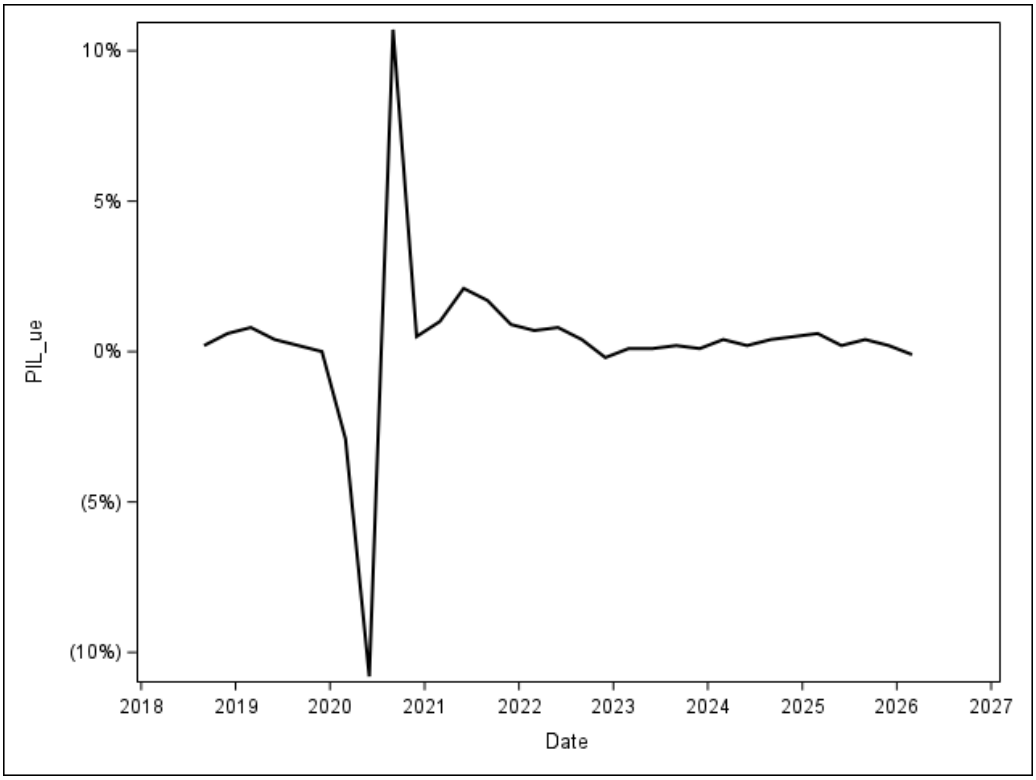


Figure 2.5: Euro Area GDP growth rate (%), 2018Q3–2026Q1

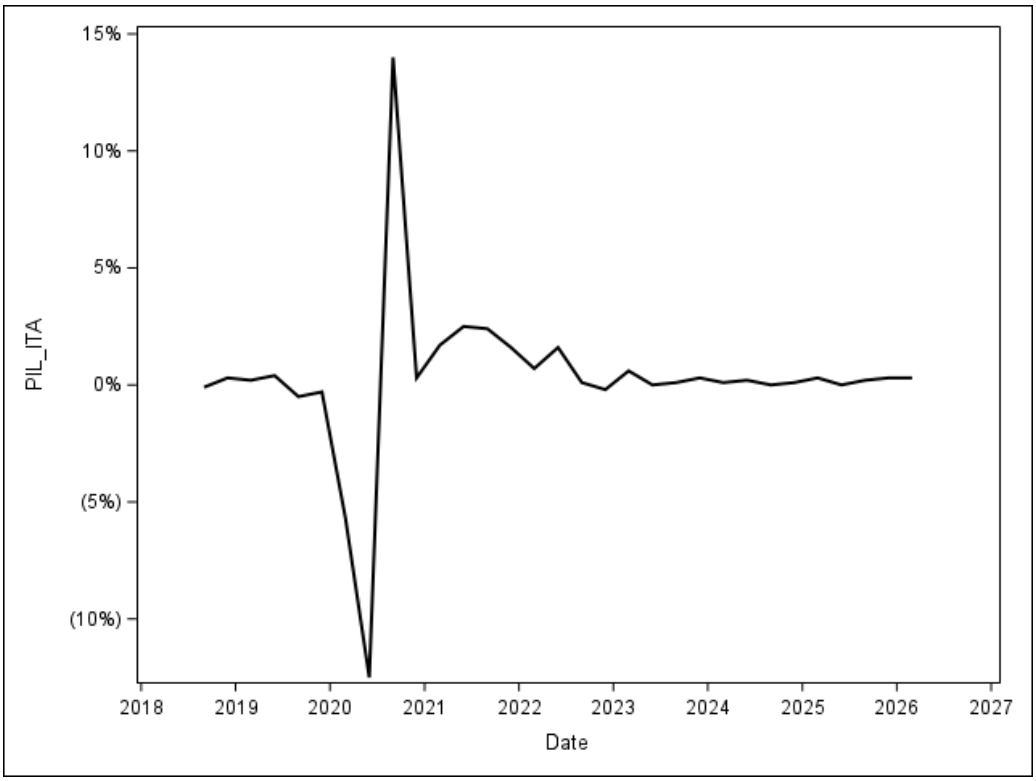


Figure 2.6: Italian GDP growth rate (%), 2018Q3–2026Q1

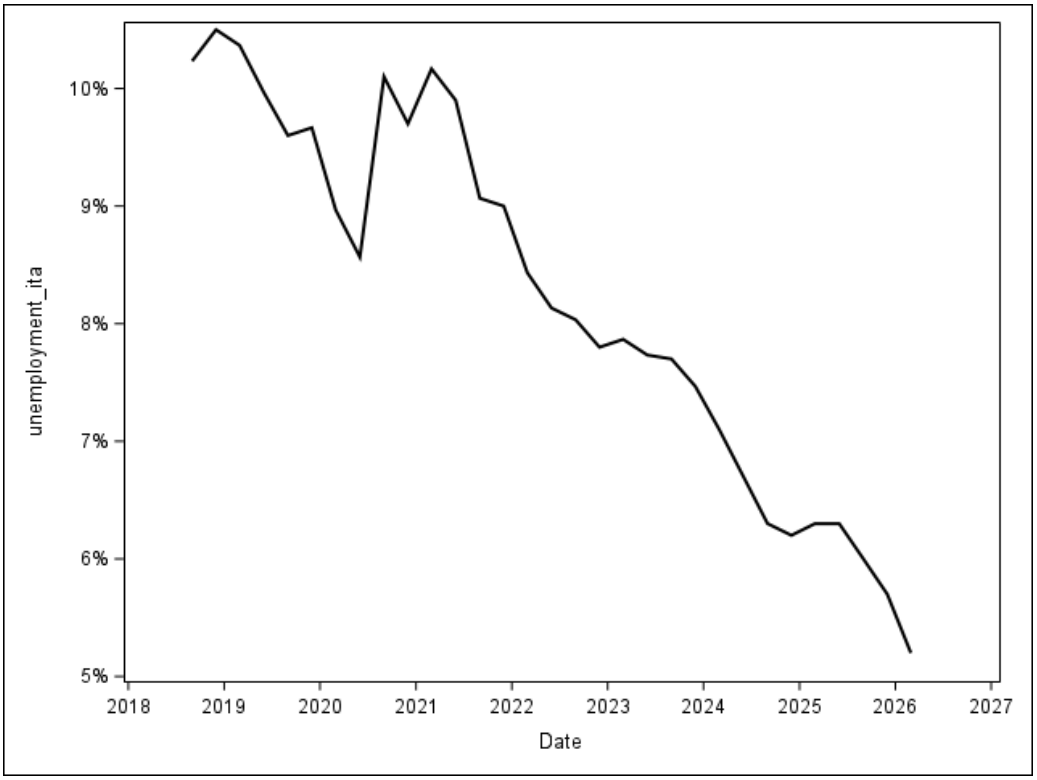


Figure 2.7: Italian unemployment rate (%), 2018Q3–2026Q1

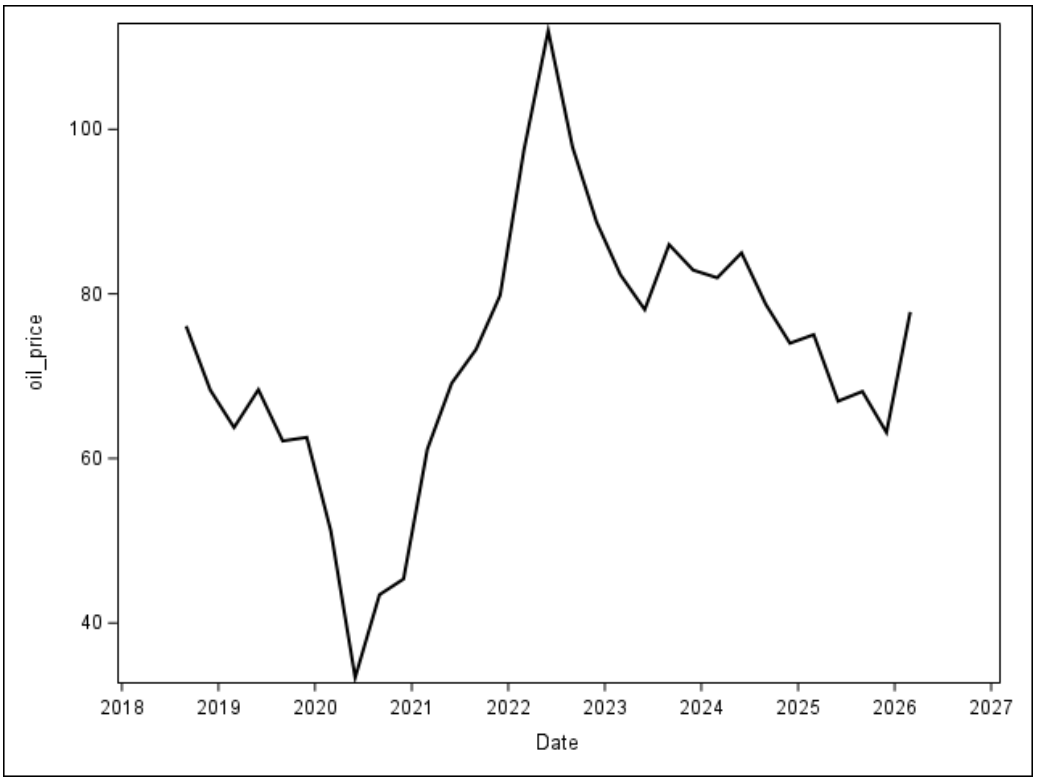


Figure 2.8: Brent crude oil price (USD per barrel), 2018Q3–2026Q1

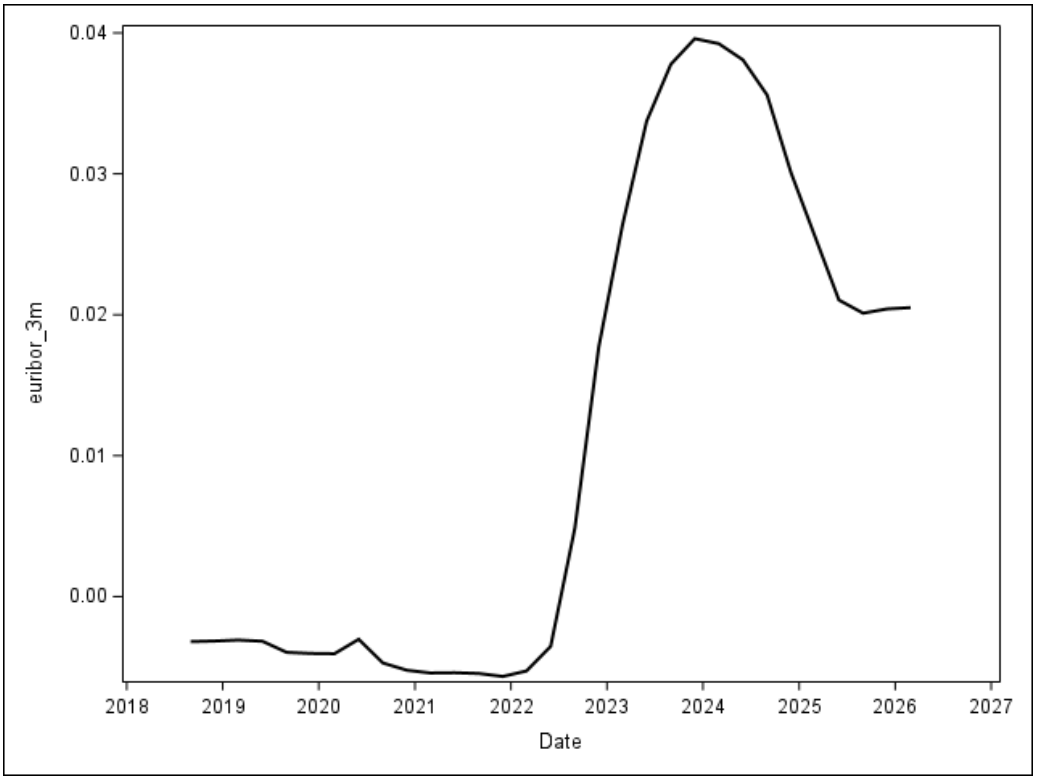


Figure 2.9: Three-month Euribor rate (%), 2018Q3–2026Q1

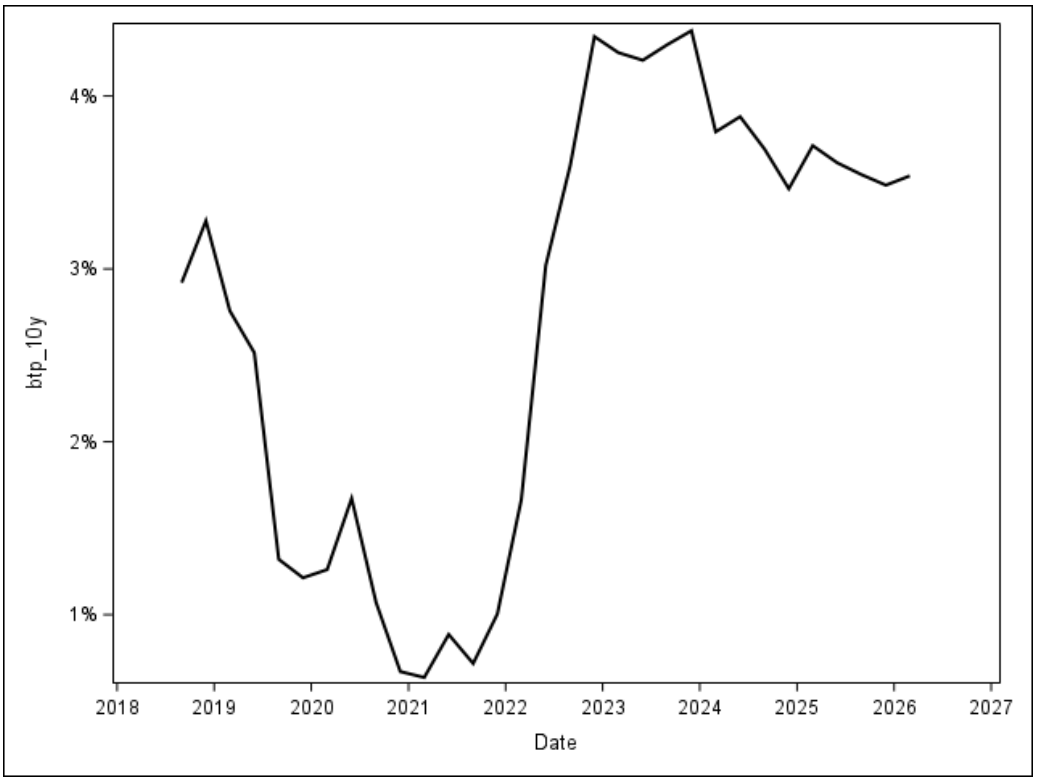


Figure 2.10: Yield on the 10-year Italian government bond (BTP) (%), 2018Q3–2026Q1

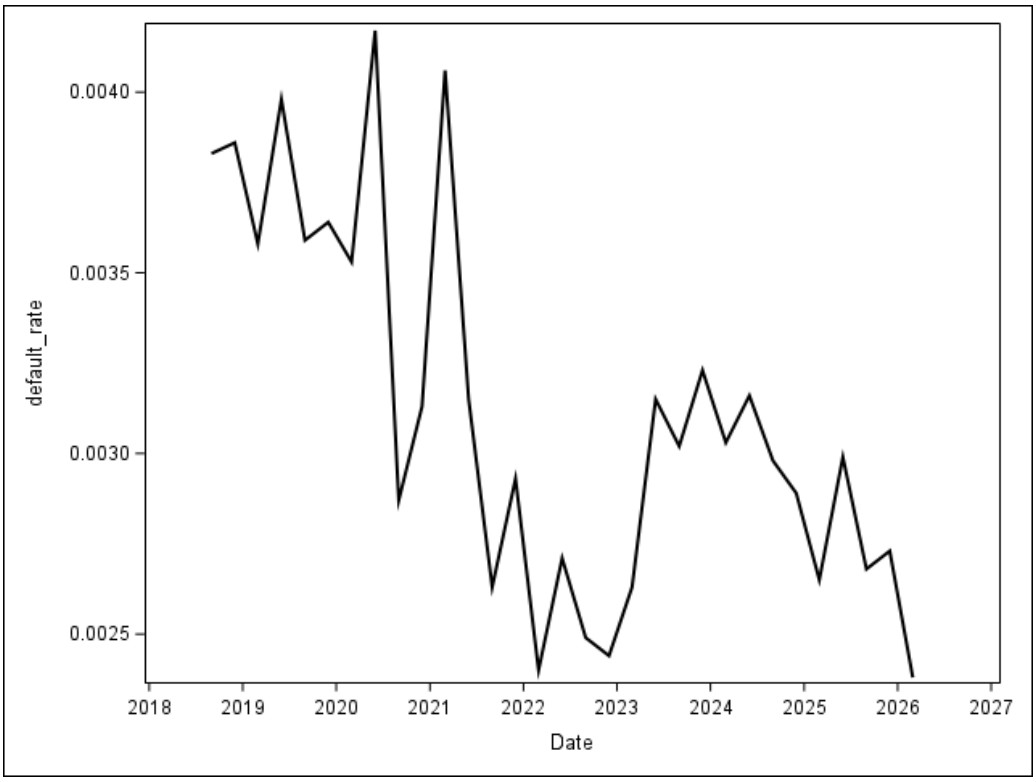


Figure 2.11: Aggregate default rate (%), 2018Q3–2026Q1

Euro Area and Italy display p-values below the 1% significance level in both tests, allowing the null hypothesis of normality to be rejected with strong statistical evidence. Similar results are obtained for the three-month Euribor rate and the 10-year BTP yield, whose p-values are well below the conventional 5% significance threshold, indicating distributions that are not consistent with normality. By contrast, different results emerge for the Italian unemployment rate, oil prices, and the aggregate default rate.

Table 2.7: Shapiro-Wilk and Kolmogorov-Smirnov normality test statistics and p-values for the seven candidate macroeconomic and financial variables

	Shapiro-Wilk		Kolmogorov-Smirnov	
Variable	Statistic (W)	p-value	Statistic (D)	p-value
EU GDP Growth	0,5458	$< 0,0001$	0,3618	$< 0,0100$
IT GDP Growth	0,5888	$< 0,0001$	0,3501	$< 0,0100$
Unemployment Rate	0,9399	0,0821	0,1302	$> 0,1500$
Oil Price	0,9780	0,7556	0,1109	$> 0,1500$
3M Euribor	0,7798	$< 0,0001$	0,3054	$< 0,0100$
10Y BTP Yield	0,8728	0,0016	0,1979	$< 0,0100$
Default Rate	0,9394	0,0797	0,1414	0,1134

To analyse the dynamic properties of the variables considered, sample autocorrelations up to the sixth lag, the Ljung-Box test, and the Durbin-Watson statistic have been examined. The results are reported in Table 2.8.

The empirical evidence reveals a clear distinction between GDP growth rates and the remaining macroeconomic and financial variables. In particular, the GDP growth rates of both the Euro Area and Italy display relatively low autocorrelations, generally close to zero across all lags considered. Moreover, the p-values associated with the Ljung-Box test are equal to 0.7442 and 0.8946, respectively, well above the conventional 5% significance level. Therefore, the null hypothesis of no serial autocorrelation cannot be rejected. This evidence is further supported by the values of the Durbin-Watson statistic, which are equal to 2.572 for Euro Area GDP and 2.412 for Italian GDP. Although these values suggest a slight tendency towards negative autocorrelation, they remain relatively close to the theoretical value of 2 and are therefore consistent with the absence of strong serial dependence.

By contrast, the Italian unemployment rate, oil prices, the three-month Euribor rate, the 10-year BTP yield, and the aggregate default rate exhibit a high degree of persistence over time. In all these cases, the first-order autocorrelation is particularly high, ranging from 0.850 to 0.958, and remains positive at subsequent lags, although its magnitude gradually declines. The Ljung-Box test statistics range from 34.61 for the aggregate default rate to 107.68 for the three-month Euribor rate, with all associated p-values below 0.0001, leading to a decisive rejection of the null hypothesis of no serial autocorrelation in every case. This evidence is consistent with the corresponding Durbin-Watson statistics, which range from 0.051 for the three-month Euribor rate to 0.734 for the aggregate default rate, all considerably below 2 and therefore indicative of strong positive serial correlation.

Table 2.8: Autocorrelation coefficients (lags 1 to 6), Ljung-Box statistics and p-values, and Durbin-Watson statistics for the seven candidate macroeconomic and financial variables

Variable	ACF(1)	ACF(2)	ACF(3)	ACF(4)	ACF(5)	ACF(6)	LB χ^2	p-value	DW
EU GDP Growth	-0,286	-0,086	0,039	-0,043	-0,086	-0,027	3,50	0,7442	2,572
IT GDP Growth	-0,206	-0,106	0,048	-0,042	-0,048	-0,065	2,26	0,8946	2,412
Unemployment Rate	0,874	0,760	0,652	0,552	0,484	0,402	89,28	$< 0,0001$	0,079
Oil Price	0,850	0,654	0,439	0,243	0,092	-0,086	49,63	$< 0,0001$	0,296
3M Euribor	0,873	0,758	0,625	0,480	0,335	0,195	107,68	$< 0,0001$	0,051
10Y BTP Yield	0,930	0,814	0,663	0,517	0,354	0,203	85,74	$< 0,0001$	0,126
Default Rate	0,958	0,770	0,599	0,445	0,315	0,135	34,61	$< 0,0001$	0,734

To assess the stationarity of the variables considered, the Augmented Dickey-Fuller (ADF) test has been applied. The results are reported in Table 2.9. As in the case of the dependent variable, the test has been carried out under three different specifications.

The results show that the GDP growth rates of both the Euro Area and Italy are clearly stationary in levels. Under all specifications considered, the test statistics are strongly negative and the corresponding p-values are below the 5% significance level.

The results for the Italian unemployment rate are less conclusive. Under the *Zero Mean* specification, the test yields a Tau statistic of -2.14 with a p-value of 0.0330, suggesting the rejection of the null hypothesis of a unit root. However, under the *Single Mean* and *Trend* specifications, the p-values are equal to 0.9523 and 0.2569, respectively. Since these specifications include deterministic components that are generally more realistic for a macroeconomic variable

such as the unemployment rate, the evidence in favour of stationarity appears weak and not robust.

For oil prices, the three-month Euribor rate, and the 10-year BTP yield, p-values are consistently above the 5% significance level across all specifications considered. These results do not allow the null hypothesis of a unit root to be rejected and suggest that these series are characterised by a significant degree of persistence and may exhibit non-stationary behaviour.

The aggregate default rate also provides mixed results. Under the *Zero Mean* and *Single Mean* specifications, the null hypothesis cannot be rejected. However, under the *Trend* specification, the test returns a Tau statistic of -3.76 with a p-value of 0.0352, suggesting that the series may be stationary around a deterministic trend. Nevertheless, this evidence is less robust than that observed for GDP growth rates.

Overall, the ADF test provides clear evidence of stationarity in levels for the GDP growth rates of the Euro Area and Italy. By contrast, for the remaining variables, the presence of a unit root cannot generally be ruled out with sufficient confidence.

Table 2.9: Augmented Dickey-Fuller test statistics and p-values (Zero Mean, Single Mean, and Trend specifications) for the seven candidate macroeconomic and financial variables in levels

Variable	Zero Mean		Single Mean		Trend	
	Tau	p-value	Tau	p-value	Tau	p-value
EU GDP Growth	-7,09	< 0, 0001	-7,10	0,0002	-6,98	0,0001
IT GDP Growth	-6,58	< 0, 0001	-6,52	0,0002	-6,42	0,0001
Unemployment Rate	-2,14	0,0330	0,01	0,9523	-2,67	0,2569
Oil price	-0,29	0,5738	-1,49	0,5261	-1,71	0,7220
3M Euribor	-0,07	0,6521	-0,75	0,8186	-0,71	0,9631
10Y BTP Yield	-0,16	0,6201	-0,86	0,7869	-1,64	0,7511
Default Rate	-2,09	0,2314	-2,60	0,1039	-3,76	0,0352

Finally, the ADF test has been applied also to the first differences of all the variables considered, and the results are reported in Table 2.10. The evidence obtained shows that almost all variables exhibit clear stationarity after first differencing. The only variable for which the results remain less conclusive is the three-month Euribor rate. Even after differencing, the p-values remain above the conventional 5% significance level under all three specifications considered, preventing a formal rejection of the null hypothesis of a unit root. This result may be attributed both to the limited sample size available, which reduces the statistical power of the test, and to the substantial changes in the European monetary policy environment observed in recent years, which have significantly altered the dynamics of the series.

Table 2.10: Augmented Dickey-Fuller test statistics and p-values (Zero Mean, Single Mean, and Trend specifications) for the first differences of the seven candidate macroeconomic and financial variables

Variable	Zero Mean		Single Mean		Trend	
	Tau	p-value	Tau	p-value	Tau	p-value
EU GDP Growth	-10,23	< 0, 0001	-10,04	0,0002	-9,86	0,0001
IT GDP Growth	-9,70	< 0, 0001	-9,53	0,0002	-9,35	0,0001
Unemployment Rate	-5,19	< 0, 0001	-6,16	0,0002	-6,10	0,0002
Oil Price	-4,10	0,0002	-4,01	0,0042	-3,93	0,0237
3M Euribor	-1,53	0,1165	-1,53	0,5029	-1,53	0,7968
10Y BTP Yield	-3,64	0,0007	-3,58	0,0124	-3,62	0,0461
Default Rate	-8,27	< 0, 0001	-8,28	< 0, 0001	-8,13	< 0, 0001

Taken together, these outputs motivate the use of quarter-on-quarter changes, rather than levels, as candidate macroeconomic regressors in the satellite models developed in the remainder of this thesis (Section 2.4.1).

2.4 Estimation

2.4.1 Specification of the Model

The objective is to develop an econometric model for Loss Given Default (LGD), a variable bounded within the interval $(0, 1)$, using macroeconomic variables as explanatory drivers. As discussed in Section 1.1.4, the empirical distribution of LGD is typically characterised by a strong concentration of observations near the two extremes, giving rise to a U-shaped bimodal pattern. It should be noted that, unlike the individual-account data typically used in the literature, the four portfolio-level series considered here do not exhibit values close to the boundaries of the $(0, 1)$ interval, a consequence of the portfolio-level aggregation underlying these series, as documented in Section 2.3.1.

A relevant feature to keep in mind, however, is that, as shown in Section 2.3.1, the LGD series analysed in this thesis are themselves non-normal and highly persistent in levels. Applying a linear regression directly to LGD would ignore both of these features.

First, since a linear model imposes no bounds on the predicted values, forecasts could fall outside the admissible range $(0, 1)$, particularly under adverse macroeconomic scenarios.

Second, a linear specification implicitly assumes a constant marginal effect of each regressor across the entire range of LGD. This assumption would be particularly inappropriate for LGD data that exhibit the bimodal, U-shaped distribution documented above: a segment whose LGD is close to one of the two extremes has little room left to move further in that direction, so the effect of a given shock should reasonably be smaller there than closer to the centre of the interval – an effect a linear model cannot represent. This is precisely the concern that motivates Crook & Bellotti (2012) to test alternative specifications (as described in Chapter 3).

Third, because LGD is bounded, the variance of the residuals from a linear model tends to shrink mechanically as fitted values approach the boundaries of the $(0, 1)$ interval, giving rise to heteroskedasticity that a standard linear specification does not account for; although this concern is less acute in the present sample, given the absence of extreme values noted above, it remains a further reason, consistent with standard practice in the literature, to prefer a bounded specification over a direct linear regression on LGD.

Therefore, a logit transformation is applied to the dependent variable, following the approach adopted, among others, by Crook & Bellotti (2012) under the name *Fractional Logit Transformation* (Chapter 3). For each portfolio segment $j \in \{\text{Corporate, SME, Retail, Financial Institutions}\}$

and period $t \in \{2018Q3, \ldots, 2026Q1\}$, the transformation used is:

$$LGD_{j,t}^{logit} = \log\left(\frac{LGD_{j,t}}{1 - LGD_{j,t}}\right) \tag{2.2}$$

In this way, the transformed variable $LGD_{j,t}^{logit}$ takes values over the entire real line, $(-\infty, +\infty)$, and can be modelled using standard linear regression techniques.

Selection of Regressors

As shown in Section 2.3.2, the macroeconomic variables considered are generally non-stationary in levels. Consistently with the choice motivated there, first differences are used throughout the analysis:

$$\Delta X_t = X_t - X_{t-1}, \tag{2.3}$$

where X_t denotes a generic macroeconomic variable.
Furthermore, since the effects of macroeconomic conditions on Loss Given Default are not necessarily immediate, several lagged specifications are considered for each macroeconomic driver, up to a maximum of eight quarters, a horizon consistent with that adopted by Konečný et al. (2017) to capture delayed responses of LGD to macroeconomic conditions (Section 1.1.4). This choice also extends the more limited lag structure tested by Crook & Bellotti (2012), who use only three fixed points in time, namely the date of default and six months before and after it (Chapter 3).
Let

$$j \in \{\text{Retail, SME, Corporate, Financial Institutions}\} \tag{2.4}$$

denote the portfolio segment and $p = 1, \ldots, 7$ the index of the seven candidate macroeconomic variables, namely EU GDP growth, IT GDP growth, the unemployment rate, oil prices, the Euribor rate, the 10-year BTP yield, and the aggregate default rate. For each macroeconomic variable, the corresponding lagged versions are constructed:

$$\Delta X_{p,t}, \Delta X_{p,t-1}, \ldots, \Delta X_{p,t-l}, \ldots, \Delta X_{p,t-8}, \tag{2.5}$$

where l denotes the number of quarters by which the macroeconomic variable is lagged.
To identify the most informative lag structure, a univariate regression of the following form is estimated for each combination of portfolio segment, macroeconomic variable, and lag $l = 0, \ldots, 8$:

$$LGD_{j,t}^{logit} = \beta_{0,j} + \beta_{j,p,l}\, \Delta X_{p,t-l} + \varepsilon_{j,t}, \tag{2.6}$$

where:

- $LGD_{j,t}^{logit}$ represents the logit-transformed LGD for portfolio segment j at time t, with $t \in \{2018Q3, \ldots, 2026Q1\}$;

- $\Delta X_{p,t-l}$ represents the change in macroeconomic variable p, observed with lag l;
- $\beta_{0,j}$ is the intercept associated with portfolio segment j;
- $\beta_{j,p,l}$ measures the effect of macroeconomic variable p, observed with lag l, on the logit-transformed LGD of segment j;
- $\varepsilon_{j,t}$ represents the error term.

For each macroeconomic driver, nine univariate regressions are therefore estimated, corresponding to the contemporaneous value and lags ranging from one to eight quarters. The objective of this stage is to identify, for each explanatory variable, the lag structure providing the greatest explanatory power for LGD across the portfolio segments considered.

The Covid Dummy, by contrast, is not subject to this lag-selection procedure, since it identifies a specific historical period rather than a time series with a meaningful lag structure; it is introduced directly, in its untransformed form, at the multivariate stage described below.

The analysis shows, however, that the only variable capable of providing a statistically robust and stable contribution across all portfolio segments is the Euribor rate. More specifically, among the nine lag specifications tested for this variable in each segment, the coefficient associated with the change in Euribor lagged by two quarters, $\Delta\text{Euribor}_{t-2}$, has been selected as it is statistically significant at the 5% level for all segments analysed (Retail, SME, Corporate, and Financial Institutions) and exhibits an economically consistent sign, as reported in Table 2.11.

Table 2.11: Univariate regression results (coefficient, standard error, t-value, p-value, and goodness-of-fit statistics) of logit-transformed LGD on $\Delta\text{Euribor}_{t-2}$, by portfolio segment

Portfolio Segment	β	Std. Error	t-value	p-value	F	R^2	Adj. R^2
Financial Institutions	-17,4918	8,2125	-2,13	0,0428	4,54	0,1486	0,1158
Corporate	-5,7944	1,7504	-3,31	0,0027	10,96	0,2965	0,2695
SME	-7,1281	1,8119	-3,93	0,0006	15,47	0,3731	0,3490
Retail	-8,9310	3,3838	-2,64	0,0139	6,97	0,2113	0,1810

For this reason, $\Delta\text{Euribor}_{t-2}$ is selected as the first regressor of the multivariate model for each portfolio segment.

Once the main macroeconomic driver has been identified, the selection of the remaining explanatory variables is carried out through an incremental procedure. For each portfolio segment j, the analysis starts from the following baseline model, which includes only the two-quarter lagged change in Euribor identified before:

$$LGD_{j,t}^{logit} = \beta_{0,j} + \beta_{1,j}\Delta\text{Euribor}_{t-2} + \varepsilon_{j,t}. \tag{2.7}$$

Starting from this baseline, each remaining candidate macroeconomic variable $p = 1, \ldots, m$ is tested one at a time, across all lags $l = 0, \ldots, 8$, giving rise to models of the form:

$$LGD_{j,t}^{logit} = \beta_{0,j} + \beta_{1,j}\Delta\text{Euribor}_{t-2} + \beta_{2,j,p,l}\Delta X_{p,t-l} + \varepsilon_{j,t}, \tag{2.8}$$

where:

- $LGD_{j,t}^{logit}$ represents the logit-transformed LGD for portfolio segment j at time t, with $t \in \{2018Q3, \ldots, 2026Q1\}$;
- $\Delta\text{Euribor}_{t-2}$ represents the main macroeconomic driver identified during the preliminary selection stage;
- $\Delta X_{p,t-l}$ represents the change in candidate macroeconomic variable $p = 1, \ldots, m$, tested at each lag $l = 0, \ldots, 8$;
- $\beta_{0,j}$ is the intercept of the model for portfolio segment j;
- $\beta_{1,j}$ measures the effect of Euribor on the logit-transformed LGD of segment j;
- $\beta_{2,j,p,l}$ measures the additional contribution of macroeconomic variable p, observed with lag l, on the logit-transformed LGD, evaluated at each step of the procedure;
- $\varepsilon_{j,t}$ represents the error term.

The objective of this stage is to assess whether the inclusion of additional macroeconomic drivers improves the explanatory power of the model relative to the specification based solely on Euribor. Consistent with the general philosophy adopted by Crook & Bellotti (2012) (Chapter 3), the choice of the most appropriate specification is not based exclusively on the statistical significance of the estimated coefficients, but places the main emphasis on improvements in overall model performance; while Crook & Bellotti (2012) apply the Adjusted R^2 to compare the overall model structures considered and rely on the Akaike Information Criterion for the underlying forward selection of regressors, this thesis uses the Adjusted R^2 directly as the selection criterion at each incremental step, together with the absence of relevant multicollinearity among the explanatory variables, as detailed below.
Applying this procedure identifies two additional explanatory variables, common to all portfolio segments:

- the change in oil prices, lagged by two quarters, ΔOil_{t-2};
- the change in the yield of Italian government bonds, lagged by seven quarters, ΔBTP_{t-7}.

In addition to these regressors, identified through the incremental selection procedure described above, a Covid Dummy is introduced directly into the multivariate specification. Together with the two-quarter lagged change in Euribor identified in the preliminary stage, these four regressors constitute the final specification adopted for each portfolio segment.
Multicollinearity among the four regressors thus identified is monitored through the Variance Inflation Factor (VIF). For each regressor X_p in the final specification, where $X_p \in \{\Delta\text{Euribor}_{t-2}, \Delta\text{Oil}_{t-2}, \Delta\text{BTP}_{t-7}, \text{Covid Dummy}\}$, an auxiliary OLS regression of X_p on the

three remaining regressors is estimated, and the resulting coefficient of determination, R_p^2, is used to compute:

$$VIF_p = \frac{1}{1 - R_p^2}. \tag{2.9}$$

The logic underlying this measure is straightforward: if X_p can be well predicted by a linear combination of the other regressors, R_p^2 will be close to one and VIF_p will be correspondingly large, signalling that X_p carries little independent information once the other variables are accounted for. Conversely, a low R_p^2, and hence a VIF close to one, indicates that X_p is largely uncorrelated with the remaining regressors. Since the four macroeconomic regressors take the same values across all portfolio segments, the VIF depends only on the correlation structure among the regressors themselves, not on the dependent variable; as a result, a single set of VIF values applies to all four segments considered in this thesis.

As reported in Table 2.12, the VIF values for all four regressors, $\Delta\text{Euribor}_{t-2}$, ΔOil_{t-2}, ΔBTP_{t-7}, and the Covid Dummy, are close to one and well below the thresholds commonly adopted in the literature, indicating the absence of relevant multicollinearity issues in the final specification.

Table 2.12: Variance Inflation Factor (VIF) values for the regressors included in the final specification

Variable	VIF
$\Delta\text{Euribor}_{t-2}$	1.166
ΔOil_{t-2}	1.071
ΔBTP_{t-7}	1.222
Covid Dummy	1.159

As a further validation of the variable selection procedure, a robustness check inspired by Konečný et al. (2017) has been conducted. The authors investigate the relationship between LGD and macroeconomic variables using automated variable selection techniques based on penalised regressions, specifically elastic net regularisation (Section 1.1.4). Rather than replicating elastic net directly, a LASSO regression has been estimated on the logit-transformed LGD as a simpler, more conservative benchmark for this additional robustness check on the specification selected in the previous stage.

LASSO performs variable selection through a penalised least-squares optimisation.

Let $p = 1, \ldots, 6$ index the remaining candidate macroeconomic variables – namely EU GDP growth, IT GDP growth, the unemployment rate, oil prices, the 10-year BTP yield, and the aggregate default rate – and $l = 0, \ldots, 8$ the corresponding lags (so that $\Delta X_{p,t-l}$ denotes the change in variable p observed with lag l, as before); let D_t additionally denote the Covid Dummy. Specifically, for each portfolio segment j, the procedure estimates the coefficients by solving:

$$\min_{\boldsymbol{\beta}_j} \left\{ \sum_{t=1}^{T} \left(LGD_{j,t}^{logit} - \beta_{0,j} - \sum_{p=1}^{6} \sum_{l=0}^{8} \beta_{j,p,l}\, \Delta X_{p,t-l} - \delta_j D_t \right)^2 + \lambda \left(\sum_{p=1}^{6} \sum_{l=0}^{8} |\beta_{j,p,l}| + |\delta_j| \right) \right\}, \tag{2.10}$$

where λ is the regularisation parameter controlling the intensity of the penalisation. As λ increases, the coefficients associated with less informative regressors, including the Covid Dummy, are progressively shrunk towards zero and may eventually become exactly equal to zero; variables with non-zero coefficients are retained in the final specification, while the remaining regressors are automatically excluded from the model. The optimal value of λ is selected via ten-fold cross-validation, exploring the entire regularisation path without an early stopping rule, consistently with standard practice for this class of procedures. To ensure full reproducibility of the results, the procedure has been estimated using a fixed random seed of 12345. The variables selected by the algorithm are reported in Table 2.13.

Table 2.13: Macroeconomic variables and lags selected by the LASSO regularisation procedure, by portfolio segment

Segment	**Selected Variables**
Retail	ΔOil_{t-1}, ΔOil_{t-4}, ΔOil_{t-5}, $\Delta\text{Euribor}_{t}$, $\Delta\text{Euribor}_{t-1}$, $\Delta\text{Euribor}_{t-2}$, $\Delta\text{Euribor}_{t-8}$, Covid Dummy
SME	$\Delta\text{Euribor}_{t-2}$, $\Delta\text{Euribor}_{t-3}$, $\Delta\text{Euribor}_{t-4}$, ΔBTP_{t}, ΔBTP_{t-1}, ΔBTP_{t-6}
Corporate	$\Delta\text{Euribor}_{t-1}$, $\Delta\text{Euribor}_{t-4}$
Financial Institutions	$\Delta\text{Unemployment}_{t-6}$, ΔOil_{t-2}, ΔOil_{t-3}, ΔOil_{t-7}, ΔOil_{t-8}, $\Delta\text{Euribor}_{t}$, $\Delta\text{Euribor}_{t-1}$, $\Delta\text{Euribor}_{t-2}$, $\Delta\text{Euribor}_{t-5}$, ΔBTP_{t}, Covid Dummy

The results provide additional support for the macroeconomic drivers identified through the econometric methodology adopted in this study. The most striking result concerns the Euribor rate, which is selected in all four portfolio segments. Although the specific lag structure differs across specifications, lagged changes in Euribor systematically emerge as the most informative variable, confirming the central role of monetary conditions in explaining LGD dynamics – consistent with the two-quarter lag identified as the single most robust driver through the incremental selection procedure described above.

Additional evidence is provided by oil prices, selected in the Retail and Financial Institutions portfolios, and by the yield on Italian government bonds, selected in the SME and Financial Institutions specifications. The Covid Dummy is retained for Retail and Financial Institutions, while unemployment is selected only for Financial Institutions.

These findings broadly confirm the informational relevance of the variables retained in the final satellite model specifications, while also highlighting some differences in the number of selected lags and the degree of model parsimony. In fact, the LASSO procedure frequently retains several lagged versions of the same macroeconomic variable simultaneously, particularly for Euribor and oil prices, resulting in comparatively more complex specifications. By contrast, the incremental selection procedure adopted in this thesis prioritises interpretability and economic plausibility, retaining a limited number of regressors on the basis of their contribution to explanatory power, statistical significance, and the absence of multicollinearity.

Overall, aside from the Covid Dummy, which is retained in the final specification for all four segments given its economic relevance rather than its LASSO selection status, this robustness

exercise supports the validity of the variable selection process and provides additional confidence in the stability and economic relevance of the final model specifications.

2.4.2 Logit Transformation and OLS Estimation

Building on the four regressors selected in the previous Section, the final Logit-OLS specification adopted for each portfolio segment $j \in \{\text{Retail, SME, Corporate, Financial Institutions}\}$ can therefore be expressed as:

$$LGD_{j,t}^{logit} = \beta_{0,j} + \beta_{1,j}\Delta\text{Euribor}_{t-2} + \beta_{2,j}\Delta\text{Oil}_{t-2} + \beta_{3,j}\Delta\text{BTP}_{t-7} + \beta_{4,j}\,\text{Dummy}_{\text{Covid},t} + \varepsilon_{j,t}. \tag{2.11}$$

Because each macroeconomic regressor is first-differenced, which requires one prior observation, and the specification includes lags of up to seven quarters, most notably the change in the BTP yield lagged by seven quarters, the earliest observations of the original sample cannot be used to construct the lagged regressors; the model is therefore estimated over $t \in \{2020Q3, \ldots, 2026Q1\}$, corresponding to 23 quarterly observations, rather than the full 31 observations available for the LGD series alone.

Table 2.14 reports the estimated coefficients of the final Logit-OLS specification in Equation (2.11). The reported parameters correspond to the estimated intercept term and the estimated sensitivities of the logit-transformed LGD to the four selected macroeconomic drivers, namely $\Delta\text{Euribor}_{t-2}$, ΔOil_{t-2}, ΔBTP_{t-7}, and the Covid Dummy, estimated separately for each portfolio segment j. The table also reports the associated p-values and the main goodness-of-fit statistics.

Table 2.14: Estimated coefficients, p-values, and goodness-of-fit statistics of the final Logit-OLS specification in Equation (2.11), by portfolio segment

Variable	Retail	SME	Corporate	Institutions
Intercept	-0.9536	-0.4763	-0.5557	-0.9071
(p-value)	(< 0.0001)	(< 0.0001)	(< 0.0001)	(< 0.0001)
$\Delta\text{Euribor}_{t-2}$	-13.8183	-6.6409	-5.1207	-24.4304
(p-value)	(0.0003)	(< 0.0001)	(0.0001)	(0.0003)
ΔOil_{t-2}	-0.0030	-0.0014	-0.0015	-0.0038
(p-value)	(0.0450)	(0.0259)	(0.0053)	(0.1446)
ΔBTP_{t-7}	4.5418	0.0819	-1.3886	-5.8964
(p-value)	(0.1042)	(0.9427)	(0.1442)	(0.2350)
Covid Dummy	-0.0691	0.0310	0.0132	-0.1902
(p-value)	(0.0373)	(0.0283)	(0.2262)	(0.0030)
R^2	0.6048	0.6897	0.7019	0.6333
Adj. R^2	0.5170	0.6208	0.6357	0.5518
F-statistic	6.89	10.00	10.60	7.77
p-value(F)	0.0015	0.0002	0.0001	0.0008

First, all models are jointly significant according to the F-test. The F-statistics range from 6.89 to 10.60, with p-values consistently below the 1% significance level. This allows the null hypothesis

that all regressor coefficients are jointly equal to zero to be rejected. Therefore, the set of selected macroeconomic variables provides a statistically significant contribution to explaining LGD dynamics across the different portfolio segments.

The explanatory power of the models is also satisfactory. The Adjusted R^2 is equal to 51.70% for the Retail segment, 62.08% for SME, 63.57% for Corporates, and 55.18% for Financial Institutions, indicating a good ability of the models to capture the relationship between LGD and macroeconomic conditions.

The estimated intercepts are negative and statistically significant at the 1% level across all portfolio segments considered. These coefficients represent the expected value of the logit-transformed LGD when all explanatory variables are equal to zero. Although the economic interpretation of the intercept is limited, its significance suggests the presence of a structural component of LGD that is not explained solely by the macroeconomic variables included in the model. This links back to the comparison with Crook & Bellotti (2012) developed in Chapter 3: the macroeconomic-only framework used here already achieves a relatively high Adjusted R^2, thanks to the reduced idiosyncratic noise of aggregated portfolio data. However, the intercept remains significant, which suggests that borrower- and exposure-specific characteristics, like those used by Crook & Bellotti, would likely help explain at least part of this structural component, were such information available.

Since the dependent variable is represented by the logit transformation of LGD, the sign of the coefficients indicates the direction of the relationship between the macroeconomic drivers and expected LGD: negative coefficients imply a reduction in LGD, whereas positive coefficients are associated with an increase in LGD. At this stage, the discussion is limited to the numerical results obtained from the Logit-OLS specification; the fact that this sign-based interpretation extends directly to the original LGD scale, since both the logit and Beta transformations are monotonic increasing functions, and the economic interpretation of these results are further discussed in Section 2.4.4.

Among all the variables considered, the two-quarter lagged change in the three-month Euribor rate, $\Delta\text{Euribor}_{t-2}$, emerges as the most robust driver in the analysis. The estimated coefficient is negative and statistically significant at the 1% level across all portfolio segments. This evidence suggests that an increase in interest rates is associated, with a two-quarter (six-month) lag, with a reduction in expected LGD. This lag was identified as the most informative during the preliminary univariate analysis (Table 2.11), and the variable remains significant even once combined with the other regressors in the multivariate specifications, confirming the robustness of the relationship observed before.

The two-quarter lagged change in oil prices, ΔOil_{t-2}, also exhibits negative coefficients across all portfolio segments. The variable is statistically significant in the Retail, SME, and Corporate segments, while it loses significance in the Financial Institutions segment.

The dummy variable capturing the pandemic period reveals heterogeneous effects across the different segments. In the Retail and Financial Institutions portfolios, the coefficient is negative

and statistically significant, indicating that during the period characterised by extraordinary economic support measures, the logit of LGD is on average lower than in other periods. In the SME segment, by contrast, the coefficient is positive and significant at the 5% level, suggesting that smaller firms are more severely affected by the economic consequences of the health crisis. For the Corporate segment, the variable is not statistically significant.

Finally, the seven-quarter lagged change in Italian government bond yields, ΔBTP_{t-7}, is not statistically significant across the portfolio segments considered. Nevertheless, its inclusion in the final specification leads to a systematic and substantial improvement in the explanatory power of the models across all portfolio segments, as measured by the Adjusted R^2 (Table 2.15), without generating multicollinearity issues, as confirmed by the VIF values already reported in Table 2.12 and as discussed before.

Table 2.15: Comparison of Adjusted R^2 between the final specification and a specification excluding ΔBTP_{t-7}, by portfolio segment

Segment	Adj. R^2 with ΔBTP_{t-7}	Adj. R^2 without ΔBTP_{t-7}
Retail	0.5170	0.3965
SME	0.6208	0.4896
Corporate	0.6357	0.4127
Financial Institutions	0.5518	0.2025

The improvement is particularly pronounced for the Financial Institutions segment, where the Adjusted R^2 more than doubles, from 20.25% to 55.18%, and remains substantial for the other three segments. Although the BTP yield does not reach statistical significance on its own, this evidence suggests that it captures a systematic component of sovereign risk and financial conditions that complements the information already transmitted by Euribor, oil prices, and the Covid Dummy, rather than duplicating it.

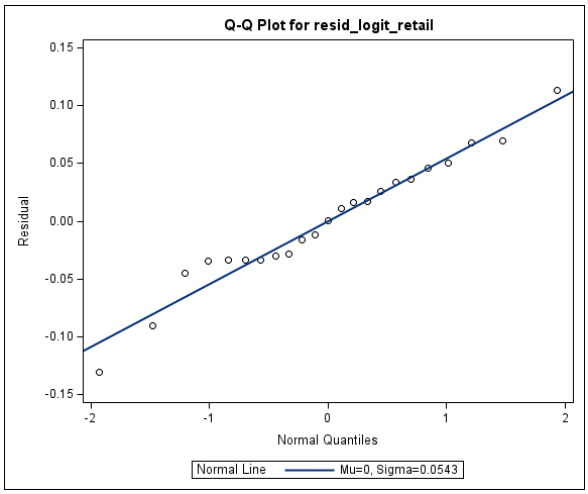
Overall, the results indicate that interest-rate dynamics, as captured by the Euribor rate, represent the main macroeconomic factor associated with LGD evolution, while oil prices, sovereign bond yields, and the effects of the pandemic help explain additional, segment-specific components of the variability observed across the different portfolio segments.

Although the regression results indicate good explanatory power for the transformed LGD, it is necessary to verify whether the main assumptions of the classical linear regression model are reasonably satisfied, particularly given the limited sample size available for estimation. In the present study, a direct assessment of the residual distribution from the Logit-OLS models is carried out.

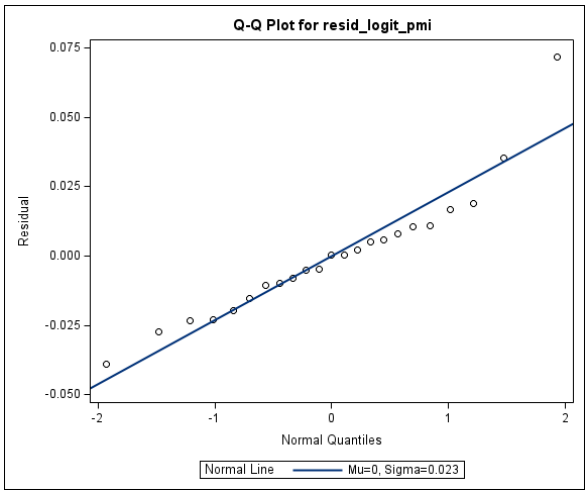
Table 2.16 reports the results of the Shapiro-Wilk test applied to the regression residuals, while Figure 2.12 provide the corresponding QQ plots. Both the statistical and graphical evidence support the assumption of residual normality. For all portfolio segments considered, the null hypothesis of normality cannot be rejected at the 5% significance level, suggesting that the residuals are compatible with a normal distribution; the p-value is lowest for the SME segment (0.0840), which nonetheless remains well above the 5% threshold. Consistently, the QQ plots show

Table 2.16: Shapiro-Wilk normality test statistics and p-values for the residuals of the final Logit-OLS specification, by portfolio segment

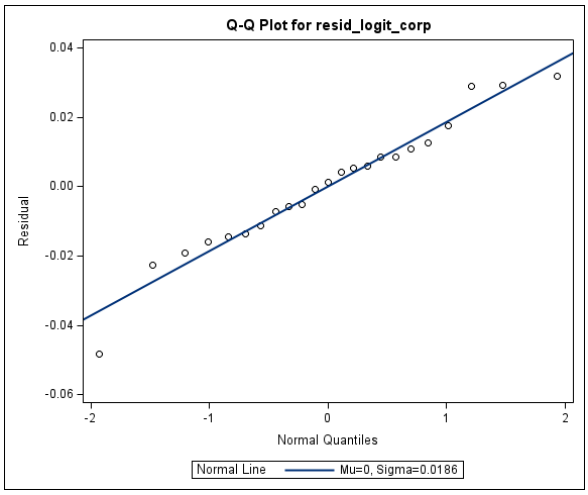
Segment	Shapiro-Wilk (W)	p-value
Retail	0.9736	0.7746
SME	0.9192	0.0840
Corporate	0.9685	0.6053
Financial Institutions	0.9750	0.8057



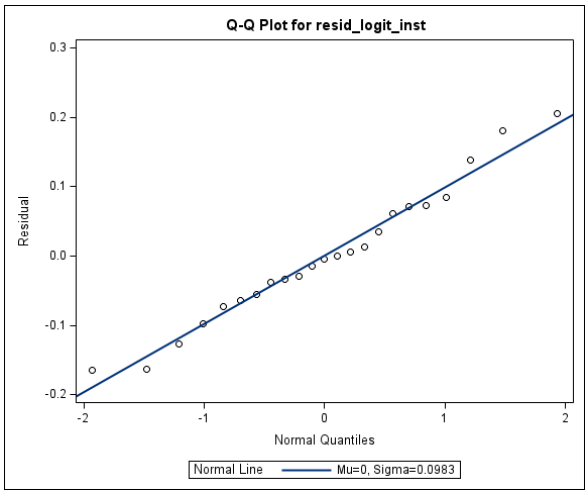
(a) Retail



(b) SME



(c) Corporate



(d) Financial Institutions

Figure 2.12: QQ plots of residuals from the Logit-OLS models, by portfolio segment

that the empirical residual quantiles are reasonably aligned with the theoretical normal quantiles. This evidence indicates that the logit transformation adequately satisfies the assumptions of the classical linear regression model within the sample analysed. As a result, although the bootstrap procedures proposed by Crook & Bellotti (2012) have not been implemented (Chapter 3), the statistical properties of the residuals suggest that inference based on OLS standard errors and coefficient significance tests remains appropriate.

Beyond residual normality, it is also important to assess whether the estimated relationships are statistically valid from a time-series perspective.

As shown in Section 2.3.1, the LGD series analysed in this thesis are non-stationary in levels and they are not differenced prior to estimation (Section 2.4.1). This raises the possibility that the relationships estimated between logit-transformed LGD and the selected macroeconomic drivers could, at least in principle, reflect spurious correlation rather than a genuine economic relationship (as anticipated in Section 2.3). For this reason, the adequacy of the specification is verified directly by testing the stationarity of the estimated residuals. Since the residuals of OLS regressions have a zero mean by construction, the Augmented Dickey-Fuller test under the *Zero Mean* specification has been applied to the residuals of the final regressions.

The results reported in Table 2.17 show that, for all portfolio segments, the null hypothesis of a unit root is rejected at the 5% significance level. The residuals can therefore be considered stationary across all four specifications. This evidence indicates that the estimated relationships between the logit-transformed LGD and the selected macroeconomic variables are not spurious, and provides the empirical confirmation that the econometric specifications adopted in this thesis are valid despite the non-stationarity of LGD in levels.

Table 2.17: Augmented Dickey-Fuller test statistics (Zero Mean specification) and p-values for the residuals of the final Logit-OLS regressions, by portfolio segment

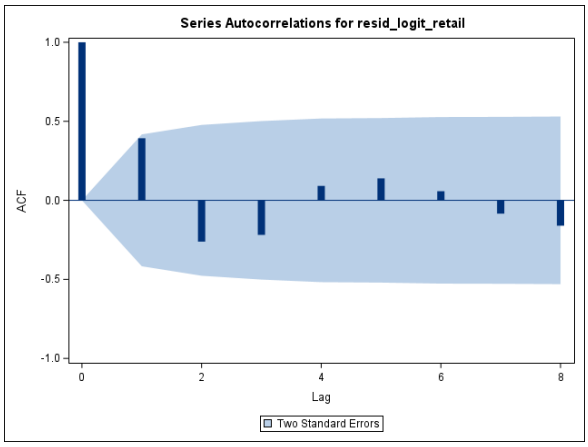
Segment	Tau	p-value
Retail	-2.97	0.0048
SME	-5.25	< 0.0001
Corporate	-4.96	< 0.0001
Financial Institutions	-2.30	0.0238

Finally, residual independence has been assessed through autocorrelation diagnostics. In particular, the Ljung-Box white-noise test has been applied to the residuals of the final regressions. As reported in Table 2.18, the null hypothesis, that the residuals do not exhibit significant serial correlation, cannot be rejected at the 5% significance level for the Retail, SME, and Corporate segments, indicating that the residuals are consistent with a white-noise process. By contrast, the statistic for the Financial Institutions segment results significant at the 5% level, indicating the presence of residual serial dependence.

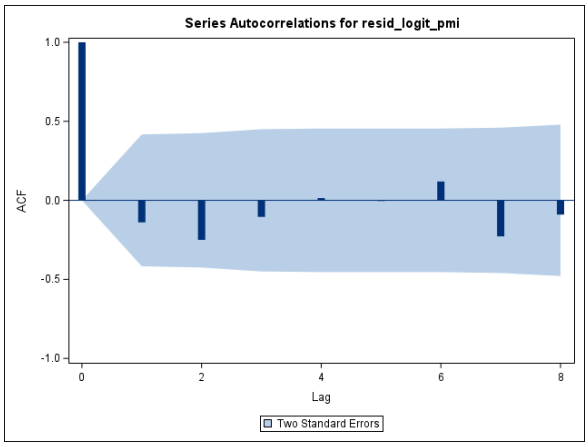
Figure 2.13 provides a graphical representation of the residual autocorrelation structure for each portfolio segment, consistent with the Ljung-Box statistics reported above. At lag 0 the autocorrelation is trivially equal to one for all four segments, while at lag 1 both Retail and

Table 2.18: Ljung-Box white-noise test statistics and p-values for the residuals of the final Logit-OLS regressions, by portfolio segment

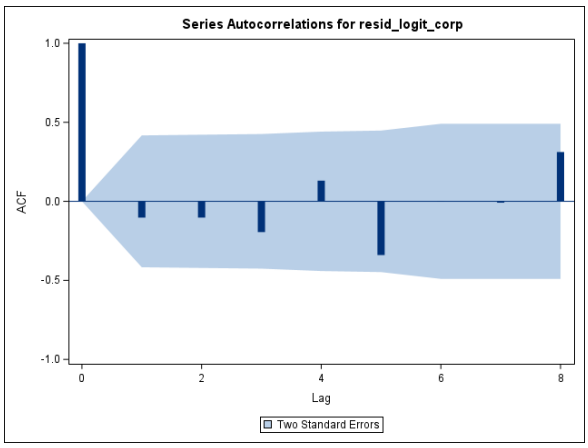
Segment	Ljung-Box Statistic	p-value
Retail	8.26	0.2199
SME	3.02	0.8062
Corporate	5.86	0.4388
Financial Institutions	16.18	0.0128



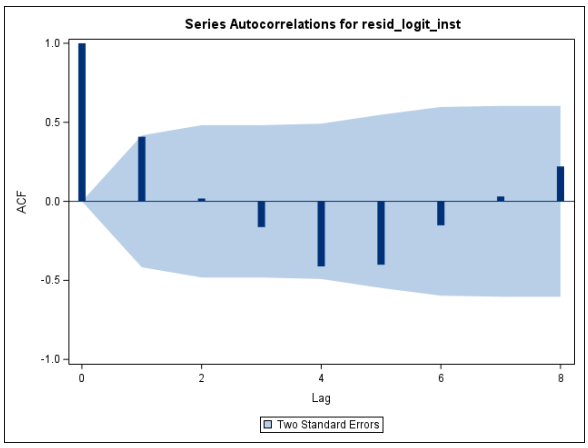
(a) Retail



(b) SME



(c) Corporate



(d) Institutions

Figure 2.13: Autocorrelation function (ACF) of the residuals from the final Logit-OLS regressions, by portfolio segment, with 95% confidence bounds

Financial Institutions display a coefficient close to the confidence bounds. Beyond lag 1, the coefficients for Retail, SME, and Corporate settle comfortably within the confidence bounds, whereas those for Financial Institutions remain somewhat closer to the bounds across several subsequent lags, although never actually crossing them.

Taken together, the residual diagnostic tests provide generally favourable evidence regarding the adequacy of the proposed Logit-OLS specifications. To summarise, the residuals are found to be approximately normally distributed and stationary across all portfolio segments, supporting the statistical validity of the estimated relationships. Residual independence is also supported for the Retail, SME, and Corporate portfolios. For the Financial Institutions portfolio, however, the Ljung-Box test reveals some remaining serial dependence, consistent with the greater historical volatility and the level shift documented for this segment in Section 2.3.1, and suggesting that the selected specification may not fully capture all of its underlying temporal dynamics.

However, for a satellite model intended for stress-testing and forward-looking applications, it is also important to evaluate the model's ability to accurately reproduce the observed dynamics of the target variable. To this end, a predictive accuracy analysis has been conducted for each portfolio segment by comparing estimated LGD values with the corresponding observed values. Once the parameters $\beta_{0,j}, \ldots, \beta_{4,j}$ of the final Logit-OLS specification in Equation (2.11) are estimated by OLS for each portfolio segment j, denoted by their estimated counterparts $\hat{\beta}_{0,j}, \ldots, \hat{\beta}_{4,j}$, the fitted value of the logit-transformed LGD is obtained as:

$$\widehat{LGD}_{j,t}^{\,logit} = \hat{\beta}_{0,j} + \hat{\beta}_{1,j}\Delta\text{Euribor}_{t-2} + \hat{\beta}_{2,j}\Delta\text{Oil}_{t-2} + \hat{\beta}_{3,j}\Delta\text{BTP}_{t-7} + \hat{\beta}_{4,j}\,\text{Dummy}_{\text{Covid},t} \tag{2.12}$$

Then, the estimated values of the logit-transformed LGD are converted back to the original $(0, 1)$ scale using the inverse of the logit transformation, namely the logistic function:

$$\widehat{LGD}_{j,t} = \frac{\exp\left(\widehat{LGD}_{j,t}^{\,logit}\right)}{1 + \exp\left(\widehat{LGD}_{j,t}^{\,logit}\right)}, \tag{2.13}$$

where $\widehat{LGD}_{j,t}^{\,logit}$ denotes the fitted value from the Logit-OLS model, on the logit scale, introduced in Equation (2.12), for portfolio segment $j \in \{\text{Retail, SME, Corporate, Financial Institutions}\}$ and $t \in \{2020Q3, \ldots, 2026Q1\}$, while $\widehat{LGD}_{j,t}$ denotes the corresponding prediction converted back to the original LGD scale. Since the logistic function in Equation (2.13) maps any real number onto the interval $(0, 1)$, the resulting predictions $\widehat{LGD}_{j,t}$ are guaranteed to lie within the admissible range for LGD; this is precisely the first reason that motivated the choice of the logit transformation.

The prediction error for segment j in period t, with $t \in \{2020Q3, \ldots, 2026Q1\}$, is then defined as the difference between the observed and the predicted LGD, on the original scale:

$$e_{j,t} = LGD_{j,t} - \widehat{LGD}_{j,t}. \tag{2.14}$$

Based on this error, two complementary summary measures of predictive accuracy are computed for each segment j, where N_j denotes the number of quarterly observations for which the dependent variable and all lagged regressors are simultaneously available. Since the most heavily lagged regressor, ΔBTP_{t-7}, requires both first-differencing and a seven-quarter lag, $N_j = 23$, spanning $t \in \{2020Q3, \ldots, 2026Q1\}$, for the estimation of the final Logit-OLS specifications and the computation of the predictive accuracy measures below.
The Mean Absolute Error (MAE) for segment j is defined as:

$$MAE_j = \frac{1}{N_j} \sum_{t=2020Q3}^{2026Q1} |e_{j,t}|, \tag{2.15}$$

and measures the average magnitude of the prediction error, in absolute value, treating over- and under-predictions symmetrically and without penalising large errors more than small ones.
A second measure is obtained by first computing the Mean Squared Error (MSE):

$$MSE_j = \frac{1}{N_j} \sum_{t=2020Q3}^{2026Q1} e_{j,t}^2, \tag{2.16}$$

which represents the average squared prediction error; because errors are squared before being averaged, larger deviations contribute disproportionately more to MSE_j than smaller ones. Taking the square root of the MSE, so as to express the measure in the same units as LGD itself, yields the Root Mean Squared Error:

$$RMSE_j = \sqrt{MSE_j} = \sqrt{\frac{1}{N_j} \sum_{t=2020Q3}^{2026Q1} e_{j,t}^2}. \tag{2.17}$$

Because of this squaring step, RMSE is more sensitive than MAE to occasional large forecast errors, and is therefore particularly useful for detecting the presence of predictions that deviate markedly from the observed values, even when the majority of predictions are accurate. Lower values of both MAE and RMSE indicate a better ability of the model to reproduce the observed dynamics of LGD within the portfolio segment considered.

Table 2.19: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) of the final Logit-OLS models, by portfolio segment, with RMSE also expressed as a percentage of the segment's historical mean LGD

Segment	MAE	MSE	RMSE	RMSE (% of mean LGD)
Retail	0.0085	0.000113	0.0106	3.9
SME	0.0038	0.000029	0.0053	1.4
Corporate	0.0033	0.000018	0.0042	1.1
Financial Institutions	0.0152	0.000391	0.0198	7.7

The results reported in Table 2.19 indicate a high level of predictive accuracy across all portfolio segments considered. In particular, the Corporate portfolio exhibits the lowest RMSE value, equal to 0.0042, followed by the SME segment (0.0053) and the Retail portfolio (0.0106). By contrast, the Financial Institutions segment displays the largest prediction errors, with an RMSE of 0.0198, more than four times that of Corporate.

Since MAE, MSE, and RMSE are expressed in the same decimal units as LGD itself, while the descriptive statistics in Table 2.1 are reported in percentage points, the last column of Table 2.19 re-expresses RMSE as a percentage of each segment's historical average LGD, to allow for a fair comparison of prediction errors across segments with different mean LGD levels. On this relative basis, the RMSE amounts to approximately 1.1% of the mean LGD for Corporate, 1.4% for SME, 3.9% for Retail, and 7.7% for Financial Institutions, indicating that prediction errors are not only larger in absolute terms for Financial Institutions, but also considerably more significant relative to the typical LGD level of this segment.

This ranking is consistent with evidence discussed elsewhere in this thesis: the segment of Financial Institutions has been already identified in Section 2.3.1 as the most volatile of the four LGD series, exhibiting a marked level shift from 2025 onwards, and its residuals are the only ones to display statistically significant serial dependence according to the Ljung-Box test (Section 2.4.2). Taken together, this evidence suggests that the relatively weaker predictive performance observed for Financial Institutions reflects genuine, segment-specific dynamics that the four macroeconomic regressors included in the final specification capture only partially, rather than a shortcoming affecting the Logit-OLS approach as a whole.

Overall, average prediction errors remain low across all segments and are broadly consistent with the Adjusted R^2 values obtained in the multivariate regressions (Table 2.14), confirming the ability of the models to adequately reproduce the historical dynamics of LGD, with the notable exception of the comparatively weaker performance observed for Financial Institutions.

To complement this quantitative evidence, Figures 2.14, 2.15, 2.16, and 2.17 compare the observed LGD values with the corresponding values estimated by the Logit-OLS model, on the original LGD scale, for each portfolio segment. This graphical representation makes it possible to visually assess the degree of fit between the estimated and observed data and to identify any potential systematic deviations over the sample period; consistent with the discussion in Section 2.4.2, the fitted series in these figures begins only in 2020Q3, reflecting the eight quarters lost to first-differencing and lagging in the construction of ΔBTP_{t-7}.

2.4.3 Beta Transformation and OLS Estimation

The Beta Transformation is intended to make the distribution of the dependent variable closer to normality before applying linear regression. This approach is adopted, among others, by Crook & Bellotti (2012) in the context of *Loss Given Default* (LGD) modelling (Chapter 3), and is

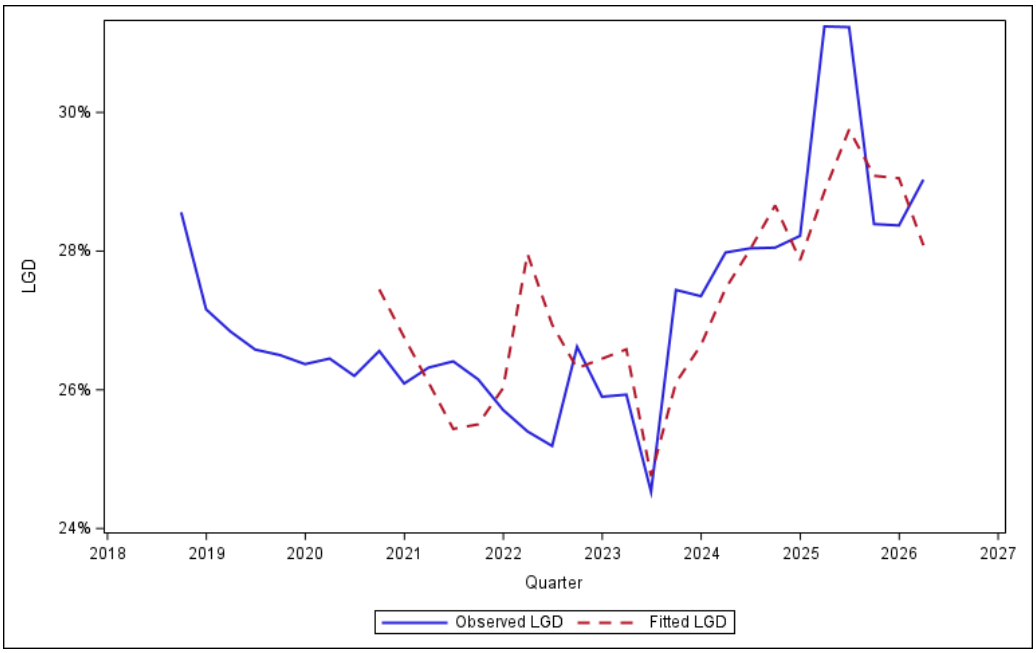


Figure 2.14: Observed and fitted LGD values for the Retail portfolio obtained from the Logit-OLS model

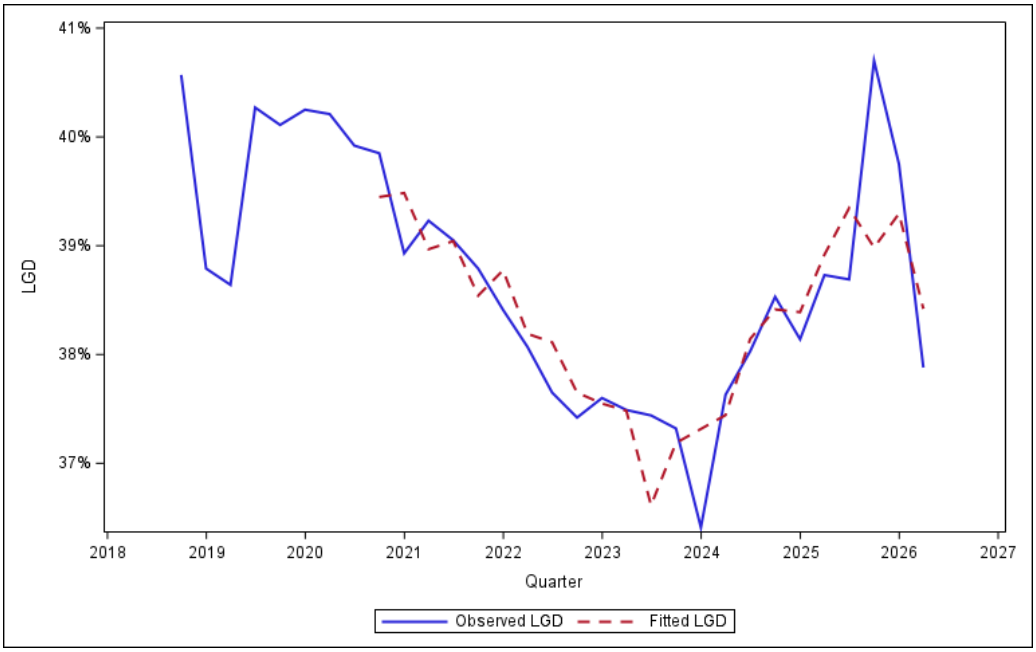


Figure 2.15: Observed and fitted LGD values for the SME portfolio obtained from the Logit-OLS model

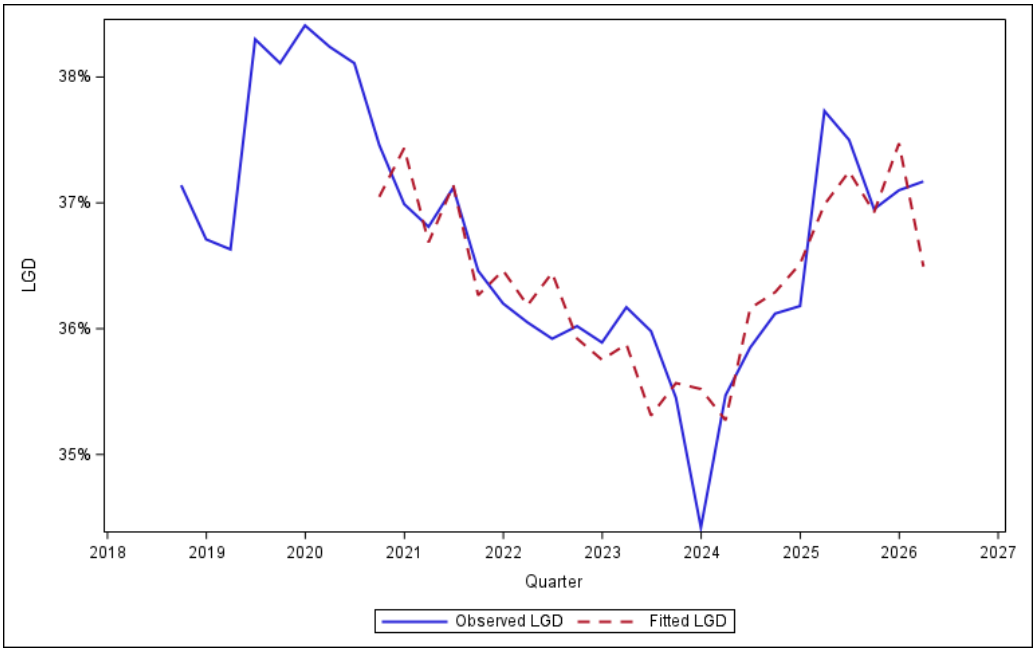


Figure 2.16: Observed and fitted LGD values for the Corporate portfolio obtained from the Logit-OLS model

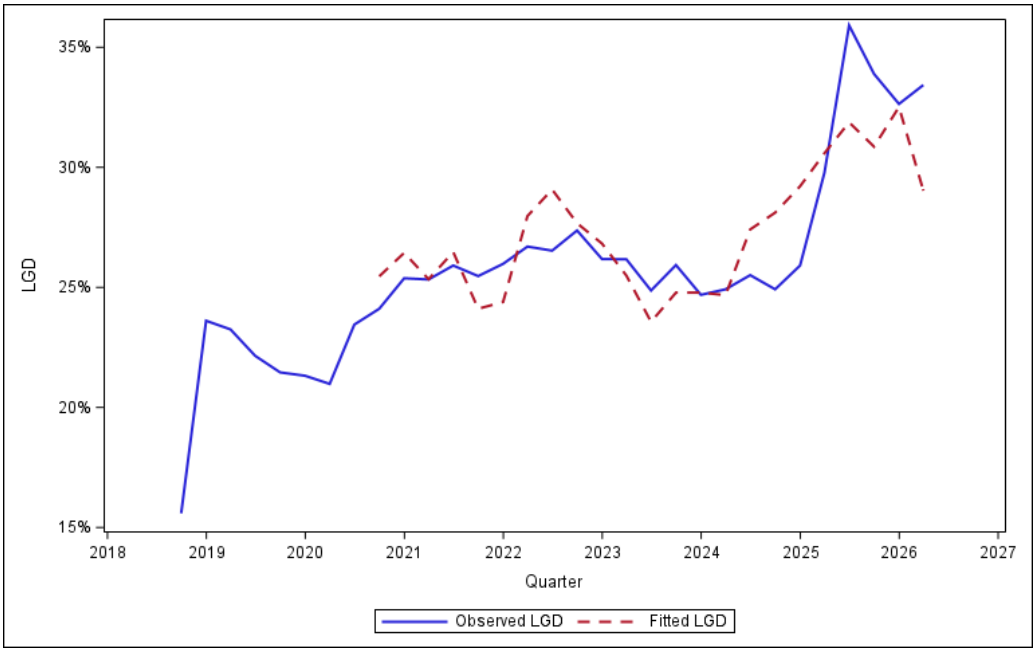


Figure 2.17: Observed and fitted LGD values for the Financial Institutions portfolio obtained from the Logit-OLS model

particularly suitable for bounded variables such as LGD, which, for the reasons already discussed in Section 2.4.2, typically exhibits a U-shaped, bimodal empirical distribution.

The Beta distribution is characterised by two shape parameters, α and β, which jointly determine the shape, skewness, and dispersion of the distribution over the interval $(0, 1)$. The initial intention was to estimate these parameters using Maximum Likelihood Estimation (MLE), by identifying the pair (α, β) that maximises the probability of observing the available empirical LGD sample, independently for each of the Corporate, SME, Retail, and Financial Institutions series.

Denoting by $LGD_{j,t}$ the Loss Given Default observed at time t for portfolio segment j (Retail, SME, Corporate, or Financial Institutions), by T_j the total number of historical LGD observations available for that segment, equal to 31 quarters for all four segments considered (Section 2.1), and by α_j and β_j the two shape parameters of the Beta distribution to be estimated for segment j, the corresponding likelihood function would take the form:

$$L(\alpha_j, \beta_j) = \prod_{t=1}^{T_j} \frac{\Gamma(\alpha_j + \beta_j)}{\Gamma(\alpha_j)\,\Gamma(\beta_j)}\, LGD_{j,t}^{\alpha_j - 1}\,(1 - LGD_{j,t})^{\beta_j - 1}, \tag{2.18}$$

where $\Gamma(\cdot)$ denotes the Gamma function, and the coefficient $\dfrac{\Gamma(\alpha_j + \beta_j)}{\Gamma(\alpha_j)\,\Gamma(\beta_j)}$ is the normalising constant that ensures the Beta density integrates to one over $(0, 1)$ for any given pair (α_j, β_j).

This approach, however, did not prove to be successful in practice: for some of the portfolio segments considered, the maximum likelihood optimisation procedure was not able to estimate both distribution parameters in a numerically stable manner, failing to reach convergent estimates. To ensure methodological consistency across all four segments analysed, the parameters of the Beta distribution are therefore estimated using the method of moments, which derives α_j and β_j directly from the sample mean and variance of the LGD_j series and is numerically more stable in the presence of small samples and highly concentrated distributions. In this way it is possible to identify, for each portfolio segment, a theoretical Beta distribution characterised by the same mean and variance as the empirical LGD distribution, indicated respectively with m_j and v_j, while, at the same time, ensuring the availability of robust and comparable parameter estimates across the different exposure categories considered.

Define the auxiliary quantity:

$$\phi_j = \frac{m_j(1 - m_j)}{v_j} - 1, \tag{2.19}$$

which is positive whenever the sample variance v_j is smaller than $m_j(1 - m_j)$ (the maximum variance attainable by a Beta-distributed variable with mean m_j); this condition is satisfied by all four LGD series considered in this thesis. The estimated Beta distribution parameters are then obtained as:

$$\hat{\alpha}_j = m_j \phi_j, \qquad \hat{\beta}_j = (1 - m_j)\phi_j. \tag{2.20}$$

Table 2.20 reports the estimated Beta distribution parameters for each portfolio segment, obtained by applying Equations (2.19) and (2.20) to the sample mean and variance of each of the four LGD series.

Table 2.20: Method-of-moments estimates of the Beta distribution parameters ($\hat{\alpha}$, $\hat{\beta}$), obtained from the sample mean and variance of each LGD series, by portfolio segment

Segment	Mean LGD (m)	Variance (v)	$\hat{\alpha}$	$\hat{\beta}$
Retail	0.2712	0.0002373	225.66	606.34
SME	0.3873	0.0001275	720.46	1139.96
Corporate	0.3673	0.0000928	918.97	1582.94
Financial Institutions	0.2579	0.0016543	29.57	85.11

The results highlight substantial heterogeneity across portfolios. In particular, the Corporate and SME segments exhibit very high values of the shape parameters $\hat{\alpha}$ and $\hat{\beta}$. By contrast, the Financial Institutions segment displays considerably lower parameter values. Since a larger sum $\hat{\alpha}_j + \hat{\beta}_j$ corresponds to a Beta distribution more tightly concentrated around its mean, this pattern indicates that the fitted Beta distribution is considerably more diffuse for Financial Institutions than for the other three segments, mirroring the same volatility ranking observed in Section 2.3.1. Once the parameters $\hat{\alpha}_j$ and $\hat{\beta}_j$ have been estimated, the cumulative probability associated with each observed LGD value is computed using the cumulative distribution function of the Beta distribution:

$$p_{j,t} = F_{\text{Beta}}\left(LGD_{j,t}; \hat{\alpha}_j, \hat{\beta}_j\right), \tag{2.21}$$

which represents the proportion of the fitted Beta distribution for segment j lying at or below the observed value $LGD_{j,t}$, and therefore takes values in the interval $(0, 1)$ by construction.
The resulting probability is then transformed through:

$$Z_{j,t} = \Phi^{-1}\left(p_{j,t}\right), \tag{2.22}$$

where $\Phi^{-1}(\cdot)$ denotes the inverse cumulative distribution function of the standard normal distribution. Since $\Phi^{-1}(\cdot)$ maps the interval $(0, 1)$ onto the entire real line, $Z_{j,t}$ is no longer bounded, and can therefore be modelled using standard linear regression technique.
This two-step procedure – first computing the Beta cumulative probability in Equation (2.21), and then applying the standard normal quantile function in Equation (2.22) – relies on a well-known statistical result known as the *Probability Integral Transform*.
The Probability Integral Transform states that, for any continuous random variable X with cumulative distribution function F_X, the transformed variable $F_X(X)$ is always uniformly distributed on $(0, 1)$, regardless of the original distribution of X. Applying this result to $LGD_{j,t}$ in Equation (2.21) therefore produces a variable that is approximately uniform on $(0, 1)$; the standard normal quantile function in Equation (2.22) is then used to convert this uniform variable into one that is approximately normally distributed.

The result is a transformed variable that mirrors, through a different route, the same objective pursued by the logit transformation for the Logit-OLS specification: making the dependent variable suitable for estimation by ordinary least squares.

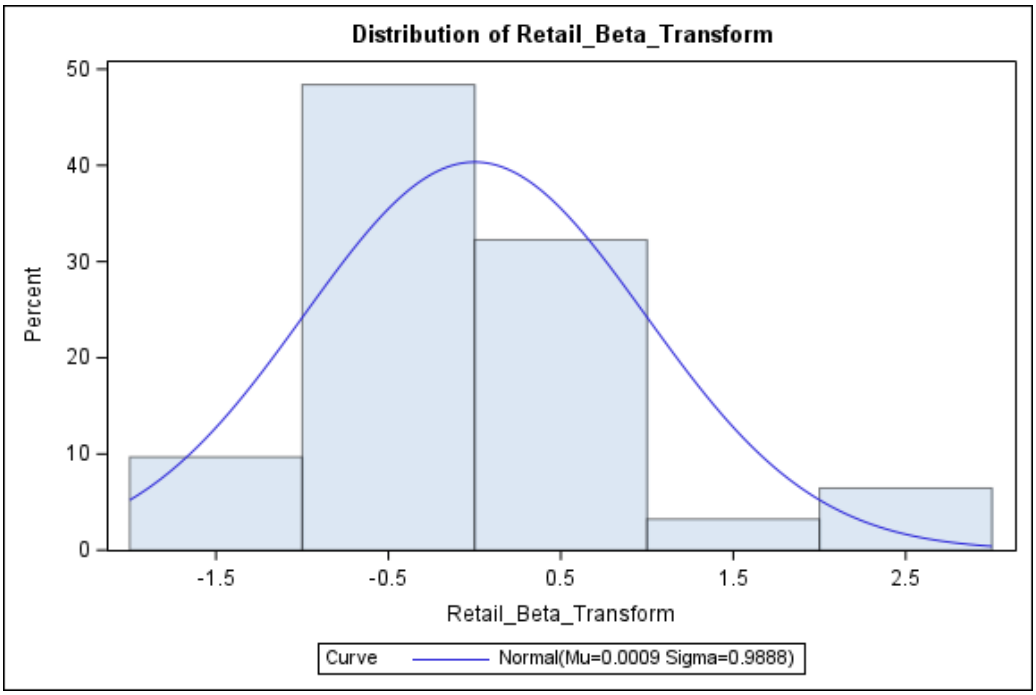


Figure 2.18: Distribution of the Beta-transformed LGD for the Retail portfolio, together with the fitted normal density (estimated mean and standard deviation)

The resulting empirical distributions of $Z_{j,t}$, shown in Figures 2.18, 2.19, 2.20, and 2.21, and summarised in Table 2.21, are assessed against the standard normal benchmark both in terms of their first two moments and, subsequently, through formal normality tests. Across all four segments, the fitted means are very close to zero and the fitted standard deviations are close to one, indicating that the transformed variable closely approximates a standard normal distribution in terms of location and scale.

Table 2.21: Estimated mean and standard deviation of the normal distribution fitted to the Beta-transformed LGD series, by portfolio segment

Segment	Mean (μ)	Std. Dev. (σ)
Retail	0.000889	0.988754
SME	0.000130	0.999533
Corporate	0.000113	1.000114
Financial Institutions	0.001771	0.996049

A close match in mean and standard deviation, however, does not by itself guarantee normality of the full distribution, since higher-order moments such as skewness and kurtosis are left unconstrained. Figure 2.22 provides a graphical assessment of this point through QQ plots, while Table 2.22 reports the corresponding Shapiro-Wilk and Kolmogorov-Smirnov normality tests applied to $Z_{j,t}$ for each portfolio segment.

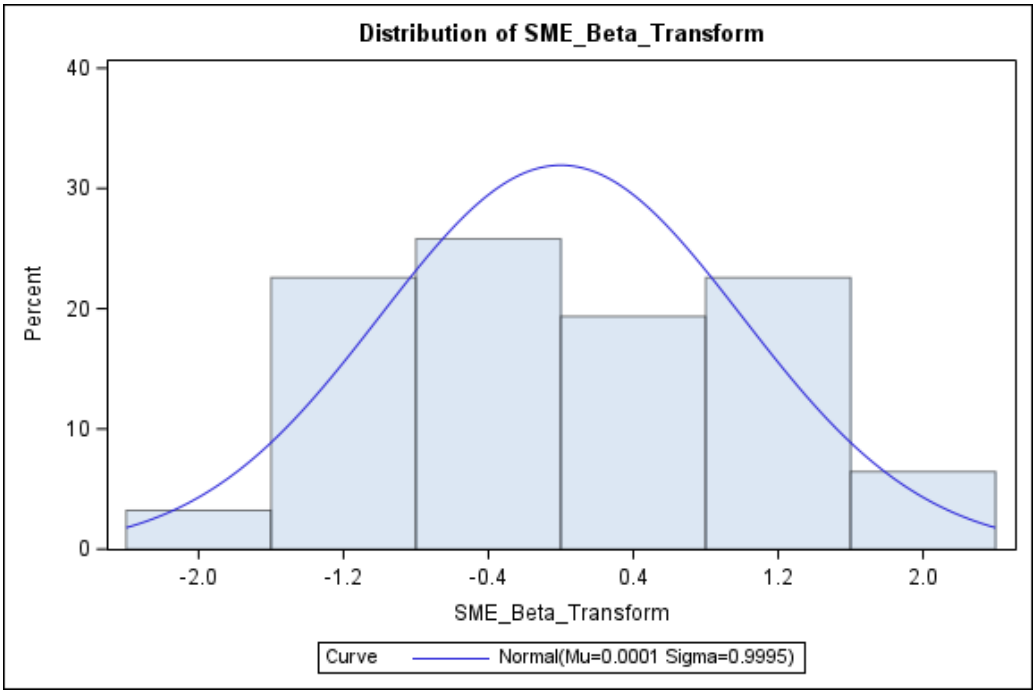


Figure 2.19: Distribution of the Beta-transformed LGD for the SME portfolio, together with the fitted normal density (estimated mean and standard deviation)

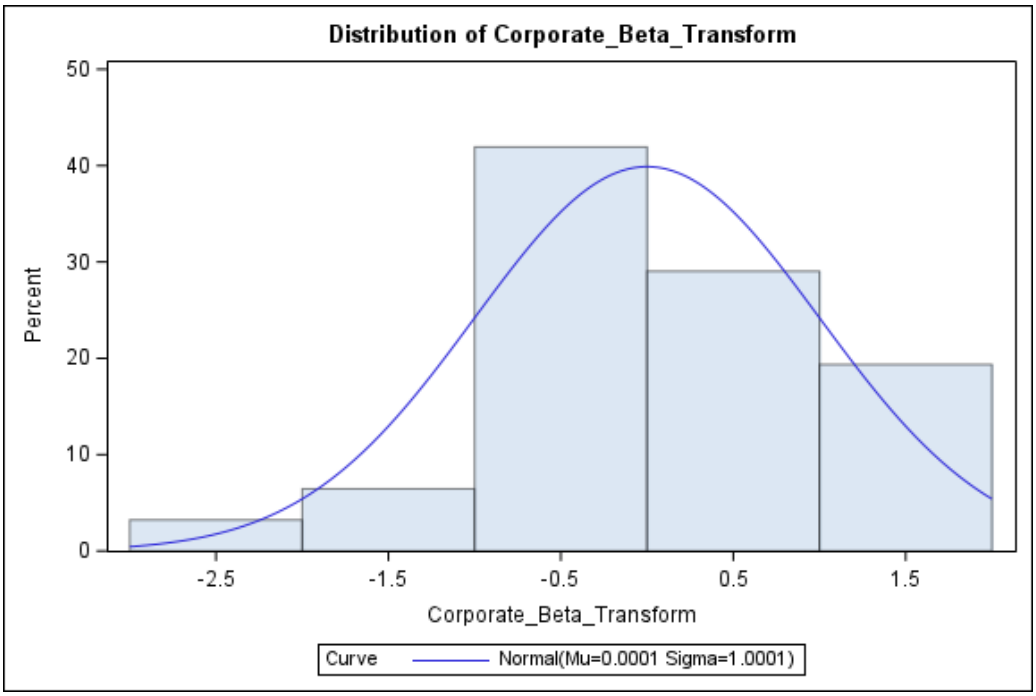


Figure 2.20: Distribution of the Beta-transformed LGD for the Corporate portfolio, together with the fitted normal density (estimated mean and standard deviation)

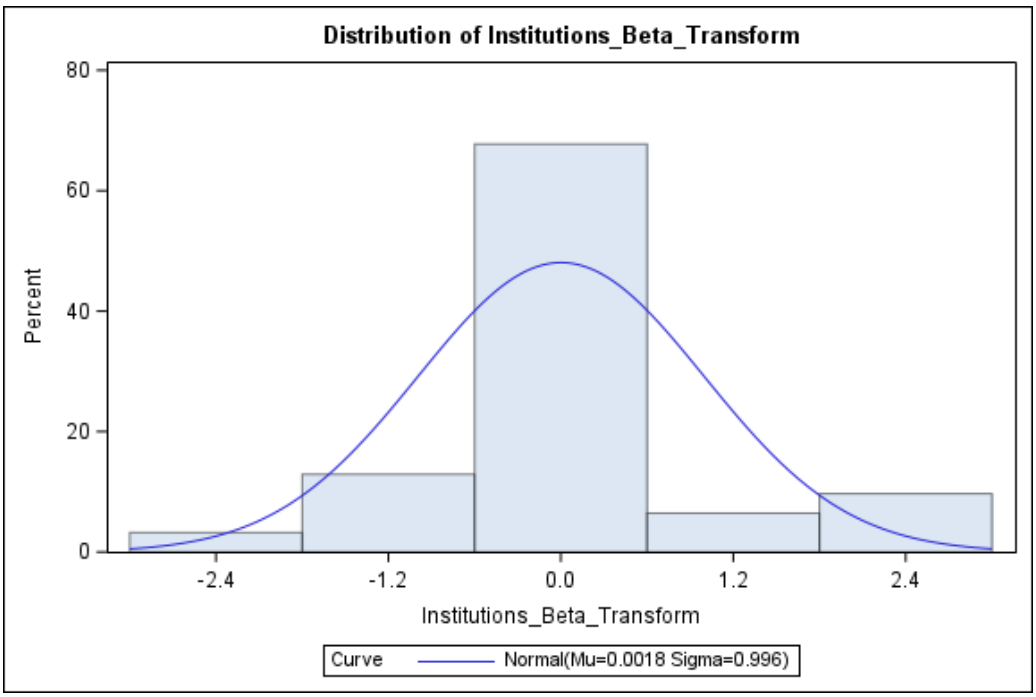
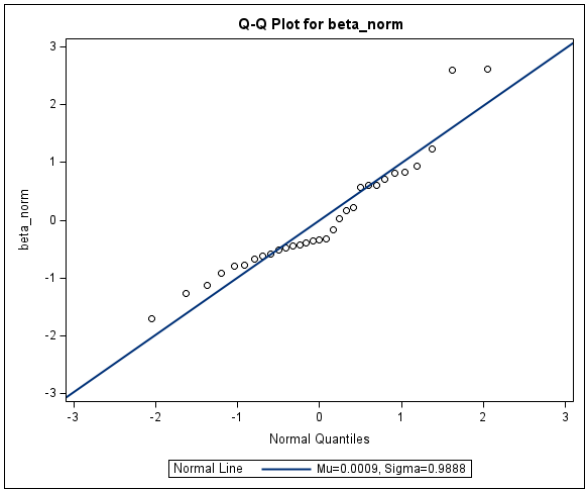


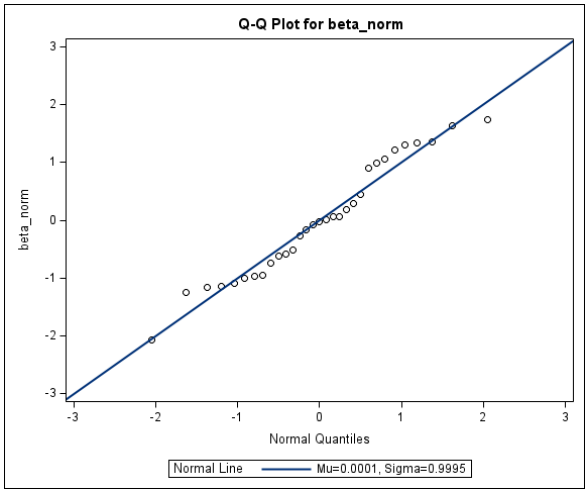
Figure 2.21: Distribution of the Beta-transformed LGD for the Financial Institutions portfolio, together with the fitted normal density (estimated mean and standard deviation)

Table 2.22: Shapiro-Wilk and Kolmogorov-Smirnov normality test statistics and p-values for the Beta-transformed LGD series, by portfolio segment

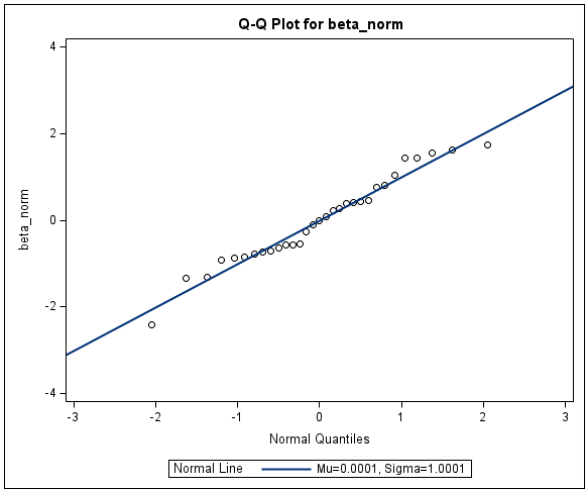
	Shapiro-Wilk		Kolmogorov-Smirnov	
Segment	Statistic (W)	p-value	Statistic (D)	p-value
Retail	0.9158	0.0181	0.1739	0.0178
SME	0.9581	0.2596	0.1083	> 0.1500
Corporate	0.9659	0.4133	0.1276	> 0.1500
Financial Institutions	0.9107	0.0135	0.2056	< 0.010



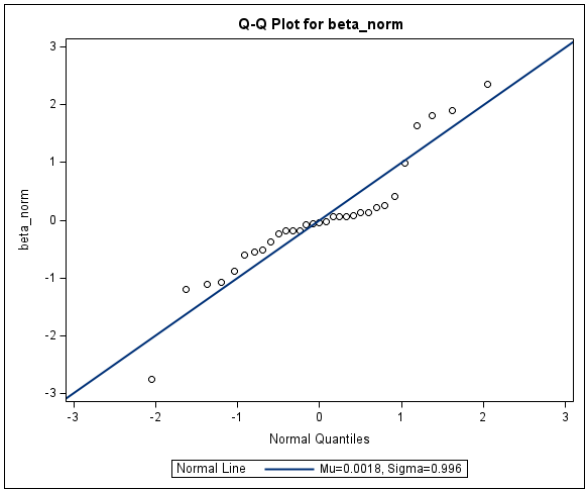
(a) Retail



(b) SME



(c) Corporate



(d) Institutions

Figure 2.22: QQ plots of the Beta-transformed LGD series against the standard normal distribution, by portfolio segment

For the SME and Corporate segments, neither test rejects the null hypothesis of normality at the 5% significance level, indicating that the Beta Transformation successfully achieves its intended objective for these two portfolios. By contrast, for the Retail and Financial Institutions segments, both tests yield p-values below the conventional 5% threshold, indicating a statistically significant deviation from normality despite the favourable mean and standard deviation reported in Table 2.21. This divergence between the first two moments and the formal test outcomes is consistent with the more markedly non-normal empirical LGD distributions already documented for these two segments in Section 2.3.1 – moderately so for Retail, and considerably more so for Financial Institutions, whose historical volatility is nearly three times higher than Retail's and more than four times that of Corporate and SME – suggesting that, for these portfolios, the Probability Integral Transform succeeds in centring and rescaling the transformed variable, but does not fully eliminate the residual skewness and kurtosis present in the original LGD series.

Despite these residual deviations from normality for Retail and Financial Institutions, the transformed variable, obtained as in Equation (2.22), is retained as the dependent variable for all four segments, consistently with the objective of preserving a uniform methodological framework across the different portfolios and of directly comparing the results with those obtained under the Logit-OLS specification. The same principle motivates the choice to retain the same four macroeconomic regressors selected in Section 2.4.1.

The final Beta-OLS specification can therefore be written as:

$$Z_{j,t} = \gamma_{0,j} + \gamma_{1,j}\Delta\text{Euribor}_{t-2} + \gamma_{2,j}\Delta\text{Oil}_{t-2} + \gamma_{3,j}\Delta\text{BTP}_{t-7} + \gamma_{4,j}\,\text{Dummy}_{\text{Covid},t} + \varepsilon_{j,t} \quad (2.23)$$

where $Z_{j,t}$ represents the Beta-transformed LGD for portfolio segment j (Retail, SME, Corporate, or Financial Institutions) at time $t \in \{2020Q3, \ldots, 2026Q1\}$, obtained as in Equation (2.22).

Finally, Table 2.23 reports the estimated coefficients of the multivariate Beta-OLS specifications, together with their corresponding p-values and goodness-of-fit statistics.

In all segments, the model is globally significant, as evidenced by the values of the F-statistic, ranging from 6.87 for Retail to 10.62 for Corporate, with p-values consistently below the 1% significance level.

The explanatory power of the models is also satisfactory. The Adjusted R^2 is equal to 51.64% for Retail, 61.98% for SME, 63.62% for Corporate, and 54.88% for Financial Institutions, with Corporate again displaying the highest explanatory power among the four portfolio segments considered.

The estimated intercepts are statistically significant at conventional levels across all portfolio segments, positive for Retail and Financial Institutions and negative for SME and Corporate. As with the Logit-OLS specification, this significance suggests the presence of a structural component of LGD that is not explained solely by the macroeconomic variables included in the model, consistent with the comparison developed in Chapter 3.

Table 2.23: Estimated coefficients, p-values, and goodness-of-fit statistics of the final Beta-OLS specification in Equation (2.23), by portfolio segment

Variable	Retail	SME	Corporate	Institutions
Intercept	0.4686	-0.3611	-0.2818	0.7552
(p-value)	(0.0310)	(0.0177)	(0.0409)	(< 0.0001)
$\Delta\text{Euribor}_{t-2}$	-177.8913	-139.3656	-123.2868	-117.4939
(p-value)	(0.0003)	(< 0.0001)	(0.0001)	(0.0004)
ΔOil_{t-2}	-0.03872	-0.03007	-0.03643	-0.01864
(p-value)	(0.0446)	(0.0260)	(0.0052)	(0.1414)
ΔBTP_{t-7}	57.8793	1.7429	-33.3508	-27.6620
(p-value)	(0.1074)	(0.9421)	(0.1447)	(0.2472)
Covid Dummy	-0.8934	0.6510	0.3174	-0.9078
(p-value)	(0.0368)	(0.0286)	(0.2259)	(0.0033)
R^2	0.6043	0.6889	0.7023	0.6308
Adj. R^2	0.5164	0.6198	0.6362	0.5488
F-statistic	6.87	9.97	10.62	7.69
p-value(F)	0.0015	0.0002	0.0001	0.0009

However, since the dependent variable is represented by the transformed variable $Z_{j,t}$ obtained through the Probability Integral Transform (Equations (2.21) and (2.22)), these coefficients do not have a direct economic interpretation in terms of LGD, and represent instead the expected value of the transformed variable in the absence of changes in the explanatory variables. At this stage, the discussion is limited to the numerical results obtained from the Beta-OLS specification; the economic implications of these results, and their consistency with the Logit-OLS specification, are discussed further in Section 2.4.4.

The coefficients associated with changes in the Euribor rate, $\Delta\text{Euribor}_{t-2}$, are negative and statistically significant across all segments considered, confirming its role as the single most robust macroeconomic driver of LGD across all four portfolios.

The lagged change in oil prices, ΔOil_{t-2}, exhibits a negative and significant coefficient in the Retail, SME, and Corporate segments, while it is not statistically significant for Financial Institutions, suggesting a lower sensitivity of this segment to conditions in the real economy.

Changes in the BTP yield, ΔBTP_{t-7}, do not show evidence of statistical significance in any of the estimated specifications; nevertheless, its inclusion in the final specification is consistent with the systematic improvement in explanatory power documented for the Logit-OLS specification (Table 2.15).

The Covid Dummy is statistically significant in the Retail, SME, and Financial Institutions segments, while no evidence of significance emerges in the Corporate segment. The coefficients show opposite signs across the four segments, negative for Retail and Financial Institutions and positive for SME and Corporate, suggesting a heterogeneous impact of the pandemic period across different business areas.

While the results obtained so far are broadly consistent with what can be expected, it is necessary to verify that the Beta-OLS models satisfy the same classical linear regression assumptions

already validated for the Logit-OLS models. To this end, a direct assessment of the residual distribution is performed, rather than through the bootstrap procedures applied by Crook & Bellotti (2012) (Chapter 3). Table 2.24 reports the results of the Shapiro-Wilk test applied to the residuals of the Beta-OLS models.

Table 2.24: Shapiro-Wilk normality test statistics and p-values for the residuals of the Beta-OLS specification, by portfolio segment

Segment	Shapiro-Wilk (W)	p-value
Retail	0.9752	0.8096
SME	0.9181	0.0608
Corporate	0.9672	0.6221
Financial Institutions	0.9738	0.7778

For all portfolio segments considered, the null hypothesis of normality cannot be rejected at the 5% significance level. These findings suggest that the residuals of the Beta-OLS regressions are compatible with a normal distribution.

As with the Logit-OLS model, a precautionary check against the risk of spurious correlation has been performed: the stationarity of the estimated residuals has been verified directly, using the Augmented Dickey-Fuller test under the *Zero Mean* specification.

The results reported in Table 2.25 show that, for all portfolio segments considered, the null hypothesis of a unit root is rejected at the 5% significance level, with the weakest rejection observed for the Financial Institutions segment ($p = 0.0234$) and the strongest for SME and Corporate ($p < 0.0001$). The residuals can therefore be regarded as stationary across all four specifications. This evidence indicates that the relationships estimated using the Beta Transformation are not spurious in nature, providing further, independent support for the econometric validity of the adopted specifications.

Table 2.25: Augmented Dickey-Fuller test statistics (Zero Mean specification) and p-values for the residuals of the final Beta-OLS regressions, by portfolio segment

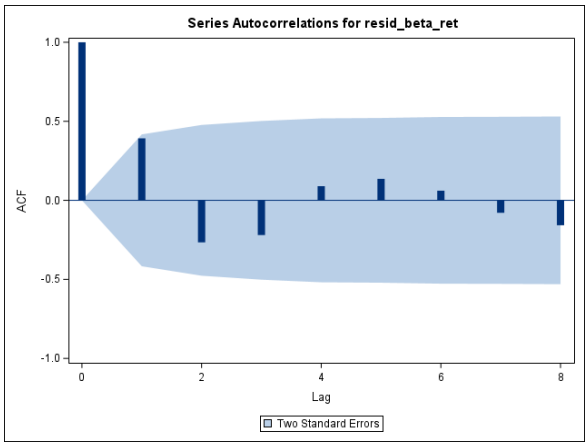
Segment	Tau	p-value
Retail	-2.98	0.0047
SME	-5.25	< 0.0001
Corporate	-4.96	< 0.0001
Financial Institutions	-2.30	0.0234

Finally, the Ljung-Box white-noise test has been applied to the residuals of the final Beta-OLS regressions to assess their independence. The results are reported in Table 2.26: the null hypothesis of no residual serial correlation cannot be rejected at the 5% significance level for the Retail, SME, and Corporate segments, suggesting that the residuals are consistent with a white-noise process. For the Financial Institutions segment, by contrast, the Ljung-Box statistic is significant at the 5% level (p-value = 0.0129), indicating the presence of residual serial dependence.

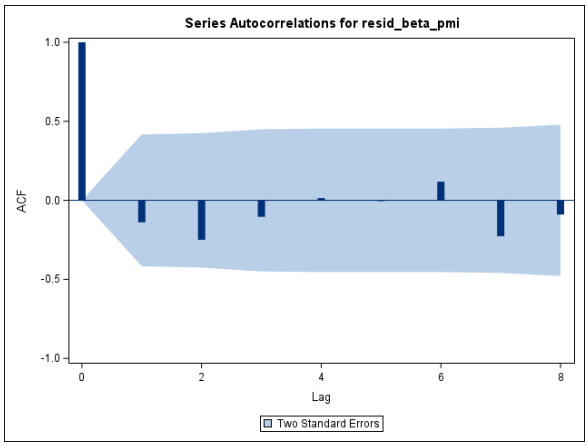
Table 2.26: Ljung-Box white-noise test statistics and p-values for the residuals of the final Beta-OLS regressions, by portfolio segment

Segment	Ljung-Box Statistic	p-value
Retail	8.30	0.2170
SME	3.00	0.8087
Corporate	5.87	0.4384
Financial Institutions	16.16	0.0129

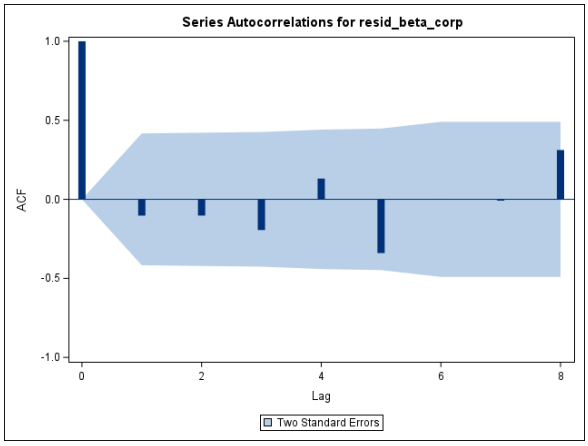
This result is consistent with the evidence discussed in Section 2.3.1, where the Financial Institutions series have shown to be the most volatile among the four portfolio segments and to exhibit a marked level shift from 2025 onwards (as discussed in Section 2.3.1); the presence of residual autocorrelation for this segment suggests that the four macroeconomic regressors included in the final specification may not fully capture the more complex dynamics of this series.



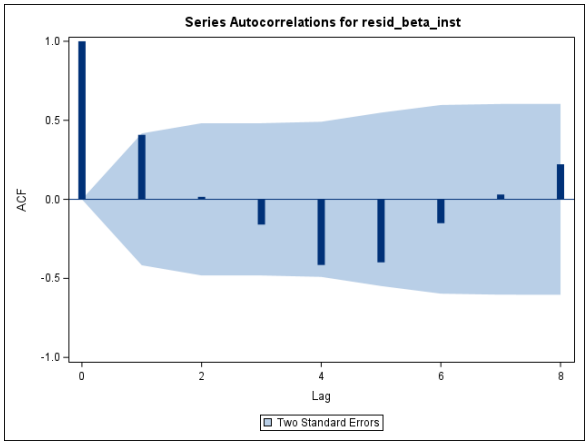
(a) Retail



(b) SME



(c) Corporate



(d) Institutions

Figure 2.23: Autocorrelation function (ACF) of the residuals from the final Beta-OLS regressions, by portfolio segment, with 95% confidence bounds

Figure 2.23 provides a graphical confirmation of these results, consistent with the Ljung-Box statistics reported above.

The overall picture that emerges from these diagnostics is encouraging for the Beta-OLS specifications proposed in this thesis. Normality and stationarity of the residuals hold across all four portfolio segments, lending statistical credibility to the estimated relationships, while residual independence is confirmed for three of the four segments, Retail, SME, and Corporate. The exception remains Financial Institutions, where the Ljung-Box test still points to some degree of serial dependence in the residuals, suggesting that the four macroeconomic regressors falling short of capturing the full temporal complexity of this particular series.

Then, the analysis proceeds to assess the accuracy of the predictions obtained on the original LGD scale. For this reason, once the parameters $\gamma_{0,j}, \ldots, \gamma_{4,j}$, which denote the regression coefficients associated with the relative regressors, are estimated by OLS in Equation (2.23), denoted by their estimated counterparts $\hat{\gamma}_{0,j}, \ldots, \hat{\gamma}_{4,j}$, the fitted value of the Beta-transformed variable is obtained, for each portfolio segment j and time $t \in \{2020Q3, \ldots, 2026Q1\}$, as:

$$\hat{Z}_{j,t} = \hat{\gamma}_{0,j} + \hat{\gamma}_{1,j}\Delta\text{Euribor}_{t-2} + \hat{\gamma}_{2,j}\Delta\text{Oil}_{t-2} + \hat{\gamma}_{3,j}\Delta\text{BTP}_{t-7} + \hat{\gamma}_{4,j}\,\text{Dummy}_{\text{Covid},t} \tag{2.24}$$

Then, the estimated values of the transformed variable $\hat{Z}_{j,t}$, obtained from Equation (2.24), are mapped back to the original $(0, 1)$ interval through the inverse of the Probability Integral Transform.

Since the Beta Transformation combines the standard normal quantile function with the Beta cumulative distribution function (Equations (2.21) and (2.22)), its inversion requires two successive steps, applied in reverse order. First, the fitted value of the transformed variable is converted into a cumulative probability through the cumulative distribution function of the standard normal distribution:

$$\hat{p}_{j,t} = \Phi\left(\hat{Z}_{j,t}\right), \tag{2.25}$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution, and $\hat{p}_{j,t}$ therefore lies in the interval $(0, 1)$ by construction.

Subsequently, the resulting probability is converted into the corresponding LGD prediction using the quantile function of the Beta distribution estimated for segment j:

$$\widehat{LGD}_{j,t} = F^{-1}_{\text{Beta}}\left(\hat{p}_{j,t}; \hat{\alpha}_j, \hat{\beta}_j\right), \tag{2.26}$$

where $F^{-1}_{\text{Beta}}(\cdot)$ denotes the quantile function of the Beta distribution, and $\hat{\alpha}_j$ and $\hat{\beta}_j$ are the parameters estimated through the method of moments for portfolio segment j (Table 2.20), the same parameters used to construct the Probability Integral Transform in Equation (2.21).

Based on the resulting prediction errors, defined as in Equation (2.14), the MAE, MSE, and RMSE summary measures of predictive accuracy, introduced in Equations (2.15), (2.16), and (2.17) for the Logit-OLS specification, are computed in the same way for the Beta-OLS specification.

The results are reported in Table 2.27.

Table 2.27: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) of the final Beta-OLS models, by portfolio segment, with RMSE also expressed as a percentage of the segment's historical mean LGD

Segment	MAE	MSE	RMSE	RMSE (% of mean LGD)
Retail	0.0085	0.000113	0.0106	3.9
SME	0.0038	0.000029	0.0053	1.4
Corporate	0.0031	0.000018	0.0042	1.1
Financial Institutions	0.0154	0.000395	0.0199	7.7

There is evidence of a high level of predictive accuracy of the Beta-OLS models across all portfolio segments considered. The Corporate portfolio exhibits the lowest RMSE value (0.0042), followed by SME and Retail with 0.0053 and 0.0106 respectively; the Financial Institutions segment again displays the largest prediction errors, with an RMSE of 0.0199, corresponding to approximately 7.7% of the segment's historical average LGD.

Taken together, the MAE and RMSE values are consistent with the good explanatory power highlighted by the multivariate regressions and confirm the ability of the Beta-OLS models to adequately reproduce the historical dynamics of LGD across the different portfolio segments, with the partial exception of Financial Institutions, whose comparatively larger prediction errors are consistent with its lower explanatory power and higher historical volatility already documented above.

The predictive performance of the Beta-OLS specification can also be assessed visually. Figures 2.24, 2.25, 2.26, and 2.27 plot the observed LGD values against those estimated by the model, on the original LGD scale, for each portfolio segment; the fitted series begins only in 2020Q3, since the construction of ΔBTP_{t-7} requires both first-differencing and a seven-quarter lag, leaving the first eight quarters of the original sample unavailable for estimation.

2.4.4 Comparison and Interpretation of the Results

Since the Logit-OLS and Beta-OLS models have been estimated using the same set of explanatory variables, the comparison focuses exclusively on the effect of the different transformations applied to the dependent variable.

The results obtained from the two different specifications reveal an extraordinary degree of convergence. Across all four portfolio segments, the estimated coefficients on the macroeconomic regressors display identical signs and the same pattern of statistical significance. The estimated intercepts, by contrast, differ in sign for the Retail and Financial Institutions segments. The only partial exception to this overall picture of convergence concerns the Financial Institutions segment, for which the gap between the two specifications, while remaining small in absolute terms, is the largest among the four segments considered, with a difference in Adjusted R^2 of approximately 0.3 percentage points (Tables 2.14 and 2.23). The predictive accuracy measures confirm this correspondence: MAE, MSE, and RMSE are nearly identical to the fourth decimal

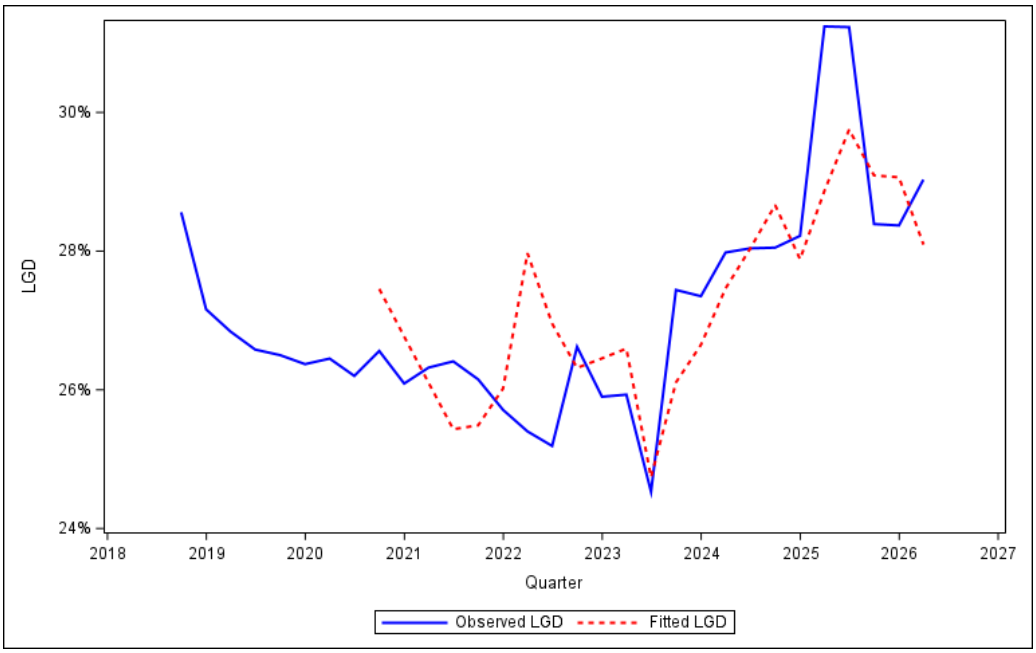


Figure 2.24: Observed and fitted LGD values for the Retail portfolio obtained from the Beta-OLS model

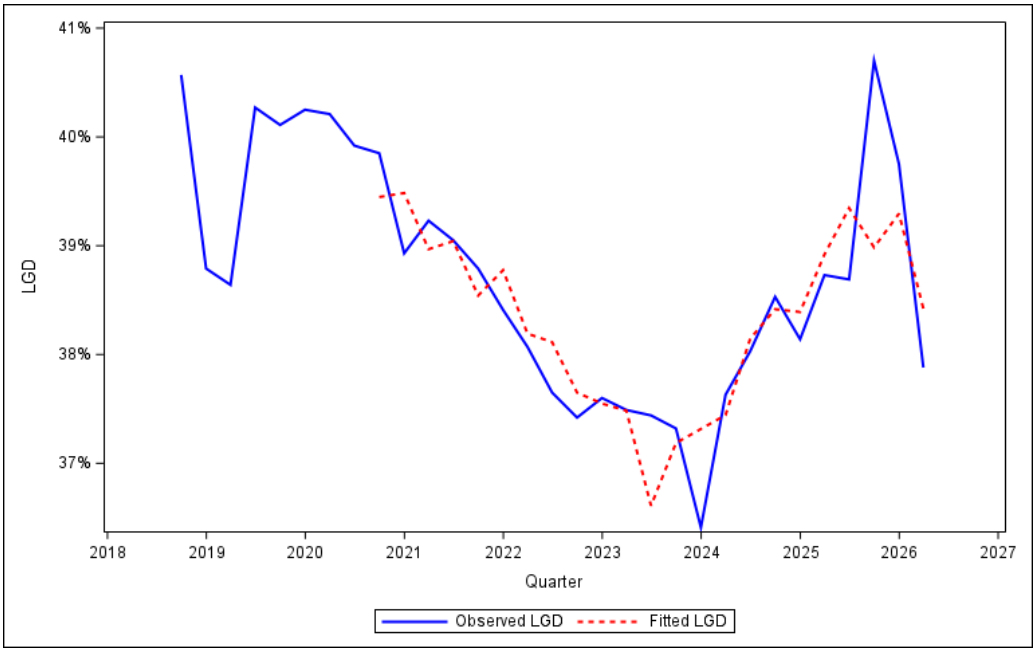


Figure 2.25: Observed and fitted LGD values for the SME portfolio obtained from the Beta-OLS model

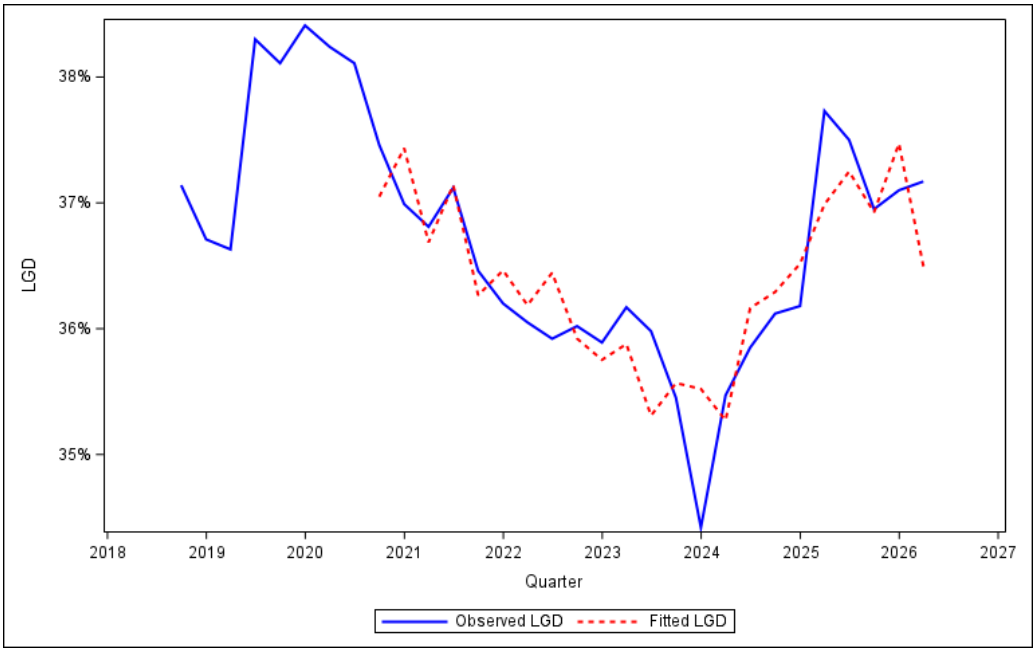


Figure 2.26: Observed and fitted LGD values for the Corporate portfolio obtained from the Beta-OLS model

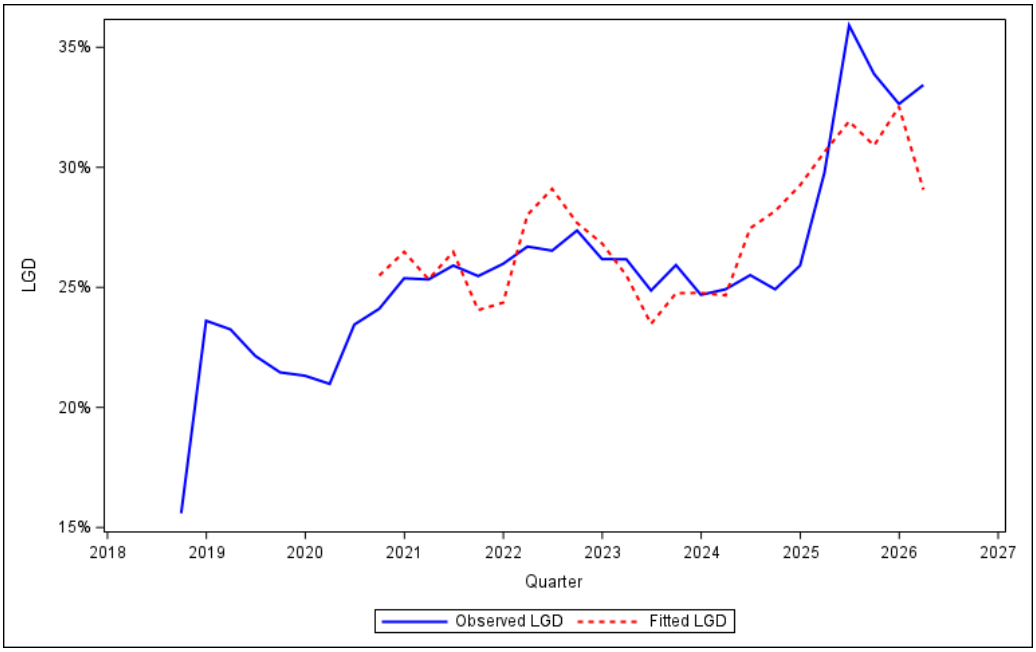


Figure 2.27: Observed and fitted LGD values for the Financial Institutions portfolio obtained from the Beta-OLS model

place in three of the four segments (Tables 2.19 and 2.27); they differ by no more than 0.0002 across all four segments, and the ranking of segments by predictive accuracy, from Corporate as the most accurate to Financial Institutions as the least, is identical under both specifications.
Therefore, the conclusion is: the relationship between LGD and the selected macroeconomic drivers is not an artefact of the specific transformation employed, but emerges in substantially identical form whether estimation is carried out on the logit or on the Beta scale.
The most plausible explanation for this general pattern lies in the behaviour of the two transformations across the support of the variable. Both the logit function and the cumulative distribution function of the Beta distribution are strongly non-linear near the boundaries of the $[0, 1]$ interval, while towards the centre of the interval they behave as approximately equivalent mappings of one another. Since the LGD series analysed in this thesis are exposure-weighted averages computed across a large number of counterparties, their values remain concentrated within a central region of the interval, between approximately 15% and 40%, as documented in Table 2.1, never approaching the boundaries where the two transformations would diverge most.
The slightly larger discrepancy observed for Financial Institutions agrees with this same logic. This segment has the highest historical volatility of the four, with a standard deviation of 4.07%, compared with 0.96%-1.54% for the others. It is also, together with Retail, one of only two segments where the Beta-transformed variable fails the normality test (Table 2.22). This suggests that its observations sit a little closer to the edges of the central region than those of the other three segments, which makes the choice of transformation slightly more important here, although it does not lead to any major difference in the results.
This explanation finds direct support in the study of Qi & Zhao (2011), who, in comparing several transformation-based LGD models, find that their performance is highly sensitive to the choice of ε, the small adjustment applied specifically to LGD observations equal to exactly 0 or 1 prior to transformation.
Also Crook & Bellotti (2012), applying the same two transformations to individual-account recovery rate data, report predictive results that differ appreciably between the two specifications. As discussed further in Chapter 3, their target variable displays a markedly bimodal empirical distribution, concentrated at the two extremes of the $[0, 1]$ interval, precisely the region where, consistently with the discussion above, the logit and Beta transformations diverge most.
Before interpreting the estimated coefficients, it is important to emphasise that both the logit transformation and the procedure based on the Beta Transformation are monotonic increasing transformations of the dependent variable. In the Logit-OLS model, the coefficients act on the logarithm of the odds ratio associated with LGD, whereas in the Beta-OLS model the effect is estimated on the transformed variable obtained through the *Probability Integral Transform*. Consequently, the sign of the estimated coefficients retains a direct qualitative interpretation on the original LGD scale: positive coefficients indicate a direct relationship between the explanatory variable and expected loss, whereas negative coefficients indicate an inverse relationship. By

contrast, the numerical magnitude of the coefficients cannot be directly interpreted in terms of changes in LGD on its original scale.

From an economic perspective, the most robust finding concerns the two-quarter lagged change in the three-month Euribor rate, which exhibits a negative and statistically significant coefficient across all portfolio segments considered, under both transformations. This significance appears consistent with the regulatory framework underlying LGD estimation (European Banking Authority (2017)), while the negative sign suggests the existence of an inverse relationship between interest-rate dynamics and LGD. From a theoretical perspective, this last finding may appear counterintuitive. According to the EBA Guidelines, future recoveries used in the calculation of economic loss for LGD estimation must be discounted using a rate based on the three-month Euribor plus a 5% add-on. Consequently, an increase in Euribor reduces the present value of future recoveries and, all else being equal, would be expected to generate a higher economic loss and therefore a higher LGD. The negative sign observed in the estimates therefore suggests that other macroeconomic channels dominate the mechanical effect associated with the discounting of recoveries: rising interest rates tend to coincide with an improving macro-financial environment, characterised by stronger economic growth, healthier labour market conditions, and higher collateral valuations, all of which make recovery procedures more effective and tend to increase recovery rates. The overall effect on LGD is therefore a reduction, even though the discounting mechanism alone would be expected to operate in the opposite direction.

Separately, it is worth noting that the estimation sample, spanning 2020Q3 to 2026Q1 (Section 2.4.2), coincides with a period in which Euribor dynamics have been shaped by two markedly distinct monetary policy regimes: an extended period of accommodative, near-zero interest rates through 2021, followed by a rapid tightening cycle initiated in 2022 in response to the inflationary pressures triggered by the outbreak of the Russia-Ukraine conflict and the associated energy price shock (AIFIRM (2025)). What is more, the two-quarter lag identified as optimal in Section 2.4.1 is broadly consistent with the time typically required for changes in policy rates to filter through to bank funding costs, loan pricing, and, ultimately, borrowers' debt-servicing capacity and collateral values, before feeding into observed recovery outcomes.

The change in oil prices also displays a negative relationship with LGD, in both specifications. The coefficient is statistically significant in the Retail, SME, and Corporate segments, while losing significance within the Financial Institutions portfolio, again consistently across both transformations. However, higher energy prices could be expected to negatively affect the profitability of firms operating in energy-intensive sectors, such as manufacturing, transportation, or other activities characterised by high production costs; this concern is not merely theoretical, as the sharp rise in energy prices triggered by the outbreak of the Russia-Ukraine conflict in early 2022 was explicitly identified as a source of heightened vulnerability for firms operating in energy-intensive sectors within the Italian banking system (AIFIRM (2025)). In such cases, one might reasonably expect a positive relationship between oil prices and LGD. However, the present analysis is based on aggregate data referring to the Italian banking system as a whole rather than

on sector-specific portfolios. Consequently, consistent with the interpretation proposed for the Euribor coefficient, oil prices may be interpreted as an indicator of global economic activity, since increases in energy demand are often associated with periods of economic expansion. Under such conditions, firms generally exhibit stronger financial positions and the value of assets used as collateral tends to remain more stable, supporting higher recovery rates and, consequently, lower LGD levels. At the aggregate, system-wide level of analysis adopted in this thesis, this demand-side channel appears therefore to dominate the sector-specific cost channel that a more granular, sector-disaggregated analysis might otherwise be expected to capture.

A different result emerges for changes in Italian government bond yields. From an economic perspective, increases in sovereign yields are generally associated with higher sovereign risk and tighter financial conditions. Under such circumstances, one might expect worsening recovery prospects and therefore higher LGD levels. The positive coefficients estimated for the Retail and SME segments are broadly consistent with this interpretation, again in both the Logit-OLS and Beta-OLS specifications; this channel is also consistent with the observation that the same tightening in financial conditions associated with rising yields has been linked, in the Italian context, to a more challenging environment for debt refinancing and a consequent softening of real estate values, particularly in the commercial segment, a development directly relevant to the collateral values underlying Retail and SME recovery processes (AIFIRM (2025)). However, the opposite sign observed for the Corporate and Financial Institutions portfolios, combined with the lack of statistical significance across all specifications, suggests that the relationship is not sufficiently stable to support a clear economic interpretation. Nevertheless, as previously discussed and shown in Table 2.15, the inclusion of this variable in the final specification improves the overall explanatory power of the model without introducing multicollinearity issues, a result that likewise holds under both transformations. This suggests that sovereign risk may contain additional information regarding the broader macro-financial environment, even though its direct impact on LGD dynamics does not appear sufficiently stable or robust.

A further interesting finding emerges from the analysis of the dummy variable associated with the pandemic period. The results reveal heterogeneous effects across the different portfolio segments, with the same pattern of signs and significance confirmed by both the Logit-OLS and Beta-OLS specifications. Within the SME segment, the coefficient is positive and statistically significant, indicating an increase in LGD during the pandemic period. This result is consistent with the greater vulnerability of small and medium-sized enterprises to economic shocks. The restrictions introduced to contain the spread of Covid-19 significantly affected numerous productive activities, leading to revenue losses, liquidity pressures, and a deterioration in the recovery of defaulted exposures. Evidence from the ECB (Fiordelisi et al. (2022)) suggests that SMEs have been among the firms most severely affected by the pandemic, particularly in sectors directly exposed to containment measures and mobility restrictions, where sharp declines in revenues translated into elevated liquidity needs and increased financial fragility.

By contrast, in the Retail and Financial Institutions portfolios, the coefficient associated with the Covid Dummy is negative and statistically significant. This suggests that observed LGD levels were, on average, lower during the pandemic period. One likely explanation is the extensive support measures introduced by public authorities and central banks. In Italy, these measures were set out in the Decreto Legge *"Cura Italia"* of 17 March 2020 (Governo Italiano (2020)), which introduced legislative payment moratoria under Article 56, later joined by further moratoria from the banking sector. Crucially, the EBA clarified in a statement of 25 March 2020 that using these moratoria should not automatically count as a sign of increased credit risk under IFRS 9 (European Banking Authority (2020)). Without this clarification, the widespread use of moratoria could have pushed many exposures into worse risk categories, simply as a mechanical effect. Loan moratoria, public guarantees, fiscal support, and extraordinary monetary policy measures helped borrowers keep up with their payments and softened the impact of the downturn. These measures likely helped limit borrower distress and improve recovery outcomes, reducing the severity of losses observed during the pandemic (Fiordelisi et al. (2022)).
Finally, in the Corporate segment, the Covid Dummy is not statistically significant, again in both specifications. This result may indicate that larger firms have been better able to absorb the effects of the crisis due to stronger financial structures, greater diversification of funding sources and easier access to capital markets. In this respect, evidence from the ECB suggests that large corporates have been generally more capable of accessing alternative forms of financing and maintaining investment activity despite the adverse shock, making them relatively less dependent on bank credit and therefore more resilient than smaller firms during the pandemic (Fiordelisi et al. (2022)).
Overall, the results demonstrate that LGD does not depend solely on the characteristics of individual exposures, but is also influenced by the evolution of the broader macroeconomic environment, and that the specific channels through which this influence operates, for example discounting, real-economy activity, sovereign and financial conditions, and extraordinary policy interventions, can be traced back to identifiable and well-documented episodes in the recent history of the Italian and European macro-financial environment.

2.5 Scenario-Based Forecasting of Loss Given Default

Once the satellite models have been estimated, they are used to generate forward-looking LGD projections, consistently with the IFRS 9 approach, according to which the assessment of expected credit losses must incorporate forward-looking information and future macroeconomic scenarios. The objective is to assess how different economic conditions may affect the evolution of Loss Given Default across the various portfolio segments. As already discussed in Section 1.1.3, the accounting standard does not prescribe a specific number of scenarios, their associated probabilities or the underlying methodology (Canton et al. (2021)). Consistently with the evidence reported by Canton et al. (2021), according to which the joint use of three macroeconomic

scenarios, baseline, adverse, and positive, represents the methodology most commonly adopted by Italian banks in practice, the present analysis proceeds at the same way.
To this end, LGD dynamics are projected over the 2026–2028 period under the following alternative macroeconomic scenarios:

- **Baseline Scenario**: represents the most likely economic outlook based on the information available at the time of the forecast and serves as the central scenario of the analysis;
- **Positive Scenario**: describes a more favourable economic environment relative to the baseline scenario, characterised by macroeconomic conditions that support a reduction in credit risk and an improvement in the recovery of defaulted exposures;
- **Adverse Scenario**: incorporates the occurrence of unfavourable macroeconomic shocks and, more generally, stressed conditions of deterioration in the economic and financial environment.

For each scenario, the corresponding projected values of the macroeconomic drivers selected during the estimation stage are obtained; these trajectories are subsequently incorporated into the final regression models in order to generate the respective forward-looking LGD projections.
It is worth noting that all data cut-off dates referenced in this section reflect the state of data availability at the time of writing and estimation of the present thesis.
In particular, the historical LGD series used in the estimation phase, available through the ECB Data Portal for IRB exposures of Significant Institutions (European Central Bank (2026a,b,c,d)), are available up to the first quarter of 2026. The macroeconomic variables considered, by contrast, are available up to the second quarter of 2026 from the respective statistical sources employed in this study, namely Eurostat for GDP (Eurostat (2026)), FRED for Brent oil prices (Federal Reserve Bank of St. Louis (2026)), and the Bank of Italy for the ten-year BTP yield (Banca d'Italia (2026d)). Consequently, the observed values of the macroeconomic drivers for the second quarter of 2026 are retained and directly incorporated into the forecasting exercise. As a result, the first projected observation of LGD still benefits from actually observed macroeconomic information.
The most recent projections for the Italian economy published by the Bank of Italy in June 2026 for the 2026–2028 period (Banca d'Italia (2026b)), based on market quotations available as of 21 May 2026, provide the baseline scenario assumptions required for the three-month Euribor rate, the yield on ten-year Italian government bonds (BTPs), and Brent crude oil prices. More specifically, under this first scenario, as outlined in Banca d'Italia (2026b), the Bank of Italy's projections incorporate a macroeconomic environment characterised by moderate economic growth, a gradual easing of inflationary pressures, and a progressive normalisation of energy prices following the tensions generated by the conflict in the Middle East.
Since the estimates published by the Bank of Italy are available only on an annual basis and provide a single expected value for each macroeconomic variable in 2026, 2027, and 2028, these

values are converted into quarterly series through linear interpolation between consecutive annual observations (described in detail in Equations 2.27 and 2.28–2.33). This procedure makes it possible to obtain a series consistent with the quarterly frequency adopted during the model estimation stage and to generate forecasts for the period from the third quarter of 2026 to the fourth quarter of 2028, preserving continuity between observed data and projected values.
In particular, the third quarter of 2026 represents a special case, as it constitutes the transition point between the historical observed component and the forward-looking component.
Denoting by $X_{Q2,2026}$ the last available observed value and by $X_{Q4,2026}$ the corresponding Bank of Italy projection for the end of 2026 – for each of the three macroeconomic variables involved in the projection exercise, namely the Euribor rate, the BTP yield, and Brent oil prices – the value for the third quarter is obtained by evenly distributing the change between the two endpoints:

$$X_{Q3,2026} = X_{Q2,2026} + \frac{1}{2}\left(X_{Q4,2026} - X_{Q2,2026}\right) \tag{2.27}$$

For the following years, the annual projections published by the Bank of Italy are assumed to represent the values of the macroeconomic variables in the fourth quarter of each year, while the observations for the first three quarters of 2027 and 2028 are obtained through:

$$X_{Q1,2027} = X_{Q4,2026} + \frac{1}{4}\left(X_{Q4,2027} - X_{Q4,2026}\right) \tag{2.28}$$

$$X_{Q2,2027} = X_{Q4,2026} + \frac{2}{4}\left(X_{Q4,2027} - X_{Q4,2026}\right) \tag{2.29}$$

$$X_{Q3,2027} = X_{Q4,2026} + \frac{3}{4}\left(X_{Q4,2027} - X_{Q4,2026}\right) \tag{2.30}$$

$$X_{Q1,2028} = X_{Q4,2027} + \frac{1}{4}\left(X_{Q4,2028} - X_{Q4,2027}\right) \tag{2.31}$$

$$X_{Q2,2028} = X_{Q4,2027} + \frac{2}{4}\left(X_{Q4,2028} - X_{Q4,2027}\right) \tag{2.32}$$

$$X_{Q3,2028} = X_{Q4,2027} + \frac{3}{4}\left(X_{Q4,2028} - X_{Q4,2027}\right) \tag{2.33}$$

The fourth-quarter observations for both years coincide directly with the annual projections published by the Bank of Italy, namely $X_{Q4,2027}$ and $X_{Q4,2028}$. Therefore, only the first three quarters are interpolated, while the year-end values remain unchanged. This procedure ensures complete consistency with the original annual projections while generating smooth quarterly trajectories of the three macroeconomic regressors required for the forecasting exercise.
The resulting quarterly trajectories are reported in Table 2.28.
For the construction of the adverse scenario, the alternative scenarios published directly by the Bank of Italy have not been used. Although the June 2026 projections include both an adverse and a severe scenario associated with a worsening of the conflict in the Middle East, the Bank of Italy explicitly presents these exercises as alternative scenarios aimed at assessing the sensitivity

Table 2.28: Quarterly trajectories of the three-month Euribor rate, the yield on ten-year Italian government bonds (BTPs), and Brent crude oil prices under the Baseline scenario, based on Bank of Italy projections (June 2026)

Date	10Y BTP (%)	Oil Price (USD/barile)	3M Euribor (%)
30/06/2026	3.797	97.048	2.250
30/09/2026	3.799	96.974	2.325
31/12/2026	3.800	96.900	2.400
31/03/2027	3.875	93.225	2.500
30/06/2027	3.950	89.550	2.600
30/09/2027	4.025	85.875	2.700
31/12/2027	4.100	82.200	2.800
31/03/2028	4.150	80.925	2.775
30/06/2028	4.200	79.650	2.750
30/09/2028	4.250	78.375	2.725
31/12/2028	4.300	77.100	2.700

of the Italian economy to specific energy and geopolitical shocks characterised by a high degree of uncertainty. More specifically, the severe scenario assumes an increase in oil prices of more than 60% relative to the baseline scenario, accompanied by a substantial increase in uncertainty and a marked weakening of international trade. Similarly it happens for the adverse scenario (Banca d'Italia (2026b)).

For this reason, in order to construct a stress scenario more consistent with the regulatory exercises commonly used in the banking sector, the shocks published as part of the 2025 stress-testing exercise conducted by the European Banking Authority (EBA) and the European Systemic Risk Board (ESRB) have been adopted as a reference (European Systemic Risk Board (2025)). In particular, the EBA shocks have been applied to the most recent baseline projections published by the Bank of Italy, thus maintaining an up-to-date macroeconomic reference framework for the Italian economy while ensuring a scenario structure consistent with the stress-testing practices adopted by supervisory authorities. Since the EBA projections refer to a forecasting horizon preceding that considered in the present analysis, the shocks embedded in the adverse scenario are shifted forward by one year. This approach reflects a common practice in stress-testing and scenario-design exercises, where the focus is placed on the magnitude of the deviation between the baseline and adverse scenarios rather than on the absolute level of the macroeconomic variables. For the yield on ten-year Italian government bonds (BTPs), the adverse scenario has obtained by applying to the Bank of Italy baseline projections the difference observed between the adverse and baseline scenarios in the EBA projections, as follows:

$$BTP^{adverse}_{Q4,2026} = BTP^{BankIT,baseline}_{Q4,2026} + \left(BTP^{EBA,adverse}_{2025} - BTP^{EBA,baseline}_{2025}\right) \tag{2.34}$$

$$BTP^{adverse}_{Q4,2027} = BTP^{BankIT,baseline}_{Q4,2027} + \left(BTP^{EBA,adverse}_{2026} - BTP^{EBA,baseline}_{2026}\right) \tag{2.35}$$

$$BTP^{adverse}_{Q4,2028} = BTP^{BankIT,baseline}_{Q4,2028} + \left(BTP^{EBA,adverse}_{2027} - BTP^{EBA,baseline}_{2027}\right) \tag{2.36}$$

where the term in parentheses represents the adverse shock, expressed in percentage points, derived from the EBA projections and shifted forward by one year.
For Brent crude oil prices, given the availability of percentage deviations between the adverse and baseline scenarios in the EBA projections, the following specification has been adopted:

$$Brent^{adverse}_{Q4,2026} = Brent^{BankIT,baseline}_{Q4,2026}\left(1 + \frac{\Delta Brent^{EBA}_{2025}}{100}\right) \tag{2.37}$$

$$Brent^{adverse}_{Q4,2027} = Brent^{BankIT,baseline}_{Q4,2027}\left(1 + \frac{\Delta Brent^{EBA}_{2026}}{100}\right) \tag{2.38}$$

$$Brent^{adverse}_{Q4,2028} = Brent^{BankIT,baseline}_{Q4,2028}\left(1 + \frac{\Delta Brent^{EBA}_{2027}}{100}\right) \tag{2.39}$$

Therefore, the adverse trajectory is obtained by applying the percentage deviation observed in the EBA stress scenario to the corresponding Bank of Italy baseline projection.
For the three-month Euribor rate, the trajectory projected under the baseline scenario has been left unchanged. This choice is consistent with the assumptions underlying the coordinated Eurosystem exercise, in which alternative scenarios maintain the same path for short-term interest rates as in the baseline scenario, assuming the absence of additional monetary policy interventions (Banca d'Italia (2026b)). Once the corresponding annual adverse values have been obtained through Equations 2.34–2.36 and 2.37–2.39, they are converted into quarterly series using the same linear interpolation procedure adopted for the baseline scenario, and applied from 2026Q3 onwards, since the second quarter of 2026 still corresponds to observed macroeconomic data and is therefore left unchanged across all three scenarios. The resulting quarterly trajectories are reported in Table 2.29.

Table 2.29: Quarterly trajectories of the three-month Euribor rate, the yield on ten-year Italian government bonds (BTPs), and Brent crude oil prices under the Adverse scenario

Date	10Y BTP (%)	Oil Price (USD/barrel)	3M Euribor (%)
30/06/2026	3.797	97.048	2.250
30/09/2026	5.089	124.633	2.325
31/12/2026	6.380	152.217	2.400
31/03/2027	6.311	139.346	2.500
30/06/2027	6.242	126.475	2.600
30/09/2027	6.173	113.604	2.700
31/12/2027	6.105	100.732	2.800
31/03/2028	6.043	104.136	2.775
30/06/2028	5.981	107.540	2.750
30/09/2028	5.919	110.945	2.725
31/12/2028	5.858	114.341	2.700

Finally, the *Positive* scenario is obtained through a symmetric transformation of the Adverse scenario around the Baseline trajectory. As discussed above, no additional shock is applied to the

three-month Euribor rate under the Adverse scenario, so its Positive-scenario trajectory is left unchanged and simply coincides with the Baseline. For the yield on ten-year Italian government bonds (BTPs) and Brent crude oil prices, by contrast, the Positive scenario is constructed as:

$$BTP_t^{positive} = BTP_t^{baseline} + 0.75\left(BTP_t^{baseline} - BTP_t^{adverse}\right), \tag{2.40}$$

$$Brent_t^{positive} = Brent_t^{baseline} + 0.75\left(Brent_t^{baseline} - Brent_t^{adverse}\right), \tag{2.41}$$

for each quarter t from 2026Q3 to 2028Q4, the horizon over which the three scenarios begin to differ, since the second quarter of 2026 is observed and therefore left identical across all three scenarios.

The idea behind Equations (2.40) and (2.41) is simple: the term in parentheses captures the size and direction of the adverse shock relative to the baseline, for each variable and horizon, since it comes directly from the regulatory stress-testing exercise applied in the construction of the Adverse scenario (European Systemic Risk Board (2025)). Mirroring this same deviation on the opposite side of the baseline, rather than building a new favourable scenario from scratch, keeps the Positive scenario tied to the same shock structure as the Adverse one, applied in the opposite direction. A scaling factor of 0.75, rather than 1, is used to avoid overly extreme assumptions for the favourable case.

Applying this procedure yields the quarterly trajectories of the Positive scenario reported in Table 2.30. Since the scenario is constructed as a reflection of the Adverse scenario around the Baseline path, both the yield on ten-year Italian government bonds (BTPs) and Brent crude oil prices, which are subject to a negative shock under the Adverse scenario, take correspondingly more favourable values relative to the Baseline from 2026Q3 onwards.

Table 2.30: Quarterly trajectories of the three-month Euribor rate, the yield on ten-year Italian government bonds (BTPs), and Brent crude oil prices under the Positive scenario, obtained by mirroring the Adverse scenario deviations around the Baseline path (Equation (2.40) and Equation (2.41)), scaled by a factor of 0.75

Date	10Y BTP (%)	Oil Price (USD/barrel)	3M Euribor (%)
30/06/2026	3.797	97.048	2.250
30/09/2026	2.831	76.230	2.325
31/12/2026	1.865	55.412	2.400
31/03/2027	2.048	58.634	2.500
30/06/2027	2.231	61.856	2.600
30/09/2027	2.414	65.079	2.700
31/12/2027	2.597	68.301	2.800
31/03/2028	2.730	63.516	2.775
30/06/2028	2.864	58.732	2.750
30/09/2028	2.998	53.948	2.725
31/12/2028	3.132	49.163	2.700

2.5.1 Forward-Looking LGD Projections under the Logit-OLS Specification

Once the macroeconomic scenarios have been defined and the corresponding quarterly trajectories of the selected regressors have been obtained, these are used to generate forward-looking projections of Loss Given Default for each portfolio segment over the horizon running from the second quarter of 2026 to the fourth quarter of 2028, consistently with the availability of the macroeconomic scenario trajectories described previously.

Since the comparison between the Logit-OLS and Beta-OLS models in Section 2.4.4 showed the two specifications to be nearly indistinguishable in terms of in-sample fit, coefficient signs, and predictive accuracy, the forward-looking exercise presented in this section is developed using only the Logit-OLS models. This choice is made purely for the sake of exposition: extending the projection exercise to both transformations would replicate every subsequent step, from the construction of scenario-specific trajectories to the calculation of the probability-weighted measure, without adding further methodological insight beyond what has already been established through the in-sample comparison.

The forecasting phase has been therefore carried out in two steps.

First, for each quarter t between 2026Q2 and 2028Q4 and each macroeconomic scenario $s \in \{\text{Baseline}, \text{Adverse}, \text{Positive}\}$, the predicted value of the transformed variable has been calculated, for each portfolio segment j, according to the following relationship:

$$\widehat{LGD}_{j,t}^{\,logit,(s)} = \hat{\beta}_{0,j} + \hat{\beta}_{1,j}\Delta\text{Euribor}_{t-2}^{(s)} + \hat{\beta}_{2,j}\Delta\text{Oil}_{t-2}^{(s)} + \hat{\beta}_{3,j}\Delta\text{BTP}_{t-7}^{(s)} + \hat{\beta}_{4,j}\,\text{Dummy}_{\text{Covid},t}, \quad (2.42)$$

where $\hat{\beta}_{0,j}$ denotes the estimated intercept and $\hat{\beta}_{1,j}, \ldots, \hat{\beta}_{4,j}$ the coefficients estimated in Section 2.4.2 for portfolio segment j (Table 2.14).

It is important to note that, although the Covid Dummy is included in the estimated model specification, it does not contribute to the forward-looking projections. In fact, it is defined as equal to one only during the pandemic period and zero in all other quarters so, in this case, its contribution to the estimated LGD is null.

Subsequently, the resulting values are transformed back to the original LGD scale through the inverse logistic function introduced in Equation (2.13):

$$\widehat{LGD}_{j,t}^{(s)} = \frac{\exp\left(\widehat{LGD}_{j,t}^{\,logit,(s)}\right)}{1 + \exp\left(\widehat{LGD}_{j,t}^{\,logit,(s)}\right)}. \quad (2.43)$$

Tables 2.31, 2.33, 2.32, and 2.34 report the projected LGD trajectories under the three macroeconomic scenarios considered.

Figures 2.28, 2.29, 2.30, and 2.31 compare the historical evolution of LGD with the forward-looking trajectories obtained under the three macroeconomic scenarios considered for each portfolio segment. It can be observed that the historical portion of these series coincides with the

Table 2.31: Projected LGD trajectories for the Retail portfolio under the Baseline, Adverse, and Positive macroeconomic scenarios, from 2026Q2 to 2028Q4

Date	Baseline	Adverse	Positive
2026Q2	26.92%	26.92%	26.92%
2026Q3	27.75%	27.75%	27.75%
2026Q4	26.14%	26.14%	26.14%
2027Q1	27.56%	25.94%	28.82%
2027Q2	27.61%	25.99%	28.87%
2027Q3	27.81%	28.37%	27.40%
2027Q4	27.76%	28.32%	27.35%
2028Q1	28.00%	28.56%	27.58%
2028Q2	27.76%	29.52%	26.49%
2028Q3	27.96%	28.71%	27.30%
2028Q4	28.03%	27.62%	28.34%

Table 2.32: Projected LGD trajectories for the Corporate portfolio under the Baseline, Adverse, and Positive macroeconomic scenarios, from 2026Q2 to 2028Q4

Date	Baseline	Adverse	Positive
2026Q2	35.93%	35.93%	35.93%
2026Q3	36.48%	36.48%	36.48%
2026Q4	35.55%	35.55%	35.55%
2027Q1	36.39%	35.43%	37.12%
2027Q2	36.37%	35.41%	37.09%
2027Q3	36.45%	36.77%	36.21%
2027Q4	36.46%	36.79%	36.22%
2028Q1	36.38%	36.70%	36.14%
2028Q2	36.46%	36.37%	36.53%
2028Q3	36.53%	35.95%	36.96%
2028Q4	36.51%	36.39%	36.59%

Table 2.33: Projected LGD trajectories for the SME portfolio under the Baseline, Adverse, and Positive macroeconomic scenarios, from 2026Q2 to 2028Q4

Date	Baseline	Adverse	Positive
2026Q2	37.80%	37.80%	37.80%
2026Q3	38.31%	38.31%	38.31%
2026Q4	37.35%	37.35%	37.35%
2027Q1	38.20%	37.27%	38.90%
2027Q2	38.20%	37.27%	38.90%
2027Q3	38.28%	38.60%	38.05%
2027Q4	38.28%	38.59%	38.05%
2028Q1	38.29%	38.60%	38.05%
2028Q2	38.28%	38.62%	38.03%
2028Q3	38.40%	38.26%	38.50%
2028Q4	38.40%	38.24%	38.52%

Table 2.34: Projected LGD trajectories for the Financial Institutions portfolio under the Baseline, Adverse, and Positive macroeconomic scenarios, from 2026Q2 to 2028Q4

Date	Baseline	Adverse	Positive
2026Q2	27.58%	27.58%	27.58%
2026Q3	28.84%	28.84%	28.84%
2026Q4	26.32%	26.32%	26.32%
2027Q1	28.46%	26.35%	30.11%
2027Q2	28.39%	26.28%	30.03%
2027Q3	28.48%	29.21%	27.95%
2027Q4	28.55%	29.27%	28.01%
2028Q1	28.24%	28.96%	27.70%
2028Q2	28.55%	27.72%	29.17%
2028Q3	28.98%	27.09%	30.46%
2028Q4	28.89%	28.70%	29.04%

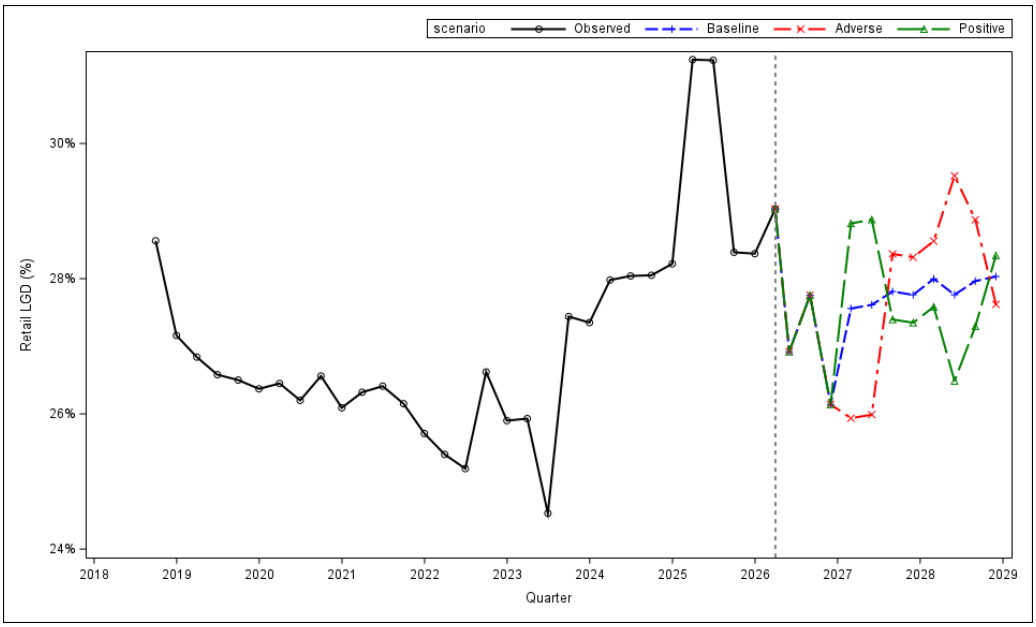


Figure 2.28: Historical and projected LGD under alternative macroeconomic scenarios for the Retail portfolio.

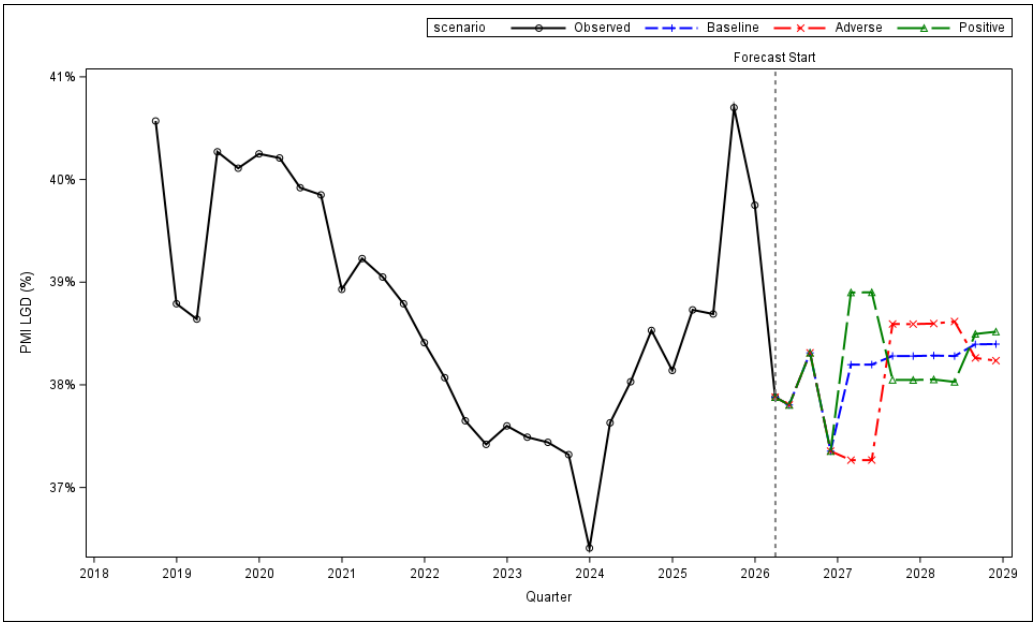


Figure 2.29: Historical and projected LGD under alternative macroeconomic scenarios for the SME portfolio.

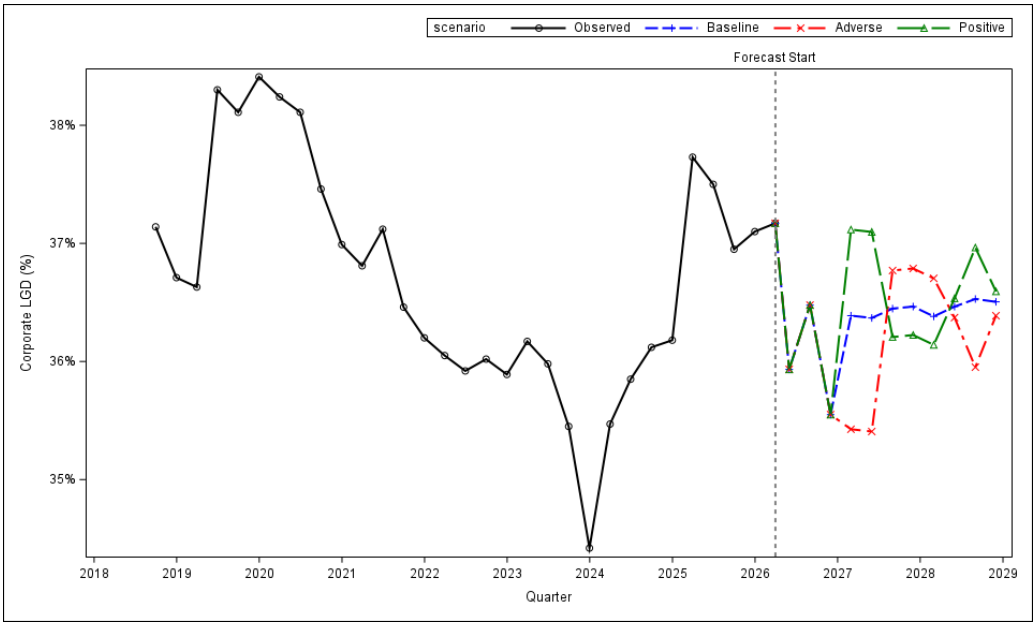


Figure 2.30: Historical and projected LGD under alternative macroeconomic scenarios for the Corporate portfolio.

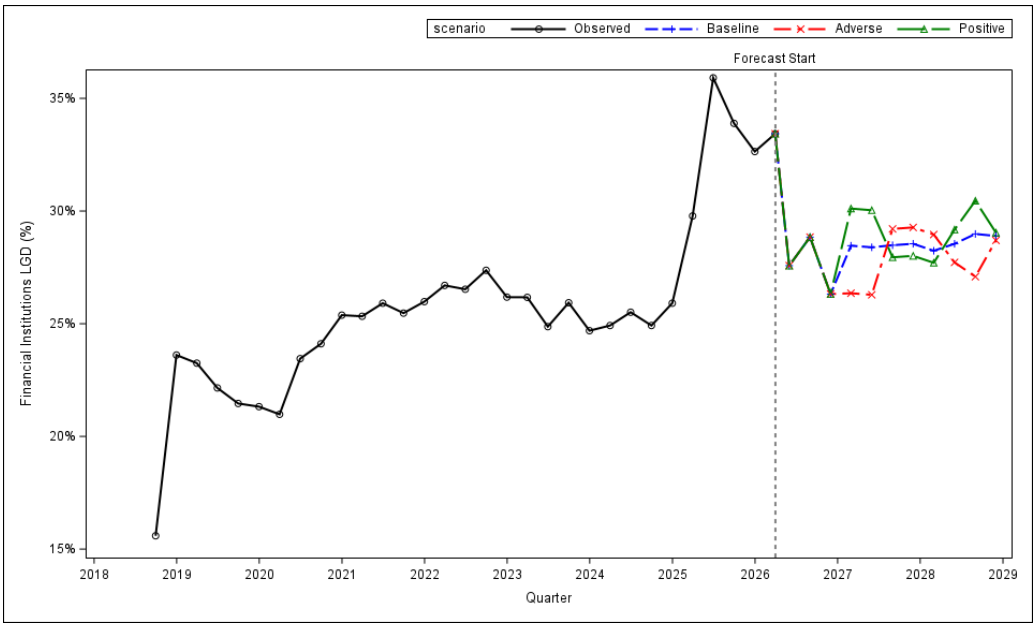


Figure 2.31: Historical and projected LGD under alternative macroeconomic scenarios for the Financial Institutions portfolio.

series documented in Section 2.3.1 (Figures 2.1, 2.2, 2.3, and 2.4), up until the first quarter of 2026, the last date for which observed values are available from the ECB Data Portal.

The analysis of the projections shows that the dynamics of Loss Given Default remain overall stable across all portfolio segments throughout the entire 2026–2028 forecasting horizon. The projected values remain close to the historical averages observed during the model calibration period, suggesting that the macroeconomic shocks considered have a relatively limited impact on the recovery component of defaulted exposures. Consistent with the lag structure embedded in the estimated specifications — changes in Euribor and oil prices lagged by two quarters, changes in the ten-year BTP yield lagged by seven quarters (Section 2.4.2) — the projected trajectories under the different scenarios initially coincide and begin to diverge only from 2027 onwards, when the macroeconomic shocks incorporated into the scenarios are progressively transmitted to the LGD estimates.

In terms of level, the SME portfolio is characterised by the highest LGD levels, with values consistently close to 38% (Table 2.33), followed by the Corporate segment, whose LGD remains around 36% (Table 2.32). The Retail and Financial Institutions portfolios exhibit lower values, generally ranging between 26% and 31% (Tables 2.31 and 2.34).

Before examining the magnitude of scenario dispersion in detail, it is worth noting that the relative ranking of the three scenarios does not remain constant across all quarters. Consider, for example, Table 2.31 (Retail portfolio): in the third quarter of 2027 the Adverse scenario does produce the highest LGD (28.37%, against 27.81% under Baseline and 27.40% under Positive), consistent with what would be expected from an adverse scenario. In the fourth quarter of 2028, however, the ranking reverses: the Adverse LGD (27.62%) is the lowest of the three, while the Positive LGD (28.34%) is the highest. This behaviour follows directly from the econometric specification

adopted, in which LGD depends on the *changes* in macroeconomic variables relative to previous quarters (lagged by two or seven quarters, Section 2.4.2), rather than on their absolute levels. The severity of a scenario, therefore, does not automatically translate into a constant ordering of projected LGD across quarters: what matters in each period is the sign and magnitude of the lagged change in the regressors, not their absolute level at that point in time.

Differences across scenarios, while generally limited relative to the average level of LGD, reach economically meaningful magnitudes in some cases and are not evenly distributed across segments. The largest deviations are observed in the Retail and Financial Institutions portfolios: in the Retail segment, during the second quarter of 2028 the projected LGD ranges from 26.49% under the Positive scenario to 29.52% under the Adverse scenario (Table 2.31), a difference exceeding three percentage points, with the Adverse scenario correctly yielding the higher value. For the Financial Institutions portfolio, the gap between the two extreme scenarios similarly exceeds three percentage points in the third quarter of 2028 (Table 2.34), although in this particular quarter it is the Positive scenario that yields the higher LGD (30.46%) relative to the Adverse scenario (27.09%), a further illustration of the scenario-ranking reversal just discussed. By contrast, the differences observed in the Corporate and SME segments are more contained (Tables 2.32 and 2.33).

This differing degree of scenario dispersion is consistent with the historical statistical properties of the four LGD series, already documented in Table 2.1. The Financial Institutions series exhibits by far the highest historical standard deviation among the four segments (4.07%, against 0.96% for Corporate, 1.13% for SME and 1.54% for Retail), largely attributable to the marked level shift recorded from 2025 onwards, when LGD moves from a historical range of 24–27% to values exceeding 35% (Section 2.3.1). The Retail series likewise displays greater variability than the Corporate and SME segments, and for both Retail and Financial Institutions the Shapiro-Wilk and Kolmogorov-Smirnov normality tests reject the null hypothesis of normality (Table 2.2), unlike Corporate and SME, whose historical distributions are instead consistent with normality. A higher historical variability and asymmetry of the target series is reflected, at the projection stage, in a wider dispersion of the trajectories across the three macroeconomic scenarios — a pattern broadly consistent with the segment-level heterogeneity in macro-sensitivity already documented for LGD by Crook & Bellotti (2012).

It should nonetheless be stressed that scenario dispersion, however relevant in some cases, remains contained relative to the historical average levels of LGD. This reversion towards historical levels, consistent with the structural nature of LGD already discussed in Table 2.1, is examined more formally below through the construction of a probability-weighted LGD measure.

In conclusion, in fact, consistently with the approach envisaged by IFRS 9, a synthetic measure of prospective LGD is also constructed as a probability-weighted average of the trajectories associated with the different macroeconomic scenarios. The accounting standard does not impose specific occurrence probabilities to be attributed to individual scenarios. As already discussed in Section 1.1.2, Canton et al. (2021) show that the methodology most commonly adopted by banks

relies on assigning probabilities of 58% to the baseline scenario and 21% to each of the adverse and positive scenarios, a pattern distinct from that observed among other financial intermediaries, for which the authors report an average weighting of 70% for the baseline scenario and 19% and 11% for the adverse and positive scenarios, respectively.

The same 58/21/21 weighting scheme is adopted in this thesis. For each portfolio segment j and quarter t, the resulting probability-weighted LGD measure is obtained as a weighted average of the scenario-specific fitted values from Equation (2.43):

$$\widehat{LGD}_{j,t}^{Weighted} = 0.58\,\widehat{LGD}_{j,t}^{Baseline} + 0.21\,\widehat{LGD}_{j,t}^{Adverse} + 0.21\,\widehat{LGD}_{j,t}^{Positive}. \tag{2.44}$$

The complete quarterly series are reported in Table 2.35. As expected by construction, the weighted LGD lies systematically between the values obtained under the three scenarios, and the segment ranking already observed in the scenario-specific projections is confirmed unchanged in the synthetic measure as well.

Table 2.35: Probability-weighted LGD projections

Date	Retail	SME	Corporate	Institutions
2026Q2	26.92%	37.80%	35.93%	27.58%
2026Q3	27.75%	38.31%	36.48%	28.84%
2026Q4	26.14%	37.35%	35.55%	26.32%
2027Q1	27.48%	38.15%	36.34%	28.37%
2027Q2	27.54%	38.15%	36.32%	28.29%
2027Q3	27.84%	38.30%	36.46%	28.52%
2027Q4	27.79%	38.30%	36.48%	28.59%
2028Q1	28.03%	38.30%	36.40%	28.28%
2028Q2	27.86%	38.29%	36.46%	28.50%
2028Q3	28.01%	38.39%	36.50%	28.90%
2028Q4	28.01%	38.39%	36.50%	28.88%

The comparison with historical averages, reported in Table 2.36, is more informative than the final level of the weighted LGD alone: it shows how far the end-2028 projection has actually moved from each segment's historical starting point.

Table 2.36: Comparison between historical average LGD and probability-weighted LGD projections

Segment	Historical Average LGD	Weighted LGD (2028Q4)
Retail	27.12%	28.01%
SME	38.73%	38.39%
Corporate	36.73%	36.50%
Financial Institutions	25.79%	28.88%

For Corporate the deviation is minimal (0.23 percentage points), for SME it is slightly higher (0.34 percentage points), for Retail it amounts to roughly 0.9 points, while for Financial Institutions it exceeds three percentage points (3.09 points), an ordering consistent with the different historical volatility of the four series already discussed above (Table 2.1).

In summary, the weighted measure simply condenses into a single figure a pattern already present in the underlying data: where historical LGD is stable, the projection converges to the mean; where the historical series is more volatile, the structural component carries relatively less weight and the projection departs more markedly from the past. This confirms that the structural component of LGD remains predominant, while macroeconomic variables modulate its dynamics around historical levels without generating persistent shifts, a result consistent with the probability-weighted logic required by IFRS 9.

3. Application: A Comparison with Crook & Bellotti (2012)

To verify the consistency of the methodology adopted in this thesis with existing academic practice, this section examines in detail the study by Crook & Bellotti (2012), from which the analysis draws inspiration. The study is useful to understand how, when a financial institution has access to granular data at the level of individual exposures, rather than aggregated historical series such as those employed in the present thesis, the modelling procedure is progressively refined and calibrated according to the specific characteristics of the portfolio under consideration.
Consistently with the objective stated in its introduction, the authors structure the analysis, related to UK retail credit cards, around four research questions: *Which variables, among those collected at the time of the card application and those relating to the account status at the time of default, constitute the main drivers of retail LGD? Which modelling approach is most appropriate? How accurately can the constructed models predict LGD? Does the inclusion of macroeconomic variables lead to an actual improvement in the predictive performance of the models?*
As anticipated in 1.1.4, the analysis specifically covers approximately 55,000 credit card accounts in default in the United Kingdom, relating to four different card products offered by a single financial institution, over the period 1999-2005. The paper does not specify the commercial characteristics of each product in detail, for reasons of commercial confidentiality towards the data provider, as explicitly stated in a footnote. It is known, however, that the four products correspond to different segments of the same issuer's credit card portfolio, likely differing in risk tier, target customer base, or product features such as standard versus premium cards, interest rates, or credit limits. Furthermore, default is defined as failure to make the minimum required payment for three or more consecutive months, consistently with Basel II.
The authors identify five conceptual categories of circumstances that may influence the amount recovered on a defaulted loan:

- the debtor's individual characteristics observable at the time of the credit application, such as age, income, employment status, housing status, and residential address – included in the model, as it represents the richest source of available explanatory variables;

- information relating to the account at the time of default, such as the account opening date or age, and the outstanding balance – included in the model, as it conditions the analysis on the occurrence of the default event;

- changes in the debtor's personal circumstances over time – excluded, since the financial institution is not readily able to track such changes after the credit application;

- macroeconomic or business conditions at the time of default, possibly leading or lagging that date – constituting the specific object of the investigation, and therefore included in all subsequent specifications;

- the financial institution's operational decisions, such as credit application acceptance policies or recovery procedures adopted – excluded due to the unavailability of data.

3.1 Methodology

Building on this conceptual framework, the authors define four alternative model structures, distinguished by the set of explanatory variables included. The first is a "simple" specification, with no covariates, which under an OLS regression is equivalent to predicting LGD as the historical mean value calculated over the estimation sample; the second is a specification including only individual variables, referred to as account variables (AV); and the third is a specification that adds macroeconomic variables to these (AV&MV). The last represents instead a specification that includes interaction terms between individual and macroeconomic variables, allowing the effect of macroeconomic conditions on LGD to vary according to the debtor's individual characteristics. Crook & Bellotti (2012) adopt the Recovery Rate (RR) as the target variable, defined as the complement to one of LGD, consistently with the same relationship already introduced in this thesis (Equation (1.6) and Equation (1.7)), i.e., the ratio between the sum of payments received over a given period following default and the outstanding balance at the date of default. The chosen recovery period is twelve months, based on an operational assessment (the authors verify, in a dedicated section, that a model so calibrated is also able to provide adequate forecasts for a longer twenty-four-month horizon, through a simple linear calibration between the two measures). Since the empirical distribution of RR is markedly bimodal and concentrated at the two extremes of the interval $[0, 1]$, the authors systematically compare four alternative estimation frameworks:

- OLS regression applied directly to RR, within which three alternative transformations of the dependent variable are further considered:

  1. the fractional logit transformation, defined as

$$T(RR) = \log(RR) - \log(1 - RR) \tag{3.1}$$

previously applied also by Dermine & Neto de Carvalho (2006) to the modelling of LGD for corporate loans;

2. the transformation based on the Beta distribution, defined as

$$T(RR) = \Phi^{-1}\left(\text{beta}(RR, \alpha, \beta, 0, 1)\right) \tag{3.2}$$

where Φ denotes the cumulative distribution function of the standard normal distribution and α, β are the shape parameters of the Beta distribution, estimated by maximum likelihood;

3. a non-parametric probit transformation, defined as

$$T(RR) = \Phi^{-1}\left(\frac{|\{i : R_i \leq RR\}|}{n}\right) \tag{3.3}$$

constructed on the basis of the empirical distribution of the sample, where $R_1, \ldots, R_n$ are the observed RR values in the sample.

- a two-limit Tobit model, based on an unobserved latent variable RR_i^* defined as

$$RR_i^* = \beta \cdot x_i + \varepsilon_i, \tag{3.4}$$

where x_i is the vector of explanatory variables for account i, β the vector of coefficients, and ε_i the error term. The observed recovery rate is then obtained through the truncation

$$RR_i = \min(1, \max(0, RR_i^*)), \tag{3.5}$$

which constrains predictions to $[0, 1]$ by assigning a value of zero or one whenever the latent variable falls outside this range. Assuming normality of ε_i, the model is estimated by maximum likelihood.

- a hybrid decision-tree-structured model, which separately treats cases of full recovery ($RR = 1$), total loss ($RR = 0$), and partial recovery ($0 < RR < 1$) through three submodels: two logistic regressions estimate, for account i, p_{1i} and p_{0i}, the probabilities of full recovery and total loss respectively, while an OLS regression estimates L_i, the expected RR conditional on partial recovery. The overall prediction for account i is then obtained as a weighted combination as follows:

$$\widehat{RR}_i = (1 - p_{0i})\, p_{1i} + (1 - p_{1i})\, L_i. \tag{3.6}$$

- a least absolute value regression, estimated on the same explanatory variables x_i, which minimises the sum of absolute residuals rather than their squares; this criterion is motivated

by the hypothesis that, for LGD, large prediction errors should not be penalised more heavily than proportionally, as instead happens under standard OLS.

It is precisely at this point that the closest point of contact between the study by Crook & Bellotti (2012) and the methodology developed in the present thesis is located: of the seven specifications tested overall by the authors, two, the fractional logit transformation and the Beta transformation, correspond exactly to the Logit-OLS and Beta-OLS models employed in this thesis (Sections 2.4.2 and 2.4.3). These are indeed the specifications regarded as the most established in the literature (Section 1.1.4) and the most directly comparable to one another, since both involve a linear OLS estimation stage following the transformation.

Beyond this foundational conceptual alignment, however, the two studies diverge in a number of operational respects. These departures do not reflect conflicting theoretical principles: they are, rather, the cumulative result of a single underlying difference in the nature of the data, from which derives each of the methodological differences described below, before the discussion turns, in Section 2.4.4, to the results obtained under these two specifications and to the specific set of variables retained by Crook & Bellotti (2012) in their final model.

First of all Crook & Bellotti (2012) have access to rich individual-level information, which allows them to build interaction terms between borrower characteristics and macroeconomic conditions; their forward selection procedure, guided by the Akaike Information Criterion, works over this larger set of candidate variables. In the present thesis, where only macroeconomic variables are available, interaction terms cannot be formed, and the incremental selection logic described in Section 2.4.1 uses the Adjusted R^2 directly as the selection criterion at each step.

The two studies also differ in the lag structure considered: Crook & Bellotti (2012) test macroeconomic variables only at three fixed points in time – the date of default, and six months before and after it – whereas this thesis systematically evaluates lags of up to eight quarters for each candidate variable, in line with the evidence on delayed transmission discussed in the literature (Section 1.1.4) and consistent with the more extensive lag structure adopted by Konečný et al. (2017). Nevertheless both studies find VIF values well below critical thresholds (Table 2.12), confirming the absence of relevant multicollinearity in the final specifications.

The distributional characteristics of the aggregated data also shape the choices made at the estimation and diagnostic stages. As regards the Beta distribution parameters, Crook & Bellotti (2012) rely on maximum likelihood, an approach that proved impractical in the present setting: for some of the portfolio segments considered, the likelihood optimisation did not reach convergent estimates, most likely due to the limited sample size combined with the high concentration of historical LGD around segment mean values. The method of moments, described in Section 2.4.3, has been therefore adopted instead, yielding stable parameter estimates for all four segments.

A parallel difference emerges in the treatment of residuals: the markedly bimodal distribution of recovery rates at the account level produces non-normal residuals in Crook & Bellotti (2012), necessitating a bootstrap procedure to construct reliable coefficient confidence intervals; in the present thesis, where aggregation smooths out the extreme observations responsible for

this non-normality, the Shapiro-Wilk test fails to reject normality for any of the four portfolio segments under either specification (Tables 2.16 and 2.24), and standard OLS inference is therefore applicable without modification.

Finally, the limited historical depth of the series available in this thesis shapes the approach to predictive validation. Crook & Bellotti (2012) use a rolling hold-out procedure over ten consecutive test quarters. For each of them, the model is estimated on all the data available before that period, using at least three years of prior data to make sure the economic cycle is properly covered. In the present thesis, the thirty-one available quarterly observations preclude an analogous technique; predictive accuracy is therefore assessed in-sample, through MAE, MSE, and RMSE computed over the full estimation window (Tables 2.19 and 2.27).

In summary, the overall modelling structure of the two studies shares the same conceptual foundation. The differences that emerge along the way are not *ad hoc*, but represent a coherent, step-by-step adaptation of the same underlying methodology to different data, and to the empirical results generated at each stage of the analysis.

3.2 Results

The specification identified by Crook & Bellotti (2012) as the reference explanatory model is the OLS regression jointly including individual and macroeconomic variables (AV&MV), estimated on the entire available sample relating to the period 1999–2005. Table 3.1 reports its coefficients, as presented by the authors.

Individual variables emerge as the primary determinants of recovery rate, accounting for the bulk of the explanatory power of the model, as will be shown later in this chapter, when macroeconomic variables are added to a specification that already includes individual variables alone.

Among these, housing status proves particularly informative: relative to the excluded category of homeowners, all alternative tenure arrangements, such as council tenancy, private rental, term-time accommodation, and living with parents, are associated with significantly lower recovery rates. This points to the role of housing status as a proxy for the borrower's financial stability.

Length of relationship with the bank carries a positive and significant coefficient, suggesting that longer-standing customers tend to be more engaged in repayment efforts, or that the bank itself, having accumulated more information on these customers over time, is better able to tailor its recovery procedures to them.

The outstanding balance at default enters with a negative sign, indicating that larger exposures are proportionally harder to recover.

The credit bureau score, which summarises the borrower's overall credit history, is positive and highly significant, confirming that borrowers with a stronger credit profile before default tend to recover more.

Table 3.1: Coefficient estimates, standard errors, and statistical significance for the OLS recovery rate model (specification AV&MV, estimated on the full 1999–2005 sample), as reported by Crook & Bellotti (2012)

Variable	**Coefficient**	**Std. Error**	**z**	$P > \|z\|$
Intercept	1.43	0.162	8.83	0
Home status (D):				
Council tenant	−0.158	0.00544	−29.0	0
Private tenant	−0.146	0.00451	−32.4	0
Term time accommodation	−0.118	0.0410	−2.87	0.004
Other	−0.116	0.00702	−16.6	0
Lives with parents	−0.0643	0.00577	−11.1	0
Excluded category: Own home				
Time with bank	0.0812	0.00716	11.4	0
No data on time with bank (D)	0.0220	0.00774	2.84	0.004
Income (log)	0.0280	0.00770	3.63	0
Income unknown (D)	0.129	0.0328	3.94	0
Number of cards	−0.0153	0.00335	−4.56	0
Years at current address	0.00175	0.000238	7.36	0
Employment status (D):				
Home maker	−0.0363	0.0109	−3.34	0.001
Unemployed	−0.0338	0.0230	−1.47	0.141
Retired	−0.00570	0.0102	−0.56	0.577
Part time	−0.00451	0.00904	−0.50	0.618
Self employed	−0.00339	0.00525	−0.65	0.519
Other	0.0230	0.00622	3.69	0
Student	0.0266	0.0120	2.21	0.027
Excluded category: Employed				
Age at default	−0.00348	0.000187	−18.6	0
Age unknown (D)	0.0586	0.0146	4.03	0
Credit bureau score	0.186	0.0117	15.9	0
Demographic group (D):				
Council or poor housing area	−0.0404	0.00385	−10.5	0
Rural area	0.0349	0.00933	3.74	0
Suburban or wealthy area	0.0355	0.00498	7.11	0
Excluded category: Other				
Credit card product (D):				
Product 1	0.00744	0.00523	1.42	0.155
Product 2	0.00803	0.00660	1.22	0.223
Product 3	0.0147	0.00505	2.90	0.004
Excluded category: Product 4				
Time on books at default (months)	0.000489	0.000133	3.68	0
Balance at default (log)	−0.165	0.00369	−44.8	0
Date of default	−0.0000161	0.000006	−2.74	0.006
annual effect (standardized)	−0.00588			
Macroeconomic variables:				
Bank interest rates	−0.0505	0.00296	−17.1	0
standardized estimate	−0.0404			
Unemployment level	−0.000191	0.000048	−3.94	0
standardized estimate	−0.0195			
Earnings growth (log)	0.839	0.484	1.73	0.083
standardized estimate	+0.0032			

Age at default, by contrast, enters with a negative sign, a finding that may reflect the greater financial complexity and accumulated obligations of older borrowers, or differences in the composition of assets available for recovery.

Among the macroeconomic variables, bank interest rates and the unemployment level are both significant at the 1% level and carry a negative sign, indicating that higher interest rates and higher unemployment are associated with lower recovery rates and therefore higher LGD, reflecting reduced borrowers' ability to service their obligations and a more adverse economic environment for debt collection. The standardised coefficient of the interest rate is more than double that of unemployment, confirming it as the quantitatively most relevant macroeconomic driver in the study.

Earnings growth, by contrast, does not reach conventional statistical significance, despite carrying the expected positive sign.

These findings are broadly consistent with the results obtained in this thesis, where the change in the three-month Euribor rate emerges as the single most robust macroeconomic driver across all portfolio segments (Section 2.4.4), in direct analogy with the bank interest rate variable in Crook & Bellotti (2012). The negative sign observed in both studies, however, works through different channels: in Crook & Bellotti (2012), higher interest rates directly increase borrowers' debt burden and reduce their chances of recovery; in the present thesis, the negative sign of $\Delta\text{Euribor}_{t-2}$ reflects instead the dominance of the macro-financial channel over the mechanical discounting effect of the EBA recovery rate formula, as discussed in Section 2.4.4. The two findings are therefore not directly comparable in magnitude or interpretation, but confirm that interest-rate dynamics are the macroeconomic variable most systematically associated with LGD across different data and portfolio types.

By contrast, the role of unemployment highlights a key empirical divergence between the two studies. While the unemployment level is a highly significant macroeconomic determinant in Crook & Bellotti (2012), which makes sense given that, in granular individual-level data, unemployment is directly connected to a borrower's ability to repay, the unemployment rate did not prove statistically significant across the aggregated LGD series analysed in this thesis (Section 2.4.1).

With respect to model goodness-of-fit, the Adjusted R^2 of the AV&MV model is equal to 0.110, up from 0.105 for the specification including only individual variables (AV), confirming that macroeconomic variables provide a statistically meaningful, but modest, incremental contribution to explanatory power to the more simple model. By comparison, the Adjusted R^2 values obtained in this thesis for the Logit-OLS and Beta-OLS models range between 51.6% and 63.6% (Tables 2.14 and 2.23). This gap fits well with a key difference in the data. Individual-account recovery rates depend mainly on factors specific to each borrower and each exposure, which a model based only on macroeconomic variables cannot capture. Portfolio-level LGD series, however, already average out much of this individual-level noise. What remains is a signal more closely tied to the macroeconomic environment, and therefore easier for a macro-only model to explain.

Table 3.2: Aggregate forecast results (MSE and Abs RR diff) for the different model structures tested through the rolling predictive validation procedure, as reported by Crook & Bellotti (2012)

Recovery period	Product	RR transform	Model	Explanatory variables	MSE	Abs RR diff
12 month	All	None	OLS regression	None (simple)	0.168	0.0384
			OLS regression	AV	0.156	0.0645
			OLS regression	AV & MV	**0.151**	**0.0130**
			OLS regression	AV & MV & interaction terms	0.152	0.0169
			OLS regression	AV & MV lag 6	0.155	0.0529
			OLS regression	AV & MV lead 6	0.154	0.0445
			Tobit	AV & MV	0.163	0.0620
			Decision tree	AV & MV	0.221	0.1766
			LAD regression	AV & MV	0.173	0.1144
		Beta	OLS regression	AV & MV	0.166	0.0748
		Logit	OLS regression	AV & MV	0.175	0.1086
		Probit	OLS regression	AV & MV	0.198	0.2041
		Beta	Tobit	AV & MV	0.188	0.1053
		Beta	Decision tree	AV & MV	0.192	0.0414
12 month	1	None	OLS regression	None (simple)	0.160	0.0383
			OLS regression	AV	0.159	0.0795
			OLS regression	AV & MV	0.155	0.0534
	2		OLS regression	None (simple)	0.175	0.0391
			OLS regression	AV	0.163	0.0732
			OLS regression	AV & MV	0.195	0.1361
	3		OLS regression	None (simple)	0.165	0.0380
			OLS regression	AV	0.155	0.0774
			OLS regression	AV & MV	0.151	0.0431
	4		OLS regression	None (simple)	0.156	0.0412
			OLS regression	AV	0.149	0.0771
			OLS regression	AV & MV	**0.142**	**0.0227**
24 month	All	None	OLS regression	AV & MV	0.194	0.0554
			OLS (12-month model)	AV & MV	**0.183**	**0.0525**

Table 3.2 reports the results of the rolling predictive validation procedure, comparing the different model structures in terms of mean squared error at the individual-account level (MSE) and absolute deviation between predicted and observed mean RR at the portfolio level (Abs RR diff). The AV&MV specification with untransformed OLS is systematically the best performing, both relative to the specification without macroeconomic variables and relative to each of the Beta, Logit, and Probit transformations, as well as relative to Tobit, the decision tree, and the LAD regression. Regarding the six-month lags or leads on the macroeconomic variables, they worsen the forecasts relative to the contemporaneous value. When the four products are analysed separately, the benefit of AV&MV appears more uncertain: it improves on the simple specification for Product 4 on both measures, but underperforms on at least one measure for Products 1, 2, and 3, and on both measures for Product 2 specifically. The authors attribute this effect to the fact that product-specific idiosyncrasies is attenuated by aggregation.

Finally, the model calibrated over twelve months provides more accurate forecasts even for the twenty-four-month horizon than a model estimated directly over that horizon. These results carry an important implication: even in a setting where individual-level information is rich and macroeconomic variables add only a modest increment to in-sample fit, their inclusion systematically improves out-of-sample forecasting performance, reinforcing the case for their use in forward-looking applications such as IFRS 9 expected credit loss estimation.

Building precisely on this forward-looking perspective, the scenario analysis constitutes a final methodological point of comparison. Crook & Bellotti (2012) conduct a preliminary stress-testing experiment by substituting historical extreme values of the interest rate into their model, obtaining changes in mean RR of -17% and $+24\%$. However, this remains a simple point-in-time sensitivity exercise, lacking a probability-weighted multi-scenario structure. The authors themselves acknowledge its limitations, noting that the linear nature of the model may generate disproportionate effects at extreme values. By contrast, the forward-looking projection exercise developed in this thesis (Section 2.5) overcomes these structural limitations, combining three scenario trajectories through a probability-weighted logic fully consistent with the IFRS 9 framework.

Returning to the four research questions posed at the outset, Crook & Bellotti (2012) conclude that the main drivers of retail LGD are the individual variables related to debtor stability, that the most effective predictive model is, somewhat surprisingly, the untransformed OLS rather than the specifications conceptually better suited to the bimodal nature of RR, that the overall fit remains modest but typical of the LGD literature, and that macroeconomic variables improve both the fit and the forecasts, despite the sample period 1999–2005 being devoid of significant recessions. In addition, Crook & Bellotti (2012) demonstrate that macroeconomic variables carry genuine explanatory and predictive value for LGD even when rich individual-level information is available: their incremental contribution is modest in-sample, but meaningful out-of-sample. In the present thesis, where no individual-level information is available, macroeconomic variables constitute the entirety of the information set, and the considerably higher Adjusted R^2 values

obtained suggest that, at the aggregated portfolio level, they capture a disproportionately larger share of the systematic variation in LGD than they do at the individual-account level.

4. Conclusion

This thesis addresses a narrow but underexplored question in the Italian literature: how much of LGD dynamics can be explained by the macroeconomic component alone, when the analysis relies on aggregated historical series rather than granular exposure-level data. This question builds on the broader context introduced from the beginning of this thesis: the transition from the *incurred loss* logic of IAS 39 to the *expected credit loss* framework of IFRS 9 requires financial institutions to estimate credit losses incorporating future macroeconomic scenarios rather than relying solely on losses already incurred. Because IFRS 9 is a *principle-based* standard that does not prescribe a single estimation methodology, institutions retain considerable discretion in designing their satellite models, a flexibility that has produced marked heterogeneity in practice across the banking sector (AIFIRM (2025); Canton et al. (2021)). It is precisely within this space of discretion that the present thesis situates itself, building its methodology step by step, from the diagnostic analysis of the LGD series, through the selection and transformation of the independent variables, to the estimation of the satellite models and, finally, to their translation into forward-looking, scenario-based projections.

The established LGD literature, from Dermine & Neto de Carvalho (2006) to Crook & Bellotti (2012), has been built predominantly on micro-level information relating to the individual borrower or exposure, with macroeconomic variables playing a complementary role. Without having access to proprietary and confidential data, the present analysis necessarily adopts the opposite perspective, relying exclusively on the historical LGD series of the Retail, SME, Corporate, and Financial Institutions portfolios published by the European Central Bank, and on seven candidate macro-financial regressors, namely Euro Area GDP growth, Italian GDP growth, the Italian unemployment rate, the price of oil, the three-month Euribor rate, the ten-year BTP yield, and the aggregate default rate, with the aim of isolating the explanatory contribution of the macroeconomic dimension alone.

The present analysis has to contend with challenges on two distinct fronts. On the data side, the historical series available are short, with only 31 quarterly observations per segment. On the modelling side, LGD itself is a notoriously difficult variable: its empirical distribution is typically U-shaped, concentrated at the two extremes of the $[0, 1]$ interval. This feature renders classical linear regression inadequate; for this reason, the Logit and Beta transformations are adopted in this thesis and, not by chance, are also among the seven tested by Crook & Bellotti (2012).

Against this backdrop, the results obtained are far from obvious: the Logit-OLS and Beta-OLS specifications converge to an extraordinary degree across all four segments, in coefficient signs, statistical significance and predictive accuracy. This convergence is consistent with the fact that the four LGD series analysed in this thesis, being aggregated at portfolio level, do not extend to the extremes of the $[0, 1]$ interval, unlike the individual-account data typically used in the literature. This idea finds further confirmation in Qi & Zhao (2011), who show that these transformations differ when observations lie exactly at the boundary values of 0 and 1.
However, the specifications both produce Adjusted R^2 values ranging from 0.5170 (Retail) to 0.6357 (Corporate) and considerably higher than those reported by Crook & Bellotti (2012), confirming the genuine explanatory power of macroeconomic variables for LGD. Already at this stage, Financial Institutions returns the lowest Adjusted R^2 among the four segments and is the only one for which the Logit-OLS and Beta-OLS specifications show a noticeable, if modest, divergence, a first hint of the segment-level heterogeneity examined more closely below.
Anyway, this strong in-sample fit does not automatically translate into large projected movements in LGD: the forward-looking projection exercise developed in this thesis, based on a structured framework of three probability-weighted scenarios (58% baseline, 21% adverse, 21% positive) calibrated on official Bank of Italy projections and EBA/ESRB stress-testing shocks, constitutes a further step relative to the simple what-if exercise conducted by Crook & Bellotti (2012), but it also reveals that a structural component of LGD, attributable to collateral, recovery processes and portfolio composition, plays an important role alongside the macroeconomic one.
The results of the analysis are examined in detail below by returning, one by one, to the research questions posed at the outset.
The first asks: *Which macroeconomic factors drive the dynamics of LGD for Corporate, SME, Retail, and Financial Institutions exposures?* The two-quarter lagged change in the three-month Euribor rate is the single driver common to all four segments, statistically significant at the 1% level under both the Logit-OLS and Beta-OLS specifications (Sections 2.4.2 and 2.4.3); its negative sign, observed consistently across segments, suggests that Euribor mainly proxies improvements in the broader macro-financial environment (stronger growth, healthier labour market conditions, higher collateral valuations), rather than the mechanical discounting effect one might expect from the regulatory framework governing the estimation of economic loss (Section 2.4.4; European Banking Authority (2017)).
Beyond this central driver, two further regressors, selected through the incremental procedure described in Section 2.4.1, help complete the picture, each capturing a distinct transmission channel. Oil prices are significant for Retail, SME, and Corporate, but not for Financial Institutions; their negative sign suggests that oil prices too proxy improvements in the broader macro-financial environment, since periods of higher energy demand tend to coincide with economic expansion, stronger corporate balance sheets, and more stable collateral values, rather than reflecting the sector-specific cost channel one might expect for energy-intensive industries,

as documented empirically following the energy price shock triggered by the Russia-Ukraine conflict (AIFIRM (2025)).

The ten-year BTP yield is significant only for Retail and SME, with a positive coefficient consistent with the role of sovereign risk in shaping financing conditions and, in the Italian context, real estate valuations, which directly affect the collateral underlying these two segments' recovery processes. For Corporate and Financial Institutions, by contrast, the relationship is not robust enough to support a clear economic reading.

Finally, the Covid dummy captures a genuinely heterogeneous shock across segments, an outcome that is itself economically meaningful. As discussed in Section 2.4.4, the pandemic period saw extraordinary support measures, including payment moratoria and the accompanying regulatory clarifications on their treatment under IFRS 9. These measures help explain why LGD is lower than usual for Retail and Financial Institutions during that period. SME, by contrast, is structurally more exposed to revenue shocks and liquidity strain, and still recorded a deterioration. Corporate, better able to draw on diversified funding sources, is essentially unaffected.

Does each of these segments exhibit its own distinct sensitivity to the economic cycle? The second question asks precisely this, and the answer is unambiguously yes, as can already be inferred from the discussion above. Corporate and SME behave as the two most structural portfolios: their explanatory power is the highest, their historical LGD volatility is the lowest, and their projected LGD reverts closely to historical averages by the end of the forecasting horizon. Retail occupies an intermediate position. Financial Institutions, by contrast, stands apart from the other three segments on every dimension considered: it has by far the highest historical volatility (4.07%, against a maximum of 1.54% elsewhere), and its Adjusted R^2 (0.5518) is second only to Retail's as the lowest among the four segments. It is also, together with Retail, one of only two segments for which normality of the Beta-transformed variable is rejected. Its gap between the projected and historical average LGD, at 3.09 percentage points, is more than ten times larger than that of Corporate. It is also the one whose scenario dispersion is the most pronounced, with differences between the adverse and positive scenarios exceeding three percentage points in some quarters. This divergence is not a statistical artefact. It follows directly from the marked level shift recorded in the Financial Institutions series from 2025 onwards, when LGD moves from a historical range of 24–27% to values exceeding 35%.

Although these discrepance may at first appears surprising, it is broadly consistent with what Crook & Bellotti (2012) document when disaggregating their analysis by credit card product, where the benefit of including macroeconomic variables varies considerably, and in one case even reverses, once idiosyncratic product-level effects are no longer smoothed out by aggregation.

The third question is: *Are macroeconomic variables alone, without any borrower-specific information, sufficient to explain a meaningful share of LGD dynamics?* The answer is certainly positive, but it is important to consider that this thesis works with aggregated, portfolio-level data, which already smooths out much of the idiosyncratic noise that would otherwise blur the relationship between LGD and the macroeconomic environment. This is precisely why the

Adjusted R^2 values obtained here, ranging from 0.5170 to 0.6357, are considerably higher than the 0.110 reported by Crook & Bellotti (2012), whose account-level data are far noisier.

The comparison with Crook & Bellotti (2012) is nonetheless instructive in the opposite direction: they show that macroeconomic variables retain genuine explanatory and predictive value even when rich borrower-specific information is already included in the model (Chapter 3), which suggests that, if such information were available here too, the explanatory power of the macroeconomic component documented in this thesis would likely be even more complete and convincing.

To conclude, the central point of this thesis is that the macroeconomic component is not a fallback relative to micro-level data, but an autonomous explanatory channel robust enough to sustain, on its own, a complete IFRS 9-consistent framework for estimating and projecting LGD, while remaining sensitive to the structural heterogeneity that distinguishes one portfolio segment from another.

The comparison with Crook & Bellotti (2012) and with their results demonstrates that a satellite model built solely on public sources is not a second-best compromise, but a methodologically valid and replicable alternative, ready to be extended along several directions: to longer historical series, to the integration of exposure-level granular data, to the other risk parameters required under IFRS 9 (namely Probability of Default and Exposure at Default), and, more broadly, to whatever macroeconomic scenario, geopolitical or environmental, the future may bring, supported by a different and suitably re-adapted set of macroeconomic variables. Among these, the growing weight of climate-related and broader ESG factors in credit risk assessment is increasingly expected by supervisory guidance to be incorporated into satellite models (AIFIRM (2025)); as these risk sources accumulate further historical depth, they may come to rival, or even surpass, interest-rate dynamics as drivers of LGD, particularly for segments such as Financial Institutions, whose already elevated sensitivity to the economic cycle suggests a correspondingly higher exposure to these emerging channels.

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